Convex Optimization

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Contents

1 Introduction **4**
1.1 Mathematical optimization . . . 4
1.2 Solving optimization problems . . . 4
1.3 Example: illumination . . . 5
1.4 Course goals and topics . . . 6

2 Convex Sets **7**
2.1 Nonlinear optimization and history . . . 7
2.2 Affine and convex sets . . . 7
2.3 Some important examples . . . 8
2.4 Operations that preserve convexity . . . 10
2.5 Generalized inequalities . . . 11

3 Convex Sets (Continued); Convex Functions **12**
3.1 Minimum and minimal elements . . . 12
3.2 Separating and supporting hyperplanes . . . 12
3.3 Dual cones and generalized inequalities . . . 13
3.4 Convex functions: definition and examples . . . 14
3.5 First- and second-order conditions . . . 15
3.6 Epigraph, sublevel sets, and Jensen's inequality . . . 16
3.7 Operations that preserve convexity . . . 17

4 Convex Functions (Continued) **18**
4.1 Pointwise maximum and supremum . . . 18
4.2 Composition . . . 18
4.3 Minimization and perspective . . . 19
4.4 The conjugate function . . . 20
4.5 Quasiconvex functions . . . 21

5 Convex Functions (Continued); Convex Optimization Problems **23**
5.1 Log-concave and log-convex functions . . . 23
5.2 Convexity with respect to generalized inequalities . . . 24
5.3 Optimization problems in standard form . . . 24
5.4 Convex optimization problems . . . 25
5.5 Equivalent convex problems . . . 27

6 Convex Optimization Problems (Continued) **28**
6.1 Quasiconvex optimization . . . 28
6.2 Linear optimization . . . 29
6.3 Quadratic optimization . . . 31
6.4 Geometric programming . . . 32
6.5 Generalized inequality constraints . . . 34
6.6 Semidefinite programming . . . 34

7 Convex Optimization Problems (Continued); Duality **36**
7.1 Vector optimization . . . 36
7.2 Scalarization . . . 37
7.3 The Lagrangian and the dual function . . . 38
7.4 Examples of dual functions . . . 38

8 Duality (Continued) **40**
8.1 Dual function and conjugate . . . 40
8.2 The dual problem . . . 40
8.3 Examples . . . 41
8.4 Geometric interpretation . . . 42
8.5 Optimality conditions . . . 43
8.6 Perturbation and sensitivity analysis . . . 44

9 Duality (Continued); Approximation and Fitting **46**
9.1 Duality and problem reformulations . . . 46
9.2 Problems with generalized inequalities . . . 47
9.3 Norm approximation . . . 48
9.4 Penalty function approximation . . . 48
9.5 Least-norm problems . . . 49
9.6 Regularized approximation . . . 50

10 Approximation and Fitting (Continued); Statistical Estimation **52**
10.1 Optimal input design . . . 52
10.2 Signal reconstruction . . . 52
10.3 Robust approximation . . . 53
10.4 Maximum likelihood estimation . . . 54
10.5 Logistic regression . . . 55

11 Statistical Estimation (Continued); Geometric Problems **56**
11.1 Binary hypothesis testing . . . 56
11.2 Experiment design . . . 57
11.3 Extremal volume ellipsoids . . . 59
11.4 Centering . . . 60

12 Geometric Problems (Continued); Numerical Linear Algebra **61**
12.1 Analytic center . . . 61
12.2 Linear discrimination . . . 62
12.3 Placement and facility location . . . 64
12.4 Matrix structure and algorithm complexity . . . 64
12.5 Linear equations that are easy to solve . . . 64
12.6 The factor-solve method . . . 65

13 Numerical Linear Algebra (Continued); Unconstrained Minimization **66**
13.1 LU, Cholesky, and LDL^T factorization . . . 66
13.2 Block elimination . . . 67
13.3 Unconstrained minimization . . . 68
13.4 Descent methods . . . 69
13.5 Gradient descent . . . 70

14 Unconstrained Minimization (Continued) **71**
14.1 Steepest descent . . . 71
14.2 Newton step . . . 72
14.3 Newton decrement . . . 73
14.4 Newton's method . . . 73

14.5 Classical convergence analysis . . . 73
14.6 Self-concordance . . . 74

15 Unconstrained Minimization (Continued); Equality Constrained Minimization **76**
15.1 Implementation of Newton's method . . . 76
15.2 Equality constrained minimization . . . 76
15.3 Newton's method with equality constraints . . . 77
15.4 Infeasible start Newton method . . . 78
15.5 Solving KKT systems . . . 79
15.6 Network flow optimization . . . 80
15.7 Analytic center of a linear matrix inequality . . . 81

16 Interior-Point Methods **82**
16.1 Inequality constrained minimization . . . 82
16.2 Logarithmic barrier and central path . . . 82
16.3 Barrier method . . . 85
16.4 Feasibility and phase I methods . . . 86

17 Interior-Point Methods (Continued); Convex-Cardinality Problems **88**
17.1 Complexity analysis via self-concordance . . . 88
17.2 Generalized inequalities . . . 89
17.3 Primal-dual interior-point methods . . . 91
17.4 Cardinality . . . 91
17.5 Applications of convex-cardinality problems . . . 92

18 Convex-Cardinality Problems (Continued); Conclusions **94**
18.1 Piecewise constant and piecewise linear fitting . . . 94
18.2 The ℓ_1-norm heuristic . . . 94
18.3 Regressor selection and sparse signal reconstruction . . . 96
18.4 Total variation reconstruction . . . 96
18.5 Iterated weighted ℓ_1 heuristic . . . 97
18.6 Extension to matrices . . . 98
18.7 Conclusions . . . 99

Lecture 1: Introduction

1.1 Mathematical optimization

Definition 1.1 [Optimization problem]. A (mathematical) optimization problem has the form

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq b_i, \quad i = 1, \ldots, m, \end{array}$$

with optimization variable $x = (x_1, \ldots, x_n)$, objective function $f_0 : \mathbb{R}^n \to \mathbb{R}$, and constraint functions $f_i : \mathbb{R}^n \to \mathbb{R}$. An optimal solution $x^\star$ has the smallest value of f_0 among all vectors that satisfy the constraints.

Application	Variables	Constraints	Objective
Portfolio optimization	amounts invested in different assets	budget, max./min. investment per asset, minimum return	overall risk or return variance
Device sizing in circuits	device widths and lengths	manufacturing limits, timing requirements, maximum area	power consumption
Data fitting	model parameters	prior information, parameter limits	misfit or prediction error

1.2 Solving optimization problems

The general optimization problem is very difficult to solve, and methods involve some compromise: very long computation time, or not always finding the solution. Certain problem classes are exceptions and can be solved efficiently and reliably: least-squares problems, linear programs, and convex optimization problems.

Least-squares. Minimize $\|Ax - b\|_2^2$. There is an analytical solution $x^\star = (A^T A)^{-1} A^T b$, reliable and efficient algorithms and software exist, and the computation time is proportional to $n^2 k$ for $A \in \mathbb{R}^{k \times n}$ (less if A is structured): a mature technology. Least-squares problems are easy to recognize, and a few standard techniques (weights, regularization terms) increase their flexibility.

Linear programming. Minimize $c^T x$ subject to $a_i^T x \leq b_i$, $i = 1, \ldots, m$. There is no analytical formula for the solution, but reliable and efficient algorithms exist, with computation time proportional to $n^2 m$ if $m \geq n$ (less with structure): also a mature technology. Linear programs are not as easy to recognize as least-squares problems; a few standard tricks convert problems into LPs, for example problems involving ℓ_1- or ℓ_∞-norms or piecewise-linear functions.

Definition 1.2 [Convex optimization problem]. A convex optimization problem is a problem

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq b_i, \quad i = 1, \ldots, m, \end{array}$$

whose objective and constraint functions are convex: $f_i(\alpha x + \beta y) \leq \alpha f_i(x) + \beta f_i(y)$ whenever $\alpha + \beta = 1$, $\alpha \geq 0$, $\beta \geq 0$.

Least-squares and linear programming are special cases. Convex problems have no analytical solution in general, but there are reliable and efficient algorithms whose computation time is roughly proportional to $\max\{n^3, n^2 m, F\}$, where F is the cost of evaluating the f_i and their first and second derivatives. Convex optimization is almost a technology. The difficulty is in modeling: convex problems are often hard to recognize, there are many tricks for transforming problems into convex form, and surprisingly many problems can be solved via convex optimization.

1.3 Example: illumination

There are m lamps illuminating n small, flat patches (Figure 1). The intensity I_k at patch k depends linearly on the lamp powers p_j:

$$I_k = \sum_{j=1}^{m} a_{kj} p_j, \qquad a_{kj} = r_{kj}^{-2} \max\{\cos\theta_{kj}, 0\},$$

where r_{kj} is the distance from lamp j to patch k and θ_{kj} the angle between the patch normal and the direction to the lamp. The problem is to achieve a desired illumination I_{des} with bounded lamp powers:

$$\begin{array}{ll} \text{minimize} & \max_{k=1,\ldots,n} |\log I_k - \log I_{\text{des}}| \\ \text{subject to} & 0 \leq p_j \leq p_{\max}, \quad j = 1, \ldots, m. \end{array}$$

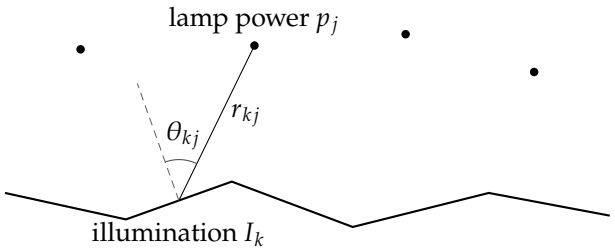


Figure 1: The illumination problem. Patch k receives power from lamp j in proportion to $\cos\theta_{kj}/r_{kj}^2$.

Approximate (suboptimal) approaches:

1. use uniform power $p_j = p$ and vary p;

2. use least-squares, minimizing $\sum_{k=1}^{n}(I_k - I_{\text{des}})^2$, then round p_j if $p_j > p_{\max}$ or $p_j < 0$;
3. use weighted least-squares, minimizing $\sum_{k=1}^{n}(I_k - I_{\text{des}})^2 + \sum_{j=1}^{m} w_j(p_j - p_{\max}/2)^2$, and iteratively adjust the weights w_j until $0 \le p_j \le p_{\max}$;
4. use linear programming, minimizing $\max_k |I_k - I_{\text{des}}|$ subject to $0 \le p_j \le p_{\max}$, which can be solved as an LP.

The fifth approach solves the problem exactly with convex optimization. Since $|\log I_k - \log I_{\text{des}}| = |\log(I_k/I_{\text{des}})|$ and $e^{|\log u|} = \max\{u, 1/u\}$, applying the increasing function exp to the objective shows the problem is equivalent to

$$\begin{array}{ll} \text{minimize} & f_0(p) = \max_{k=1,\ldots,n} h(I_k/I_{\text{des}}) \\ \text{subject to} & 0 \le p_j \le p_{\max}, \quad j = 1, \ldots, m, \end{array} \qquad h(u) = \max\{u, 1/u\}.$$

Here h is convex on $u > 0$ (Figure 2) and each I_k is linear in p, so f_0 is convex because a maximum of convex functions is convex. The exact solution costs about a modest factor times the effort of least-squares.

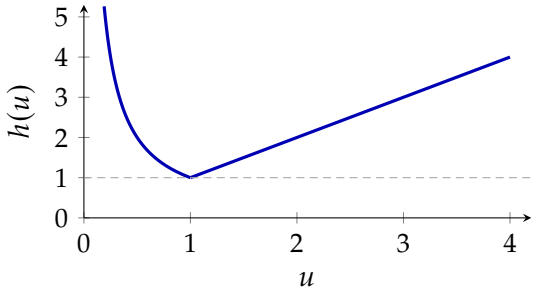


Figure 2: The function $h(u) = \max\{u, 1/u\}$, minimized at $u = 1$, that is, at $I_k = I_{\text{des}}$.

Adding constraints shows how unreliable untrained intuition is. With the constraint "no more than half of the total power is in any 10 lamps" the problem is still easy to solve; with "no more than half of the lamps are on ($p_j > 0$)" it becomes extremely difficult. Without the proper background, very easy problems can appear quite similar to very difficult ones.

1.4 Course goals and topics

The goals are to (1) recognize and formulate problems, such as the illumination problem, as convex optimization problems; (2) develop code for problems of moderate size (1000 lamps, 5000 patches); and (3) characterize the optimal solution (the optimal power distribution), give limits of performance, and so on. The topics are convex sets, functions, and optimization problems; examples and applications; and algorithms.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 1 (2023); original slides, *Introduction*.

Lecture 2: Convex Sets

2.1 Nonlinear optimization and history

Traditional techniques for general nonconvex problems involve compromises. Local optimization methods (nonlinear programming) find a point that minimizes f_0 among feasible points near it; they are fast and handle large problems, but require an initial guess and give no information about the distance to the global optimum. Global optimization methods find the global solution, but their worst-case complexity grows exponentially with problem size. Both are often based on solving convex subproblems.

The theory of convex analysis dates from about 1900 to 1970. On the algorithmic side: the simplex algorithm for linear programming (Dantzig, 1947); early interior-point methods in the 1960s (Fiacco and McCormick, Dikin); the ellipsoid method and other subgradient methods in the 1970s; polynomial-time interior-point methods for linear programming in the 1980s (Karmarkar, 1984); and polynomial-time interior-point methods for nonlinear convex optimization since the late 1980s (Nesterov and Nemirovski, 1994). Before 1990, applications were mostly in operations research, with few in engineering; since then there have been many new applications in engineering (control, signal processing, communications, circuit design) and new problem classes (semidefinite and second-order cone programming, robust optimization).

2.2 Affine and convex sets

Definition 2.1 [Affine and convex sets]. The line through x_1, x_2 is the set of points $x = \theta x_1 + (1-\theta)x_2$, $\theta \in \mathbb{R}$; the line segment between them restricts to $0 \le \theta \le 1$. A set is affine if it contains the line through any two distinct points in it, and convex if it contains the line segment between any two points in it:

$$x_1, x_2 \in C,\ 0 \le \theta \le 1 \implies \theta x_1 + (1-\theta)x_2 \in C.$$

The solution set $\{x \mid Ax = b\}$ of linear equations is affine, and conversely every affine set can be expressed as the solution set of a system of linear equations.

Definition 2.2 [Convex combination, hull, and cone]. A convex combination of $x_1, \ldots, x_k$ is a point $x = \theta_1 x_1 + \cdots + \theta_k x_k$ with $\theta_1 + \cdots + \theta_k = 1$ and $\theta_i \ge 0$. The convex hull $\operatorname{conv} S$ is the set of all convex combinations of points in S. A conic (nonnegative) combination of x_1, x_2 is a point $\theta_1 x_1 + \theta_2 x_2$ with $\theta_1, \theta_2 \ge 0$, and a convex cone is a set that contains all conic combinations of its points.

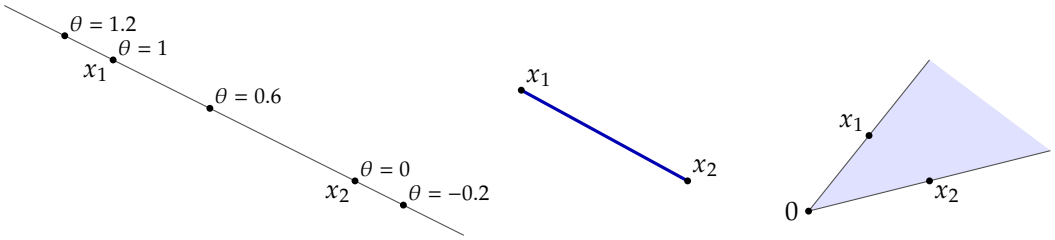


Figure 3: Combinations $\theta x_1 + (1-\theta)x_2$ of two points. Arbitrary θ traces the line, $\theta \in [0,1]$ the segment, and nonnegative combinations $\theta_1 x_1 + \theta_2 x_2$ fill a cone.

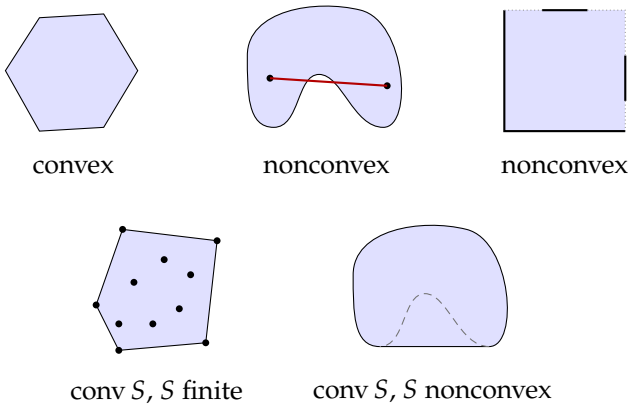


Figure 4: Top: a convex set, a set that contains two points but not the segment between them, and a square that contains only part of its boundary (solid edges). Bottom: the convex hull of a finite set and of a nonconvex set.

2.3 Some important examples

Definition 2.3 [Hyperplanes and halfspaces]. A hyperplane is a set $\{x \mid a^T x = b\}$ and a halfspace a set $\{x \mid a^T x \leq b\}$, with $a \neq 0$ the normal vector. Hyperplanes are affine and convex; halfspaces are convex (Figure 5).

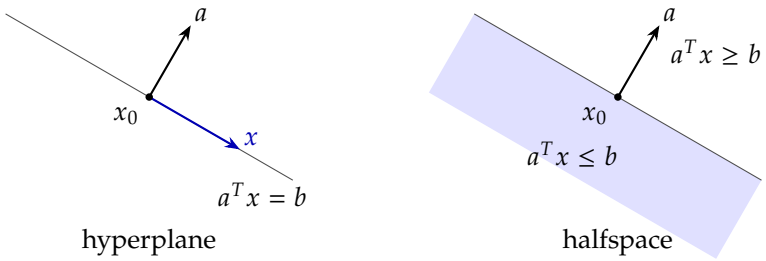


Figure 5: A hyperplane through x_0 consists of the points x with $a^T(x - x_0) = 0$; the halfspace $a^T x \leq b$ lies on the side opposite to a.

Balls and ellipsoids. The (Euclidean) BALL with center x_c and radius r is $B(x_c, r) = \{x \mid \|x - x_c\|_2 \leq r\} = \{x_c + ru \mid \|u\|_2 \leq 1\}$. An ELLIPSOID is a set

$$\{x \mid (x - x_c)^T P^{-1}(x - x_c) \leq 1\}, \qquad P \in \mathbf{S}^n_{++}$$

(P symmetric positive definite), or equivalently $\{x_c + Au \mid \|u\|_2 \leq 1\}$ with A square and nonsingular.

Norm balls and norm cones. A NORM is a function $\|\cdot\|$ with $\|x\| \geq 0$ and $\|x\| = 0$ iff $x = 0$; $\|tx\| = |t|\,\|x\|$ for $t \in \mathbb{R}$; and $\|x + y\| \leq \|x\| + \|y\|$. We write $\|\cdot\|$ for a general norm and $\|\cdot\|_{\text{symb}}$ for a particular one. The NORM BALL with center x_c and radius r is $\{x \mid \|x - x_c\| \leq r\}$ and the NORM CONE is $\{(x, t) \mid \|x\| \leq t\} \subseteq \mathbb{R}^{n+1}$. The Euclidean norm cone is the SECOND-ORDER CONE. Norm balls and cones are convex.

Definition 2.4 [POLYHEDRON]. A POLYHEDRON is the solution set of finitely many linear inequalities and equalities,

$$Ax \leq b, \qquad Cx = d \qquad (A \in \mathbb{R}^{m\times n},\ C \in \mathbb{R}^{p\times n}),$$

where $\leq$ is componentwise inequality. It is the intersection of a finite number of halfspaces and hyperplanes (Figure 6).

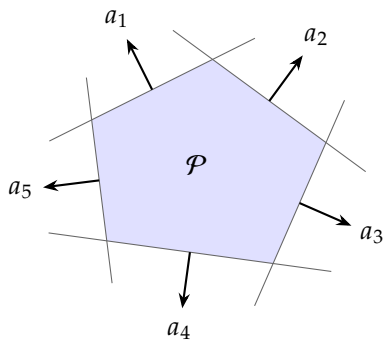


Figure 6: A polyhedron $\mathcal{P} = \{x \mid a_i^T x \leq b_i,\ i = 1, \ldots, 5\}$ with the outward normals a_i of its faces.

Positive semidefinite cone. $\mathbf{S}^n$ is the set of symmetric $n \times n$ matrices,

$$\mathbf{S}^n_+ = \{X \in \mathbf{S}^n \mid X \succeq 0\}, \qquad X \in \mathbf{S}^n_+ \iff z^T X z \geq 0 \text{ for all } z,$$

is the set of positive semidefinite matrices, and $\mathbf{S}^n_{++} = \{X \in \mathbf{S}^n \mid X \succ 0\}$ the positive definite ones. $\mathbf{S}^n_+$ is a convex cone, since $z^T X z \geq 0$ is a linear inequality in X for each z. For example,

$$\begin{bmatrix} x & y \\ y & z \end{bmatrix} \in \mathbf{S}^2_+ \iff x \geq 0,\ z \geq 0,\ xz \geq y^2,$$

a solid cone in $\mathbb{R}^3$ with a curved boundary.

2.4 Operations that preserve convexity

To establish convexity of a set C, either apply the definition directly, or show that C is obtained from simple convex sets (hyperplanes, halfspaces, norm balls, ...) by operations that preserve convexity: intersection, affine functions, the perspective function, and linear-fractional functions.

Intersection. The intersection of any number of convex sets is convex. For example, with $p(t) = x_1 \cos t + x_2 \cos 2t + \cdots + x_m \cos mt$,

$$S = \{x \in \mathbb{R}^m \mid |p(t)| \le 1 \text{ for } |t| \le \pi/3\} = \bigcap_{|t| \le \pi/3} \{x \mid |a(t)^T x| \le 1\}, \qquad a(t) = (\cos t, \ldots, \cos mt),$$

is an intersection of infinitely many slabs, hence convex (Figure 7).

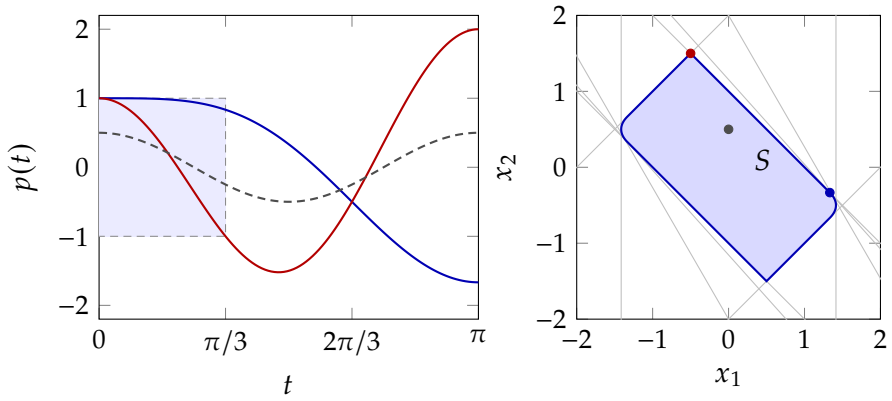


Figure 7: The intersection example for $m = 2$. Left: $p(t)$ for three points of S, each satisfying $|p(t)| \le 1$ on $[0, \pi/3]$ (the box). Right: S, with the boundaries of the slabs for $t = 0, \pi/12, \ldots, \pi/3$.

Affine functions. Let $f(x) = Ax + b$ with $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$. The image of a convex set under f is convex, and so is the inverse image:

$$S \subseteq \mathbb{R}^n \text{ convex} \implies f(S) = \{f(x) \mid x \in S\} \text{ convex},$$
$$C \subseteq \mathbb{R}^m \text{ convex} \implies f^{-1}(C) = \{x \in \mathbb{R}^n \mid f(x) \in C\} \text{ convex}.$$

Examples are scaling, translation, and projection; the solution set of a LINEAR MATRIX INEQUALITY $\{x \mid x_1 A_1 + \cdots + x_m A_m \preceq B\}$ with $A_i, B \in \mathbf{S}^p$ (the inverse image of $\mathbf{S}^p_+$ under $x \mapsto B - \sum_i x_i A_i$); and the hyperbolic cone $\{x \mid x^T P x \le (c^T x)^2,\ c^T x \ge 0\}$ with $P \in \mathbf{S}^n_+$ (the inverse image of the second-order cone under $x \mapsto (P^{1/2} x, c^T x)$).

Definition 2.5 [PERSPECTIVE AND LINEAR-FRACTIONAL FUNCTIONS]. The PERSPECTIVE FUNCTION $P : \mathbb{R}^{n+1} \to \mathbb{R}^n$ is $P(x, t) = x/t$ with $\operatorname{dom} P = \{(x, t) \mid t > 0\}$. A LINEAR-FRACTIONAL FUNCTION $f : \mathbb{R}^n \to \mathbb{R}^m$ is

$$f(x) = \frac{Ax + b}{c^T x + d}, \qquad \operatorname{dom} f = \{x \mid c^T x + d > 0\}.$$

Images and inverse images of convex sets under perspective and linear-fractional functions are convex.

For example, $f(x) = x/(x_1 + x_2 + 1)$ maps a convex set C in the region $x_1 + x_2 + 1 > 0$ to a convex set $f(C)$.

2.5 Generalized inequalities

Definition 2.6 [Proper cone]. A convex cone $K \subseteq \mathbb{R}^n$ is a proper cone if it is closed (contains its boundary), solid (has nonempty interior), and pointed (contains no line).

Examples: the nonnegative orthant $\mathbb{R}^n_+ = \{x \in \mathbb{R}^n \mid x_i \geq 0,\ i = 1, \ldots, n\}$; the positive semidefinite cone $\mathbf{S}^n_+$; and the nonnegative polynomials on $[0, 1]$, $K = \{x \in \mathbb{R}^n \mid x_1 + x_2 t + x_3 t^2 + \cdots + x_n t^{n-1} \geq 0 \text{ for } t \in [0, 1]\}$.

Definition 2.7 [Generalized inequality]. A proper cone K defines the generalized inequality

$$x \preceq_K y \iff y - x \in K, \qquad x \prec_K y \iff y - x \in \operatorname{int} K.$$

The two most common cases are componentwise inequality ($K = \mathbb{R}^n_+$), $x \preceq_{\mathbb{R}^n_+} y \iff x_i \leq y_i$ for all i, and matrix inequality ($K = \mathbf{S}^n_+$), $X \preceq_{\mathbf{S}^n_+} Y \iff Y - X$ is positive semidefinite; for these we drop the subscript K. Many properties of $\preceq_K$ are similar to those of $\leq$ on $\mathbb{R}$; for example, $x \preceq_K y,\ u \preceq_K v \implies x + u \preceq_K y + v$.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 2 (2023); original slides, *Introduction* and *Convex sets*.

Lecture 3: Convex Sets (Continued); Convex Functions

3.1 Minimum and minimal elements

A generalized inequality $\preceq_K$ is not in general a linear ordering: we can have $x \not\preceq_K y$ and $y \not\preceq_K x$.

Definition 3.1 [Minimum and minimal elements]. A point $x \in S$ is the minimum element of S with respect to $\preceq_K$ if $y \in S \implies x \preceq_K y$. It is a minimal element of S with respect to $\preceq_K$ if $y \in S,\ y \preceq_K x \implies y = x$.

Equivalently, x is the minimum element if $S \subseteq x + K$, and minimal if $(x - K) \cap S = \{x\}$ (Figure 8).

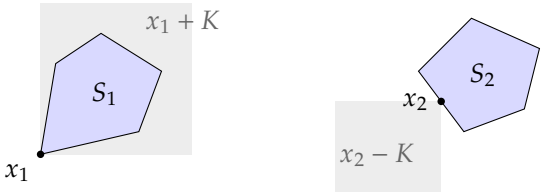


Figure 8: With $K = \mathbb{R}^2_+$, x_1 is the minimum element of S_1, and x_2 is a minimal element of S_2.

3.2 Separating and supporting hyperplanes

Theorem 3.2 [Separating hyperplane theorem]. *If C and D are nonempty disjoint convex sets, there exist $a \neq 0$ and b such that*

$$a^T x \leq b \text{ for } x \in C, \qquad a^T x \geq b \text{ for } x \in D,$$

that is, the hyperplane $\{x \mid a^T x = b\}$ separates C and D.

Strict separation requires additional assumptions, for example C closed and D a singleton.

Definition 3.3 [Supporting hyperplane]. A supporting hyperplane to a set C at a boundary point x_0 is a hyperplane $\{x \mid a^T x = a^T x_0\}$ with $a \neq 0$ and $a^T x \leq a^T x_0$ for all $x \in C$.

Theorem 3.4 [Supporting hyperplane theorem]. *If C is convex, there exists a supporting hyperplane at every boundary point of C.*

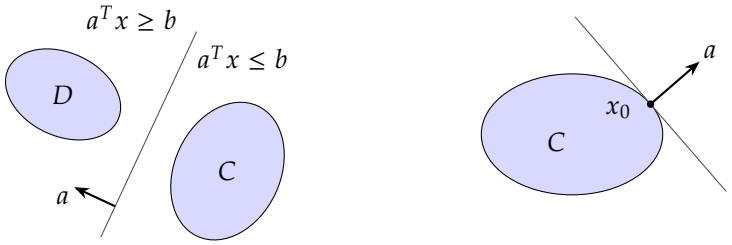


Figure 9: Left: a hyperplane separating disjoint convex sets C and D. Right: a supporting hyperplane to C at the boundary point x_0.

3.3 Dual cones and generalized inequalities

Definition 3.5 [DUAL CONE]. The DUAL CONE of a cone K is $K^* = \{y \mid y^T x \geq 0 \text{ for all } x \in K\}$.

Examples:

- $K = \mathbb{R}^n_+$: $K^* = \mathbb{R}^n_+$;
- $K = \mathbf{S}^n_+$: $K^* = \mathbf{S}^n_+$ (with the inner product $\mathrm{tr}(XY)$);
- $K = \{(x,t) \mid \|x\|_2 \leq t\}$: $K^* = \{(x,t) \mid \|x\|_2 \leq t\}$;
- $K = \{(x,t) \mid \|x\|_1 \leq t\}$: $K^* = \{(x,t) \mid \|x\|_\infty \leq t\}$.

The first three are SELF-DUAL cones. Dual cones of proper cones are proper, hence define generalized inequalities:

$$y \succeq_{K^*} 0 \iff y^T x \geq 0 \text{ for all } x \succeq_K 0.$$

Dual inequalities characterize minimum and minimal elements through linear functions.

- **Minimum element.** x is the minimum element of S with respect to $\preceq_K$ if and only if, for all $\lambda \succ_{K^*} 0$, x is the unique minimizer of $\lambda^T z$ over $z \in S$.
- **Minimal element.** If x minimizes $\lambda^T z$ over S for some $\lambda \succ_{K^*} 0$, then x is minimal. Conversely, if x is a minimal element of a *convex* set S, then there exists a nonzero $\lambda \succeq_{K^*} 0$ such that x minimizes $\lambda^T z$ over S.

Geometrically, the hyperplane $\{z \mid \lambda^T z = \lambda^T x\}$ supports S at x. For a nonconvex S, some minimal points minimize no such linear function.

Example: *Optimal production frontier.* Different production methods use different amounts of resources $x \in \mathbb{R}^n$. The production set P consists of the resource vectors of all possible production methods, and the efficient (Pareto optimal) methods correspond to resource vectors that are minimal with respect to $\mathbb{R}^n_+$. In Figure 10 ($n = 2$), x_1, x_2, x_3 are efficient, while x_4 and x_5 are not; x_2 minimizes $\lambda^T x$ over P, where λ weights (prices) labor and fuel.

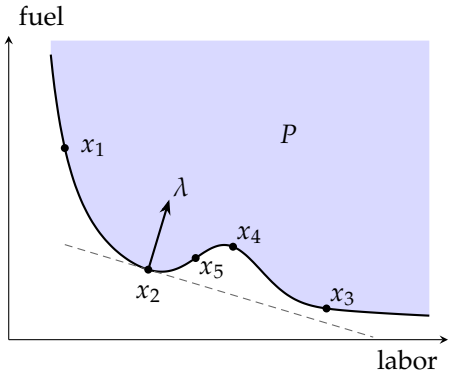


Figure 10: A production set P (schematic). The points x_1, x_2, x_3 are minimal; x_4 and x_5 use more of both resources than x_2. The dashed line $\{x \mid \lambda^T x = \lambda^T x_2\}$ supports P at x_2.

3.4 Convex functions: definition and examples

Definition 3.6 [Convex function]. A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if $\operatorname{dom} f$ is a convex set and

$$f(\theta x + (1-\theta)y) \le \theta f(x) + (1-\theta) f(y) \qquad \text{for all } x, y \in \operatorname{dom} f,\ 0 \le \theta \le 1.$$

It is concave if $-f$ is convex, and strictly convex if $\operatorname{dom} f$ is convex and the inequality is strict for $x \neq y, 0 < \theta < 1$.

Geometrically, the chord between $(x, f(x))$ and $(y, f(y))$ lies above the graph (Figure 11, left).

Examples on $\mathbb{R}$. Convex: affine $ax + b$; exponential e^{ax} for any $a \in \mathbb{R}$; powers x^α on $\mathbb{R}_{++}$ for $\alpha \ge 1$ or $\alpha \le 0$; powers of absolute value $|x|^p$ for $p \ge 1$; negative entropy $x \log x$ on $\mathbb{R}_{++}$. Concave: affine $ax + b$; powers x^α on $\mathbb{R}_{++}$ for $0 \le \alpha \le 1$; logarithm $\log x$ on $\mathbb{R}_{++}$.

Examples on $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$. Affine functions are convex and concave, and all norms are convex. On $\mathbb{R}^n$: affine $f(x) = a^T x + b$; norms $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ for $p \ge 1$ and $\|x\|_\infty = \max_k |x_k|$. On $\mathbb{R}^{m \times n}$: the affine function

$$f(X) = \operatorname{tr}(A^T X) + b = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} X_{ij} + b,$$

and the spectral (maximum singular value) norm $f(X) = \|X\|_2 = \sigma_{\max}(X) = (\lambda_{\max}(X^T X))^{1/2}$.

Proposition 3.7 [Restriction to a line]. *$f : \mathbb{R}^n \to \mathbb{R}$ is convex if and only if, for every $x \in \operatorname{dom} f$ and $v \in \mathbb{R}^n$, the function $g(t) = f(x + tv)$ with $\operatorname{dom} g = \{t \mid x + tv \in \operatorname{dom} f\}$ is convex in t.*

This reduces checking convexity to functions of one variable.

Example: $f(X) = \log\det X$ with $\operatorname{dom} f = \mathbf{S}^n_{++}$ is concave. For $X \succ 0$ and $V \in \mathbf{S}^n$,

$$g(t) = \log\det(X + tV) = \log\det X + \log\det(I + tX^{-1/2}VX^{-1/2}) = \log\det X + \sum_{i=1}^{n} \log(1 + t\lambda_i),$$

where λ_i are the eigenvalues of $X^{-1/2}VX^{-1/2}$. Each term is concave in t, so g is concave for every choice of $X \succ 0$ and V, hence f is concave.

Definition 3.8 [Extended-value extension]. The extended-value extension $\tilde{f}$ of f is $\tilde{f}(x) = f(x)$ for $x \in \operatorname{dom} f$ and $\tilde{f}(x) = \infty$ for $x \notin \operatorname{dom} f$.

It often simplifies notation. For example, the single condition $0 \le \theta \le 1 \implies \tilde{f}(\theta x + (1-\theta)y) \le \theta\tilde{f}(x) + (1-\theta)\tilde{f}(y)$, as an inequality in $\mathbb{R} \cup \{\infty\}$, means the same as the two conditions that $\operatorname{dom} f$ is convex and that the convexity inequality holds for $x, y \in \operatorname{dom} f$.

3.5 First- and second-order conditions

f is differentiable if $\operatorname{dom} f$ is open and the gradient $\nabla f(x) = (\partial f(x)/\partial x_1, \ldots, \partial f(x)/\partial x_n)$ exists at each $x \in \operatorname{dom} f$.

Theorem 3.9 [First-order condition]. *A differentiable f with convex domain is convex if and only if*

$$f(y) \ge f(x) + \nabla f(x)^T(y - x) \qquad \text{for all } x, y \in \operatorname{dom} f.$$

The first-order approximation of f is a global underestimator (Figure 11, right). Local information gives global information: in particular, if $\nabla f(x) = 0$ then $f(y) \ge f(x)$ for all y, so x is a global minimizer.

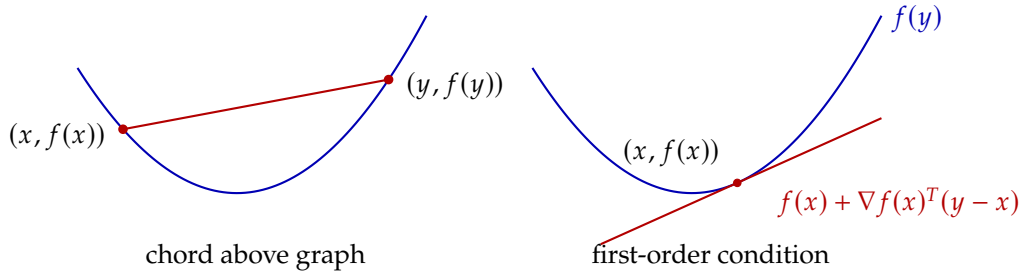


Figure 11: For convex f the chord lies above the graph (left), and the first-order approximation at x underestimates f everywhere (right).

f is twice differentiable if $\operatorname{dom} f$ is open and the Hessian $\nabla^2 f(x) \in \mathbf{S}^n$, $\nabla^2 f(x)_{ij} = \partial^2 f(x)/\partial x_i \partial x_j$, exists at each $x \in \operatorname{dom} f$.

Theorem 3.10 [Second-order conditions]. *Let f be twice differentiable with convex domain. Then f is convex if and only if $\nabla^2 f(x) \succeq 0$ for all $x \in \operatorname{dom} f$. If $\nabla^2 f(x) \succ 0$ for all $x \in \operatorname{dom} f$, then f is strictly*

convex.

Example:

- Quadratic function $f(x) = (1/2)x^T Px + q^T x + r$ with $P \in \mathbf{S}^n$: $\nabla f(x) = Px + q$, $\nabla^2 f(x) = P$; convex if $P \succeq 0$.
- Least-squares objective $f(x) = \|Ax - b\|_2^2$: $\nabla f(x) = 2A^T(Ax - b)$, $\nabla^2 f(x) = 2A^T A$; convex for any A.
- Quadratic-over-linear $f(x, y) = x^2/y$:

$$\nabla^2 f(x, y) = \frac{2}{y^3} \begin{bmatrix} y \\ -x \end{bmatrix} \begin{bmatrix} y \\ -x \end{bmatrix}^T \succeq 0,$$

  so f is convex for $y > 0$.
- Log-sum-exp $f(x) = \log \sum_{k=1}^n \exp x_k$ is convex. With $z_k = \exp x_k$,

$$\nabla^2 f(x) = \frac{1}{\mathbf{1}^T z} \operatorname{diag}(z) - \frac{1}{(\mathbf{1}^T z)^2} z z^T,$$

  and for every v,

$$v^T \nabla^2 f(x) v = \frac{(\sum_k z_k v_k^2)(\sum_k z_k) - (\sum_k v_k z_k)^2}{(\sum_k z_k)^2} \geq 0,$$

  since $(\sum_k v_k z_k)^2 \leq (\sum_k z_k v_k^2)(\sum_k z_k)$ by the Cauchy–Schwarz inequality applied to the vectors $(v_k \sqrt{z_k})$ and $(\sqrt{z_k})$.
- Geometric mean $f(x) = (\prod_{k=1}^n x_k)^{1/n}$ on $\mathbb{R}^n_{++}$ is concave, by a similar proof.

3.6 Epigraph, sublevel sets, and Jensen's inequality

Definition 3.11 [Sublevel set and epigraph]. The α-sublevel set of $f : \mathbb{R}^n \to \mathbb{R}$ is $C_\alpha = \{x \in \operatorname{dom} f \mid f(x) \leq \alpha\}$, and the epigraph is $\operatorname{epi} f = \{(x, t) \in \mathbb{R}^{n+1} \mid x \in \operatorname{dom} f,\ f(x) \leq t\}$.

Sublevel sets of convex functions are convex (the converse is false), and f is convex if and only if $\operatorname{epi} f$ is a convex set.

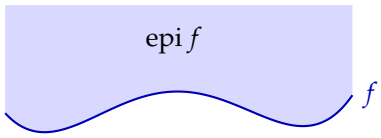


Figure 12: The epigraph of a nonconvex f, the set of points on or above its graph.

Theorem 3.12 [Jensen's inequality]. *If f is convex, then $f(\mathbb{E}z) \leq \mathbb{E}f(z)$ for any random variable z.*

The basic inequality $f(\theta x + (1-\theta)y) \le \theta f(x) + (1-\theta)f(y)$ is the special case of the discrete distribution with $\text{prob}(z = x) = \theta$, $\text{prob}(z = y) = 1 - \theta$.

3.7 Operations that preserve convexity

Practical methods for establishing convexity of a function: (1) verify the definition, often simplified by restricting to a line; (2) for twice differentiable functions, show $\nabla^2 f(x) \succeq 0$; (3) show that f is obtained from simple convex functions by operations that preserve convexity: nonnegative weighted sum, composition with an affine function, pointwise maximum and supremum, composition, minimization, and perspective.

Nonnegative weighted sums and affine composition. αf is convex if f is convex and $\alpha \ge 0$; $f_1 + f_2$ is convex if f_1, f_2 are convex (this extends to infinite sums and integrals); and $f(Ax + b)$ is convex if f is convex. Examples:

- the LOG BARRIER for linear inequalities,

$$f(x) = -\sum_{i=1}^{m} \log(b_i - a_i^T x), \qquad \text{dom } f = \{x \mid a_i^T x < b_i,\ i = 1, \ldots, m\};$$

- any norm of an affine function, $f(x) = \|Ax + b\|$.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 3 (2023); original slides, *Convex sets* and *Convex functions*.

Lecture 4: Convex Functions (Continued)

4.1 Pointwise maximum and supremum

Proposition 4.1 [Pointwise maximum]. *If $f_1, \ldots, f_m$ are convex, then $f(x) = \max\{f_1(x), \ldots, f_m(x)\}$ is convex.*

Examples:

- piecewise-linear functions $f(x) = \max_{i=1,\ldots,m}(a_i^T x + b_i)$;
- the sum of the r largest components of $x \in \mathbb{R}^n$, $f(x) = x_{[1]} + x_{[2]} + \cdots + x_{[r]}$, where $x_{[i]}$ is the ith largest component. It is convex because

$$f(x) = \max\{x_{i_1} + x_{i_2} + \cdots + x_{i_r} \mid 1 \le i_1 < i_2 < \cdots < i_r \le n\}$$

  is the maximum of $\binom{n}{r}$ linear functions.

Proposition 4.2 [Pointwise supremum]. *If $f(x, y)$ is convex in x for each $y \in \mathcal{A}$, then $g(x) = \sup_{y \in \mathcal{A}} f(x, y)$ is convex.*

Examples:

- the support function of a set C, $S_C(x) = \sup_{y \in C} y^T x$;
- the distance to the farthest point in a set C, $f(x) = \sup_{y \in C} \|x - y\|$;
- the maximum eigenvalue of a symmetric matrix, for $X \in \mathbf{S}^n$, $\lambda_{\max}(X) = \sup_{\|y\|_2 = 1} y^T X y$, a supremum of linear functions of X.

4.2 Composition

Proposition 4.3 [Composition with scalar functions]. *Let $g : \mathbb{R}^n \to \mathbb{R}$, $h : \mathbb{R} \to \mathbb{R}$, and $f(x) = h(g(x))$. Then f is convex if*

- *g is convex, h is convex, and $\tilde{h}$ is nondecreasing; or*
- *g is concave, h is convex, and $\tilde{h}$ is nonincreasing.*

Proof. For $n = 1$ and differentiable g, h,

$$f''(x) = h''(g(x))g'(x)^2 + h'(g(x))g''(x).$$

Both terms are nonnegative under either set of conditions. □

The monotonicity must hold for the extended-value extension $\tilde{h}$, not just on $\operatorname{dom} h$. Examples: $\exp g(x)$ is convex if g is convex, and $1/g(x)$ is convex if g is concave and positive.

Proposition 4.4 [Vector composition]. *Let $g : \mathbb{R}^n \to \mathbb{R}^k$, $h : \mathbb{R}^k \to \mathbb{R}$, and $f(x) = h(g(x)) = h(g_1(x), g_2(x), \ldots, g_k(x))$. Then f is convex if*

- *each g_i is convex, h is convex, and $\tilde{h}$ is nondecreasing in each argument; or*
- *each g_i is concave, h is convex, and $\tilde{h}$ is nonincreasing in each argument.*

Proof. For $n = 1$ and differentiable g, h,

$$f''(x) = g'(x)^T \nabla^2 h(g(x)) g'(x) + \nabla h(g(x))^T g''(x),$$

and again both terms are nonnegative. □

Examples: $\sum_{i=1}^m \log g_i(x)$ is concave if the g_i are concave and positive, and $\log \sum_{i=1}^m \exp g_i(x)$ is convex if the g_i are convex.

4.3 Minimization and perspective

Proposition 4.5 [Minimization]. *If $f(x, y)$ is convex in (x, y) and C is a convex set, then $g(x) = \inf_{y \in C} f(x, y)$ is convex.*

Example:

- Let $f(x, y) = x^T A x + 2x^T B y + y^T C y$ with

$$\begin{bmatrix} A & B \\ B^T & C \end{bmatrix} \succeq 0, \qquad C \succ 0.$$

  Minimizing over y (at $y = -C^{-1}B^T x$) gives $g(x) = \inf_y f(x, y) = x^T (A - BC^{-1}B^T)x$. Since g is convex, the Schur complement satisfies $A - BC^{-1}B^T \succeq 0$.

- The distance to a set, $\operatorname{dist}(x, S) = \inf_{y \in S} \|x - y\|$, is convex if S is convex.

Definition 4.6 [Perspective of a function]. The perspective of $f : \mathbb{R}^n \to \mathbb{R}$ is the function

$g : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$,

$$g(x,t) = tf(x/t), \qquad \operatorname{dom} g = \{(x,t) \mid x/t \in \operatorname{dom} f,\ t > 0\}.$$

If f is convex, then g is convex.

Examples:

- $f(x) = x^T x$ is convex, hence $g(x,t) = x^T x/t$ is convex for $t > 0$;
- the negative logarithm $f(x) = -\log x$ is convex, hence the RELATIVE ENTROPY $g(x,t) = t\log t - t\log x$ is convex on $\mathbb{R}^2_{++}$;
- if f is convex, then $g(x) = (c^T x + d)\, f\big((Ax+b)/(c^T x + d)\big)$ is convex on $\{x \mid c^T x + d > 0,\ (Ax+b)/(c^T x + d) \in \operatorname{dom} f\}$.

4.4 The conjugate function

Definition 4.7 [CONJUGATE FUNCTION]. The CONJUGATE of a function f is

$$f^*(y) = \sup_{x \in \operatorname{dom} f} \big(y^T x - f(x)\big).$$

For each y, $f^*(y)$ is the largest gap between the linear function $y^T x$ and $f(x)$; equivalently, the line of slope y that supports the graph of f crosses the vertical axis at $-f^*(y)$ (Figure 13). f^* is convex even if f is not, being a pointwise supremum of affine functions of y. The conjugate reappears in duality (Lecture 8).

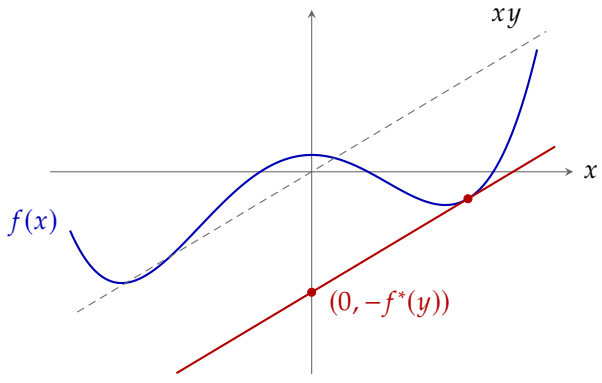


Figure 13: The conjugate of a nonconvex f at a fixed slope y. The line $yx - f^*(y)$, parallel to xy, supports the graph of f and meets the vertical axis at $(0, -f^*(y))$.

Example: The negative logarithm $f(x) = -\log x$ has

$$f^*(y) = \sup_{x>0}(xy + \log x) = \begin{cases} -1 - \log(-y) & y < 0, \\ \infty & \text{otherwise,} \end{cases}$$

since for $y < 0$ the supremum is attained at $x = -1/y$, and for $y \geq 0$ the objective is unbounded. For the strictly convex quadratic $f(x) = (1/2)x^T Q x$ with $Q \in \mathbf{S}^n_{++}$, the supremum of $y^T x - (1/2)x^T Q x$ is attained at $x = Q^{-1}y$, so $f^*(y) = \frac{1}{2}y^T Q^{-1} y$.

4.5 Quasiconvex functions

Definition 4.8 [Quasiconvex function]. $f : \mathbb{R}^n \to \mathbb{R}$ is quasiconvex if $\operatorname{dom} f$ is convex and the sublevel sets $S_\alpha = \{x \in \operatorname{dom} f \mid f(x) \leq \alpha\}$ are convex for all α. It is quasiconcave if $-f$ is quasiconvex, and quasilinear if it is both quasiconvex and quasiconcave.

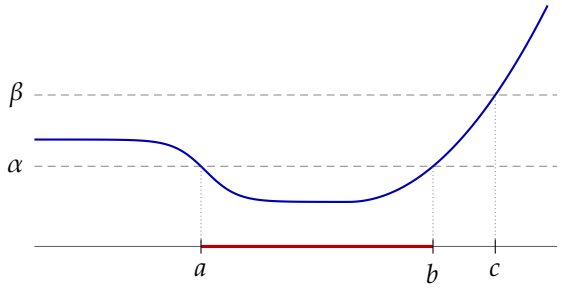


Figure 14: A quasiconvex function on $\mathbb{R}$ that is not convex. Its α-sublevel set is the interval $[a, b]$ and its β-sublevel set is $(-\infty, c]$.

Examples:

- $\sqrt{|x|}$ is quasiconvex on $\mathbb{R}$;
- $\operatorname{ceil}(x) = \inf\{z \in \mathbb{Z} \mid z \geq x\}$ is quasilinear;
- $\log x$ is quasilinear on $\mathbb{R}_{++}$;
- $f(x_1, x_2) = x_1 x_2$ is quasiconcave on $\mathbb{R}^2_{++}$;
- the linear-fractional function $f(x) = (a^T x + b)/(c^T x + d)$ with $\operatorname{dom} f = \{x \mid c^T x + d > 0\}$ is quasilinear, since each sublevel set $\{x \mid a^T x + b \leq \alpha(c^T x + d)\}$ and superlevel set is a halfspace intersected with $\operatorname{dom} f$;
- the distance ratio $f(x) = \|x - a\|_2 / \|x - b\|_2$ with $\operatorname{dom} f = \{x \mid \|x - a\|_2 \leq \|x - b\|_2\}$ is quasiconvex.

Example: *Internal rate of return.* Let $x = (x_0, \ldots, x_n)$ be a cash flow, with x_i the payment in period i (to us if $x_i > 0$), and assume $x_0 < 0$ and $x_0 + x_1 + \cdots + x_n > 0$. The present value at interest rate r is

$\mathrm{PV}(x, r) = \sum_{i=0}^{n}(1+r)^{-i}x_i$, and the internal rate of return is the smallest interest rate for which it vanishes, $\mathrm{IRR}(x) = \inf\{r \geq 0 \mid \mathrm{PV}(x, r) = 0\}$. IRR is quasiconcave, because its superlevel sets are intersections of open halfspaces:

$$\mathrm{IRR}(x) \geq R \iff \sum_{i=0}^{n}(1+r)^{-i}x_i > 0 \text{ for } 0 \leq r < R.$$

Properties. A quasiconvex f satisfies the MODIFIED JENSEN INEQUALITY

$$0 \leq \theta \leq 1 \implies f(\theta x + (1-\theta)y) \leq \max\{f(x), f(y)\}.$$

FIRST-ORDER CONDITION: a differentiable f with convex domain is quasiconvex if and only if

$$f(y) \leq f(x) \implies \nabla f(x)^T(y - x) \leq 0,$$

that is, $\nabla f(x)$ defines a supporting hyperplane to the sublevel set $\{y \mid f(y) \leq f(x)\}$ at x. Sums of quasiconvex functions are not necessarily quasiconvex.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 4 (2023); original slides, *Convex functions*.

Lecture 5: Convex Functions (Continued); Convex Optimization Problems

5.1 Log-concave and log-convex functions

Definition 5.1 [Log-concave function]. A positive function f is log-concave if $\log f$ is concave:

$$f(\theta x + (1-\theta)y) \geq f(x)^\theta f(y)^{1-\theta} \qquad \text{for } 0 \leq \theta \leq 1,$$

and log-convex if $\log f$ is convex.

Examples:

- powers x^a on $\mathbb{R}_{++}$ are log-convex for $a \leq 0$ and log-concave for $a \geq 0$;
- many common probability densities are log-concave, for example the normal density

$$f(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} e^{-\frac{1}{2}(x-\bar{x})^T \Sigma^{-1}(x-\bar{x})};$$

- the cumulative Gaussian distribution function $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-u^2/2}\, du$.

Properties.

- A twice differentiable f with convex domain is log-concave if and only if $f(x)\nabla^2 f(x) \preceq \nabla f(x) \nabla f(x)^T$ for all $x \in \operatorname{dom} f$.
- Products of log-concave functions are log-concave; sums are not always log-concave.
- *Integration*: if $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is log-concave, then $g(x) = \int f(x,y)\, dy$ is log-concave (not easy to show).

Consequences of the integration property:

- the convolution $(f * g)(x) = \int f(x-y)g(y)\, dy$ of log-concave functions f, g is log-concave;
- if $C \subseteq \mathbb{R}^n$ is convex and y is a random variable with log-concave pdf p, then $f(x) = \text{prob}(x + y \in C)$ is log-concave. Indeed, $f(x) = \int g(x+y)p(y)\, dy$ with g the indicator ($g(u) = 1$ for $u \in C$, 0 otherwise), which is log-concave, so the integrand is a product of log-concave functions.

Example: *Yield function.* Let $x \in \mathbb{R}^n$ be the nominal parameter values of a product, $w \in \mathbb{R}^n$ the random variations of the parameters in the manufactured product, and S the set of acceptable

values. The yield is $Y(x) = \text{prob}(x + w \in S)$. If S is convex and w has a log-concave pdf, then Y is log-concave, and the yield regions $\{x \mid Y(x) \geq \alpha\}$ are convex.

5.2 Convexity with respect to generalized inequalities

Definition 5.2 [K-convexity]. $f : \mathbb{R}^n \to \mathbb{R}^m$ is K-convex if $\text{dom}\, f$ is convex and

$$f(\theta x + (1-\theta)y) \preceq_K \theta f(x) + (1-\theta)f(y) \qquad \text{for } x, y \in \text{dom}\, f,\ 0 \leq \theta \leq 1.$$

Example: $f : \mathbf{S}^m \to \mathbf{S}^m$, $f(X) = X^2$ is $\mathbf{S}^m_+$-convex. For fixed $z \in \mathbb{R}^m$, $z^T X^2 z = \|Xz\|_2^2$ is convex in X, so $z^T(\theta X + (1-\theta)Y)^2 z \leq \theta z^T X^2 z + (1-\theta) z^T Y^2 z$ for $X, Y \in \mathbf{S}^m$, $0 \leq \theta \leq 1$. Since this holds for every z, $(\theta X + (1-\theta)Y)^2 \preceq \theta X^2 + (1-\theta)Y^2$.

5.3 Optimization problems in standard form

Definition 5.3 [Standard form]. An optimization problem in standard form is

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p, \end{array}$$

with optimization variable $x \in \mathbb{R}^n$, objective (cost) function $f_0 : \mathbb{R}^n \to \mathbb{R}$, inequality constraint functions f_i, and equality constraint functions h_i. The optimal value is

$$p^\star = \inf\{f_0(x) \mid f_i(x) \leq 0,\ i = 1, \ldots, m,\ h_i(x) = 0,\ i = 1, \ldots, p\},$$

with $p^\star = \infty$ if the problem is infeasible (no x satisfies the constraints) and $p^\star = -\infty$ if it is unbounded below.

Definition 5.4 [Optimal and locally optimal points]. x is feasible if $x \in \text{dom}\, f_0$ and it satisfies the constraints. A feasible x is optimal if $f_0(x) = p^\star$; X_{opt} is the set of optimal points. x is locally optimal if there is an $R > 0$ such that x is optimal for the problem with the extra constraint $\|z - x\|_2 \leq R$ on the variable z.

Examples with $n = 1$, $m = p = 0$:

- $f_0(x) = 1/x$, $\text{dom}\, f_0 = \mathbb{R}_{++}$: $p^\star = 0$, but there is no optimal point;
- $f_0(x) = -\log x$, $\text{dom}\, f_0 = \mathbb{R}_{++}$: $p^\star = -\infty$;
- $f_0(x) = x \log x$, $\text{dom}\, f_0 = \mathbb{R}_{++}$: $p^\star = -1/e$, and $x = 1/e$ is optimal;
- $f_0(x) = x^3 - 3x$: $p^\star = -\infty$, with a local optimum at $x = 1$.

Implicit constraints. The standard form problem has the implicit constraint

$$x \in \mathcal{D} = \bigcap_{i=0}^{m} \operatorname{dom} f_i \cap \bigcap_{i=1}^{p} \operatorname{dom} h_i,$$

where $\mathcal{D}$ is the DOMAIN of the problem. The constraints $f_i(x) \leq 0$, $h_i(x) = 0$ are the EXPLICIT CONSTRAINTS, and a problem is UNCONSTRAINED if it has none ($m = p = 0$). For example, minimizing $f_0(x) = -\sum_{i=1}^{k} \log(b_i - a_i^T x)$ is an unconstrained problem with implicit constraints $a_i^T x < b_i$.

Feasibility problems. The problem "find x subject to $f_i(x) \leq 0$, $h_i(x) = 0$" is the special case $f_0(x) = 0$ of the general problem: $p^\star = 0$ if the constraints are feasible, in which case any feasible x is optimal, and $p^\star = \infty$ if they are infeasible.

5.4 Convex optimization problems

Definition 5.5 [CONVEX OPTIMIZATION PROBLEM]. A CONVEX OPTIMIZATION PROBLEM in standard form is

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & a_i^T x = b_i, \quad i = 1, \ldots, p, \end{array}$$

often written with $Ax = b$, where $f_0, f_1, \ldots, f_m$ are convex and the equality constraints are affine. The problem is QUASICONVEX if f_0 is quasiconvex (and $f_1, \ldots, f_m$ are convex).

The feasible set of a convex optimization problem is convex.

Example: The problem

$$\begin{array}{ll} \text{minimize} & f_0(x) = x_1^2 + x_2^2 \\ \text{subject to} & f_1(x) = x_1/(1 + x_2^2) \leq 0 \\ & h_1(x) = (x_1 + x_2)^2 = 0 \end{array}$$

has convex f_0 and convex feasible set $\{(x_1, x_2) \mid x_1 = -x_2 \leq 0\}$, but it is not a convex problem by our definition: f_1 is not convex and h_1 is not affine. It is equivalent (but not identical) to the convex problem of minimizing $x_1^2 + x_2^2$ subject to $x_1 \leq 0$, $x_1 + x_2 = 0$.

Theorem 5.6 [LOCAL OPTIMA ARE GLOBAL]. *Any locally optimal point of a convex problem is (globally) optimal.*

Proof. Suppose x is locally optimal but some feasible y has $f_0(y) < f_0(x)$. Local optimality gives an $R > 0$ with z feasible, $\|z - x\|_2 \leq R \implies f_0(z) \geq f_0(x)$. Consider $z = \theta y + (1 - \theta)x$ with $\theta = R/(2\|y - x\|_2)$. Since $\|y - x\|_2 > R$, $0 < \theta < 1/2$. As a convex combination of two

feasible points, z is feasible; $\|z - x\|_2 = R/2$; and

$$f_0(z) \leq \theta f_0(y) + (1-\theta) f_0(x) < f_0(x),$$

contradicting local optimality of x. □

Theorem 5.7 [Optimality criterion]. *For a convex problem with differentiable f_0 and feasible set X, x is optimal if and only if it is feasible and*

$$\nabla f_0(x)^T (y - x) \geq 0 \qquad \text{for all feasible } y.$$

Sufficiency follows from the first-order condition, $f_0(y) \geq f_0(x) + \nabla f_0(x)^T (y - x) \geq f_0(x)$. Geometrically, if $\nabla f_0(x) \neq 0$, it defines a supporting hyperplane to X at x (Figure 15).

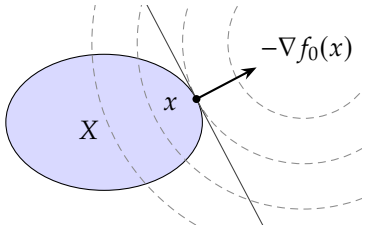


Figure 15: The optimality criterion. At the optimal x, the feasible set X lies on the side of the supporting hyperplane where $\nabla f_0(x)^T (y - x) \geq 0$; dashed curves are level sets of f_0.

Special cases:

- *Unconstrained problem*: x is optimal if and only if $x \in \operatorname{dom} f_0$ and $\nabla f_0(x) = 0$.
- *Equality constrained problem*, minimize $f_0(x)$ subject to $Ax = b$: x is optimal if and only if there exists a ν such that

  $$x \in \operatorname{dom} f_0, \qquad Ax = b, \qquad \nabla f_0(x) + A^T \nu = 0.$$

  (Feasible y are $x + v$ with $Av = 0$, so the criterion requires $\nabla f_0(x) \perp \mathcal{N}(A)$, that is, $\nabla f_0(x) \in \mathcal{R}(A^T)$.)
- *Minimization over the nonnegative orthant*, minimize $f_0(x)$ subject to $x \geq 0$: x is optimal if and only if

  $$x \in \operatorname{dom} f_0, \qquad x \geq 0, \qquad \begin{cases} \nabla f_0(x)_i \geq 0 & x_i = 0, \\ \nabla f_0(x)_i = 0 & x_i > 0. \end{cases}$$

  (The criterion $\nabla f_0(x)^T y \geq \nabla f_0(x)^T x$ for all $y \geq 0$ holds iff $\nabla f_0(x) \geq 0$ and $\nabla f_0(x)^T x = 0$.)

5.5 Equivalent convex problems

Two problems are (informally) EQUIVALENT if the solution of one is readily obtained from the solution of the other, and vice versa. Common transformations that preserve convexity:

- *Eliminating equality constraints.* Minimizing $f_0(x)$ subject to $f_i(x) \leq 0$, $Ax = b$ is equivalent to minimizing $f_0(Fz + x_0)$ over z subject to $f_i(Fz + x_0) \leq 0$, where F and x_0 are such that $Ax = b \iff x = Fz + x_0$ for some z.
- *Introducing equality constraints.* Minimizing $f_0(A_0x + b_0)$ subject to $f_i(A_ix + b_i) \leq 0$ is equivalent to minimizing $f_0(y_0)$ over x, y_i subject to $f_i(y_i) \leq 0$ and $y_i = A_ix + b_i$, $i = 0, 1, \ldots, m$.
- *Introducing slack variables* for linear inequalities. Minimizing $f_0(x)$ subject to $a_i^Tx \leq b_i$ is equivalent to minimizing $f_0(x)$ over x, s subject to $a_i^Tx + s_i = b_i$, $s_i \geq 0$.
- *Epigraph form.* The standard form convex problem is equivalent to

$$\begin{array}{ll} \text{minimize (over } x, t) & t \\ \text{subject to} & f_0(x) - t \leq 0 \\ & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & Ax = b. \end{array}$$

- *Minimizing over some variables.* Minimizing $f_0(x_1, x_2)$ subject to $f_i(x_1) \leq 0$ is equivalent to minimizing $\tilde{f}_0(x_1)$ subject to $f_i(x_1) \leq 0$, where $\tilde{f}_0(x_1) = \inf_{x_2} f_0(x_1, x_2)$.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 5 (2023); original slides, *Convex functions* and *Convex optimization problems*.

Lecture 6: Convex Optimization Problems (Continued)

6.1 Quasiconvex optimization

A quasiconvex optimization problem is

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \le 0, \quad i = 1, \dots, m \\ & Ax = b \end{array}$$

with $f_0 : \mathbb{R}^n \to \mathbb{R}$ quasiconvex and $f_1, \dots, f_m$ convex. Unlike a convex problem, it can have locally optimal points that are not globally optimal (Figure 16).

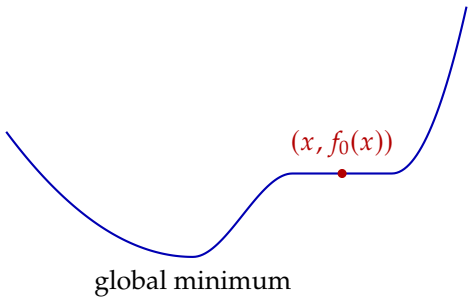


Figure 16: A quasiconvex f_0 on $\mathbb{R}$. The marked point lies on a flat stretch, so it is locally optimal, but it is not globally optimal.

Convex representation of sublevel sets. If f_0 is quasiconvex, there is a family of functions ϕ_t such that $\phi_t(x)$ is convex in x for fixed t, and the t-sublevel set of f_0 is the 0-sublevel set of ϕ_t:

$$f_0(x) \le t \iff \phi_t(x) \le 0.$$

For example, if $f_0(x) = p(x)/q(x)$ with p convex, q concave, and $p(x) \ge 0$, $q(x) > 0$ on $\operatorname{dom} f_0$, we can take $\phi_t(x) = p(x) - tq(x)$: for $t \ge 0$ it is convex in x, and $p(x)/q(x) \le t$ if and only if $\phi_t(x) \le 0$.

For fixed t, the convex feasibility problem

$$\phi_t(x) \le 0, \qquad f_i(x) \le 0, \quad i = 1, \dots, m, \qquad Ax = b \tag{6.1}$$

decides on which side of $p^\star$ the value t lies: if it is feasible, $t \ge p^\star$; if infeasible, $t \le p^\star$. Bisection on t then solves the quasiconvex problem.

Algorithm 6.1: Bisection for quasiconvex optimization

Input: $l \le p^\star$, $u \ge p^\star$, tolerance $\epsilon > 0$.

repeat
- $t \leftarrow (l+u)/2$
- Solve the convex feasibility problem (6.1)
- **if** (6.1) is feasible **then**
  - $u \leftarrow t$
- **else**
  - $l \leftarrow t$

until $u - l \le \epsilon$

Each iteration halves the interval $[l, u]$, which always contains $p^\star$, so the method requires exactly $\lceil \log_2((u-l)/\epsilon) \rceil$ iterations, where u, l are the initial values.

6.2 Linear optimization

Definition 6.1 [Linear program]. A linear program (LP) is

$$\begin{array}{ll} \text{minimize} & c^T x + d \\ \text{subject to} & Gx \preceq h \\ & Ax = b, \end{array}$$

a convex problem with affine objective and constraint functions. Its feasible set is a polyhedron.

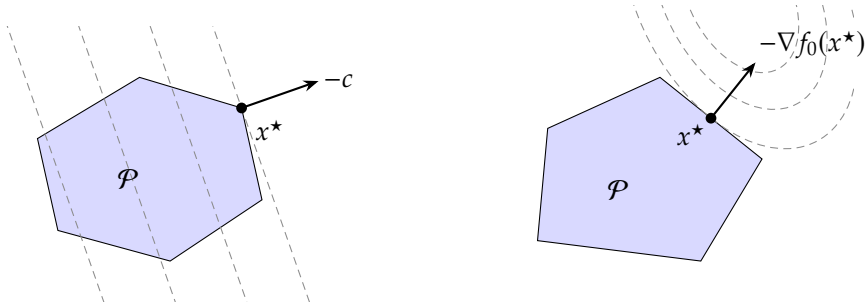


Figure 17: Left: an LP. The level sets of $c^T x$ are hyperplanes (dashed), and the optimum $x^\star$ is where the last of them touches $\mathcal{P}$, here a vertex. Right: a QP. The level sets of the convex quadratic objective are ellipsoids (dashed), and at the optimum $x^\star$ the vector $-\nabla f_0(x^\star)$ is an outward normal of $\mathcal{P}$.

Example:

- *Diet problem.* Choose quantities $x_1, \dots, x_n$ of n foods. One unit of food j costs c_j and contains an amount a_{ij} of nutrient i, and a healthy diet requires nutrient i in quantity at least b_i. The cheapest healthy diet solves the LP: minimize $c^T x$ subject to $Ax \succeq b$, $x \succeq 0$.
- *Piecewise-linear minimization.* Minimizing $\max_{i=1,\dots,m}(a_i^T x + b_i)$ is equivalent to the LP: minimize t subject to $a_i^T x + b_i \le t$, $i = 1, \dots, m$.

Example: *Chebyshev center of a polyhedron.* The CHEBYSHEV CENTER of $\mathcal{P} = \{x \mid a_i^T x \le b_i,\ i = 1, \dots, m\}$ is the center of the largest inscribed ball $\mathcal{B} = \{x_c + u \mid \|u\|_2 \le r\}$ (Figure 18). Since

$$a_i^T x \le b_i \text{ for all } x \in \mathcal{B} \iff \sup\{a_i^T(x_c + u) \mid \|u\|_2 \le r\} = a_i^T x_c + r\|a_i\|_2 \le b_i,$$

x_c and r can be determined by solving the LP: maximize r subject to $a_i^T x_c + r\|a_i\|_2 \le b_i$, $i = 1, \dots, m$.

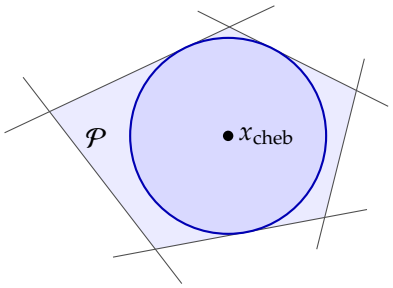


Figure 18: The Chebyshev center x_{cheb} of a polyhedron and the largest inscribed ball, which touches three of the five faces.

Definition 6.2 [LINEAR-FRACTIONAL PROGRAM]. A LINEAR-FRACTIONAL PROGRAM minimizes

$$f_0(x) = \frac{c^T x + d}{e^T x + f}, \qquad \text{dom } f_0 = \{x \mid e^T x + f > 0\},$$

subject to $Gx \preceq h$, $Ax = b$.

It is a quasiconvex optimization problem and can be solved by bisection. It is also equivalent to the LP in the variables y, z

$$\begin{array}{ll} \text{minimize} & c^T y + dz \\ \text{subject to} & Gy \preceq hz, \quad Ay = bz \\ & e^T y + fz = 1, \quad z \ge 0, \end{array}$$

via $y = x/(e^T x + f)$, $z = 1/(e^T x + f)$.

The GENERALIZED LINEAR-FRACTIONAL PROGRAM minimizes $f_0(x) = \max_{i=1,\dots,r}(c_i^T x + d_i)/(e_i^T x + f_i)$ with $\text{dom } f_0 = \{x \mid e_i^T x + f_i > 0,\ i = 1, \dots, r\}$; it is quasiconvex and can be solved by bisection.

Example: *Von Neumann model of a growing economy.* Let $x, x^+ \in \mathbb{R}^n$ be the activity levels of n sectors in the current and next period, $(Ax)_i$ and $(Bx^+)_i$ the produced and consumed amounts of good i, and x_i^+/x_i the growth rate of sector i. To allocate activity to maximize the growth rate of the slowest growing sector, solve

$$\begin{array}{ll} \text{maximize (over } x, x^+) & \min_{i=1,\dots,n} x_i^+/x_i \\ \text{subject to} & x^+ \ge 0, \quad Bx^+ \preceq Ax. \end{array}$$

6.3 Quadratic optimization

Definition 6.3 [Quadratic program]. A quadratic program (QP) is

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P x + q^T x + r \\ \text{subject to} & Gx \preceq h \\ & Ax = b \end{array}$$

with $P \in \mathbf{S}_+^n$: it minimizes a convex quadratic function over a polyhedron (Figure 17, right).

Example:

- *Least-squares*, minimize $\|Ax - b\|_2^2$: analytical solution $x^\star = A^\dagger b$ ($A^\dagger$ is the pseudo-inverse); linear constraints such as $l \preceq x \preceq u$ can be added.
- *Linear program with random cost*:

$$\begin{array}{ll} \text{minimize} & \bar{c}^T x + \gamma x^T \Sigma x = \mathbb{E} c^T x + \gamma \operatorname{Var}(c^T x) \\ \text{subject to} & Gx \preceq h, \quad Ax = b. \end{array}$$

  Here c is a random vector with mean $\bar{c}$ and covariance Σ, so $c^T x$ has mean $\bar{c}^T x$ and variance $x^T \Sigma x$; $\gamma > 0$ is the risk aversion parameter, which controls the trade-off between expected cost and variance (risk).

Definition 6.4 [QCQP]. A quadratically constrained quadratic program (QCQP) is

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P_0 x + q_0^T x + r_0 \\ \text{subject to} & (1/2)x^T P_i x + q_i^T x + r_i \le 0, \quad i = 1, \ldots, m \\ & Ax = b \end{array}$$

with $P_i \in \mathbf{S}_+^n$, so the objective and constraints are convex quadratic.

If $P_1, \ldots, P_m \in \mathbf{S}_{++}^n$, the feasible region is the intersection of m ellipsoids and an affine set.

Definition 6.5 [Second-order cone program]. A second-order cone program (SOCP) is

$$\begin{array}{ll} \text{minimize} & f^T x \\ \text{subject to} & \|A_i x + b_i\|_2 \le c_i^T x + d_i, \quad i = 1, \ldots, m \\ & Fx = g \end{array}$$

with $A_i \in \mathbb{R}^{n_i \times n}$, $F \in \mathbb{R}^{p \times n}$. The inequalities are second-order cone (SOC) constraints: $(A_i x + b_i, c_i^T x + d_i)$ lies in the second-order cone in $\mathbb{R}^{n_i+1}$.

For $n_i = 0$ an SOCP reduces to an LP, and if $c_i = 0$ it reduces to a QCQP; it is more general than both.

Robust linear programming. The parameters of optimization problems are often uncertain; in an LP (minimize $c^T x$ subject to $a_i^T x \le b_i$) there can be uncertainty in c, a_i, b_i. Two common approaches handle uncertainty in a_i:

- *deterministic model*: the constraints must hold for all $a_i \in \mathcal{E}_i$, that is, $a_i^T x \le b_i$ for all $a_i \in \mathcal{E}_i$;
- *stochastic model*: a_i is a random variable, and the constraints must hold with probability η, $\mathrm{prob}(a_i^T x \le b_i) \ge \eta$.

Deterministic approach via SOCP. Choose ellipsoids $\mathcal{E}_i = \{\bar{a}_i + P_i u \mid \|u\|_2 \le 1\}$ ($\bar{a}_i \in \mathbb{R}^n$, $P_i \in \mathbb{R}^{n\times n}$), with center $\bar{a}_i$ and semi-axes determined by the singular values and vectors of P_i. Since $\sup_{\|u\|_2\le 1}(\bar{a}_i + P_i u)^T x = \bar{a}_i^T x + \|P_i^T x\|_2$, the robust LP is equivalent to the SOCP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & \bar{a}_i^T x + \|P_i^T x\|_2 \le b_i, \quad i = 1, \dots, m. \end{array}$$

Stochastic approach via SOCP. Assume $a_i \sim \mathcal{N}(\bar{a}_i, \Sigma_i)$. Then $a_i^T x$ is Gaussian with mean $\bar{a}_i^T x$ and variance $x^T \Sigma_i x$, so

$$\mathrm{prob}(a_i^T x \le b_i) = \Phi\left(\frac{b_i - \bar{a}_i^T x}{\|\Sigma_i^{1/2} x\|_2}\right),$$

where $\Phi(x) = (1/\sqrt{2\pi}) \int_{-\infty}^{x} e^{-t^2/2}\, dt$ is the CDF of $\mathcal{N}(0,1)$. Hence the robust LP with $\eta \ge 1/2$ is equivalent to the SOCP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & \bar{a}_i^T x + \Phi^{-1}(\eta)\|\Sigma_i^{1/2} x\|_2 \le b_i, \quad i = 1, \dots, m; \end{array}$$

$\eta \ge 1/2$ makes $\Phi^{-1}(\eta) \ge 0$, so the constraints are convex.

6.4 Geometric programming

Definition 6.6 [Monomials, posynomials, and geometric programs]. A monomial is a function $f(x) = c x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}$ with $\mathrm{dom}\, f = \mathbb{R}^n_{++}$, $c > 0$, and arbitrary real exponents a_i. A posynomial is a sum of monomials, $f(x) = \sum_{k=1}^{K} c_k x_1^{a_{1k}} x_2^{a_{2k}} \cdots x_n^{a_{nk}}$. A geometric program (GP) is

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \le 1, \quad i = 1, \dots, m \\ & h_i(x) = 1, \quad i = 1, \dots, p \end{array}$$

with f_i posynomial and h_i monomial.

Convex form. Change variables to $y_i = \log x_i$ and take logarithms of the cost and constraints. A monomial $f(x) = c x_1^{a_1} \cdots x_n^{a_n}$ becomes $\log f(e^{y_1}, \dots, e^{y_n}) = a^T y + b$ with $b = \log c$, and a posynomial

becomes $\log f(e^{y_1}, \ldots, e^{y_n}) = \log \sum_{k=1}^K e^{a_k^T y + b_k}$ with $b_k = \log c_k$. The GP becomes the convex problem

$$\begin{array}{ll} \text{minimize} & \log\left(\sum_{k=1}^K \exp(a_{0k}^T y + b_{0k})\right) \\ \text{subject to} & \log\left(\sum_{k=1}^K \exp(a_{ik}^T y + b_{ik})\right) \leq 0, \quad i = 1, \ldots, m \\ & Gy + d = 0. \end{array}$$

Example: *Design of a cantilever beam.* The beam has N segments of unit length with rectangular cross-sections of size $w_i \times h_i$, and a given vertical force F is applied at the right end (Figure 19). We minimize the total weight subject to upper and lower bounds on w_i, h_i, upper and lower bounds on the aspect ratios h_i/w_i, an upper bound on the stress in each segment, and an upper bound on the vertical deflection at the end of the beam; the variables are w_i, h_i for $i = 1, \ldots, N$.

- The total weight $w_1 h_1 + \cdots + w_N h_N$ is a posynomial.
- The aspect ratio h_i/w_i and inverse aspect ratio w_i/h_i are monomials.
- The maximum stress in segment i, $6iF/(w_i h_i^2)$, is a monomial.
- The vertical deflection y_i and slope v_i of the central axis at the right end of segment i are defined recursively, for $i = N, N-1, \ldots, 1$ with $v_{N+1} = y_{N+1} = 0$ and Young's modulus E, by

$$v_i = 12(i - 1/2)\frac{F}{Ew_i h_i^3} + v_{i+1}, \qquad y_i = 6(i - 1/3)\frac{F}{Ew_i h_i^3} + v_{i+1} + y_{i+1};$$

  v_i and y_i are posynomial functions of w, h.

Writing $w_{\min} \leq w_i \leq w_{\max}$ as $w_{\min}/w_i \leq 1$, $w_i/w_{\max} \leq 1$ (similarly for h_i) and $S_{\min} \leq h_i/w_i \leq S_{\max}$ as $S_{\min} w_i/h_i \leq 1$, $h_i/(w_i S_{\max}) \leq 1$, the problem is the GP

$$\begin{array}{ll} \text{minimize} & w_1 h_1 + \cdots + w_N h_N \\ \text{subject to} & w_{\max}^{-1} w_i \leq 1, \quad w_{\min} w_i^{-1} \leq 1, \quad i = 1, \ldots, N \\ & h_{\max}^{-1} h_i \leq 1, \quad h_{\min} h_i^{-1} \leq 1, \quad i = 1, \ldots, N \\ & S_{\max}^{-1} w_i^{-1} h_i \leq 1, \quad S_{\min} w_i h_i^{-1} \leq 1, \quad i = 1, \ldots, N \\ & 6iF\sigma_{\max}^{-1} w_i^{-1} h_i^{-2} \leq 1, \quad i = 1, \ldots, N \\ & y_{\max}^{-1} y_1 \leq 1. \end{array}$$

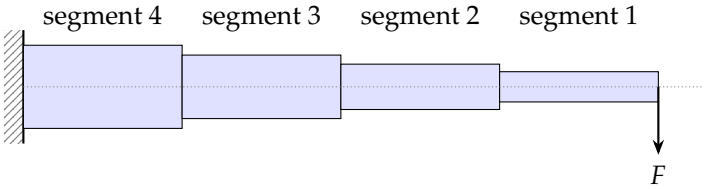


Figure 19: A cantilever beam with $N = 4$ segments, fixed at the left and loaded by a vertical force F at the right end.

Example: *Minimizing the spectral radius of a nonnegative matrix.* The Perron–Frobenius eigenvalue $\lambda_{\mathrm{pf}}(A)$ exists for (elementwise) positive $A \in \mathbb{R}^{n\times n}$. It is a real, positive eigenvalue of A, equal to the spectral radius $\max_i |\lambda_i(A)|$, and determines the asymptotic growth (decay) rate of A^k: $A^k \sim \lambda_{\mathrm{pf}}^k$ as $k \to \infty$. Alternatively, $\lambda_{\mathrm{pf}}(A) = \inf\{\lambda \mid Av \preceq \lambda v \text{ for some } v \succ 0\}$. To minimize $\lambda_{\mathrm{pf}}(A(x))$, where the elements $A(x)_{ij}$ are posynomials of x, solve the equivalent GP in the variables λ, v, x: minimize λ subject to $\sum_{j=1}^n A(x)_{ij} v_j/(\lambda v_i) \leq 1$, $i = 1, \ldots, n$.

6.5 Generalized inequality constraints

A convex problem with generalized inequality constraints is

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \preceq_{K_i} 0, \quad i = 1, \ldots, m \\ & Ax = b, \end{array}$$

with $f_0 : \mathbb{R}^n \to \mathbb{R}$ convex and $f_i : \mathbb{R}^n \to \mathbb{R}^{k_i}$ K_i-convex with respect to proper cones K_i. It has the same properties as a standard convex problem (convex feasible set, local optima are global, and so on). The conic form problem is the special case with affine objective and constraints:

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Fx + g \preceq_K 0 \\ & Ax = b, \end{array}$$

which extends linear programming ($K = \mathbb{R}^m_+$) to nonpolyhedral cones.

6.6 Semidefinite programming

Definition 6.7 [Semidefinite program]. A semidefinite program (SDP) is

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & x_1 F_1 + x_2 F_2 + \cdots + x_n F_n + G \preceq 0 \\ & Ax = b \end{array}$$

with $F_i, G \in \mathbf{S}^k$. The inequality constraint is a linear matrix inequality (LMI).

Multiple LMI constraints reduce to one: $x_1\hat{F}_1 + \cdots + x_n\hat{F}_n + \hat{G} \preceq 0$ and $x_1\tilde{F}_1 + \cdots + x_n\tilde{F}_n + \tilde{G} \preceq 0$ together are equivalent to the single block-diagonal LMI

$$x_1 \begin{bmatrix} \hat{F}_1 & 0 \\ 0 & \tilde{F}_1 \end{bmatrix} + \cdots + x_n \begin{bmatrix} \hat{F}_n & 0 \\ 0 & \tilde{F}_n \end{bmatrix} + \begin{bmatrix} \hat{G} & 0 \\ 0 & \tilde{G} \end{bmatrix} \preceq 0.$$

LP and SOCP as SDP. The LP "minimize $c^T x$ subject to $Ax \preceq b$" is equivalent to the SDP "minimize

$c^T x$ subject to $\mathrm{diag}(Ax - b) \preceq 0$" (note the different interpretations of $\preceq$). The SOCP "minimize $f^T x$ subject to $\|A_i x + b_i\|_2 \le c_i^T x + d_i$" is equivalent to the SDP

$$\begin{array}{ll} \text{minimize} & f^T x \\ \text{subject to} & \begin{bmatrix} (c_i^T x + d_i) I & A_i x + b_i \\ (A_i x + b_i)^T & c_i^T x + d_i \end{bmatrix} \succeq 0, \quad i = 1, \ldots, m. \end{array}$$

Eigenvalue minimization. To minimize $\lambda_{\max}(A(x))$, where $A(x) = A_0 + x_1 A_1 + \cdots + x_n A_n$ with given $A_i \in \mathbf{S}^k$, solve the SDP in $x \in \mathbb{R}^n$, $t \in \mathbb{R}$: minimize t subject to $A(x) \preceq tI$. This follows from $\lambda_{\max}(A) \le t \iff A \preceq tI$.

Matrix norm minimization. To minimize $\|A(x)\|_2 = (\lambda_{\max}(A(x)^T A(x)))^{1/2}$, where $A(x) = A_0 + x_1 A_1 + \cdots + x_n A_n$ with given $A_i \in \mathbb{R}^{p \times q}$, solve the SDP in $x \in \mathbb{R}^n$, $t \in \mathbb{R}$:

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & \begin{bmatrix} tI & A(x) \\ A(x)^T & tI \end{bmatrix} \succeq 0. \end{array}$$

The constraint follows from

$$\|A\|_2 \le t \iff A^T A \preceq t^2 I,\ t \ge 0 \iff \begin{bmatrix} tI & A \\ A^T & tI \end{bmatrix} \succeq 0.$$

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 6 (2023); original slides, *Convex optimization problems*.

Lecture 7: Convex Optimization Problems (Continued); Duality

7.1 Vector optimization

Definition 7.1 [Vector optimization problem]. A general vector optimization problem is

$$\begin{array}{ll} \text{minimize (w.r.t. } K) & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p, \end{array}$$

with vector objective $f_0 : \mathbb{R}^n \to \mathbb{R}^q$ minimized with respect to a proper cone $K \subseteq \mathbb{R}^q$. It is a convex vector optimization problem if f_0 is K-convex, $f_1, \ldots, f_m$ are convex, and the equality constraints are affine, $Ax = b$.

Definition 7.2 [Optimal and Pareto optimal points]. Let $\mathcal{O} = \{f_0(x) \mid x \text{ feasible}\}$ be the set of achievable objective values. A feasible x is optimal if $f_0(x)$ is the minimum value of $\mathcal{O}$, and Pareto optimal if $f_0(x)$ is a minimal value of $\mathcal{O}$.

Multicriterion optimization is vector optimization with $K = \mathbb{R}^q_+$ and $f_0(x) = (F_1(x), \ldots, F_q(x))$: there are q different objectives F_i, and roughly speaking we want all of them to be small. A feasible $x^\star$ is optimal if

$$y \text{ feasible} \implies f_0(x^\star) \leq f_0(y);$$

if an optimal point exists, the objectives are noncompeting. A feasible x^{po} is Pareto optimal if

$$y \text{ feasible}, \ f_0(y) \leq f_0(x^{\text{po}}) \implies f_0(x^{\text{po}}) = f_0(y);$$

if there are multiple Pareto optimal values, there is a trade-off between the objectives.

Example: *Regularized least-squares* is the bi-criterion problem

$$\text{minimize (w.r.t. } \mathbb{R}^2_+) \quad \left(\|Ax - b\|_2^2, \ \|x\|_2^2\right).$$

The Pareto optimal points trace the boundary of $\mathcal{O}$ between the point with the smallest residual and the point $x = 0$: the optimal trade-off curve between fit and size.

Example: *Risk-return trade-off in portfolio optimization.*

$$\begin{array}{ll} \text{minimize (w.r.t. } \mathbb{R}^2_+) & (-\bar{p}^T x, \ x^T \Sigma x) \\ \text{subject to} & \mathbf{1}^T x = 1, \quad x \succeq 0. \end{array}$$

Here $x \in \mathbb{R}^n$ is the investment portfolio, x_i the fraction invested in asset i; $p \in \mathbb{R}^n$ is the vector of relative asset price changes, modeled as a random variable with mean $\bar{p}$ and covariance Σ; the

return is $r = p^T x$, with expected return $\bar{p}^T x = \mathbb{E} r$ and variance $x^T \Sigma x = \operatorname{Var} r$. The Pareto optimal portfolios give the optimal trade-off between mean return and its standard deviation (risk).

7.2 Scalarization

To find Pareto optimal points, choose $\lambda \succ_{K^*} 0$ and solve the scalar problem

$$\begin{array}{ll} \text{minimize} & \lambda^T f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p. \end{array}$$

If x is optimal for the scalar problem, then it is Pareto optimal for the vector problem (Lecture 3: a minimizer of $\lambda^T z$ over $\mathcal{O}$ with $\lambda \succ_{K^*} 0$ is minimal). For convex vector optimization problems we can find (almost) all Pareto optimal points by varying $\lambda \succ_{K^*} 0$ (Figure 20).

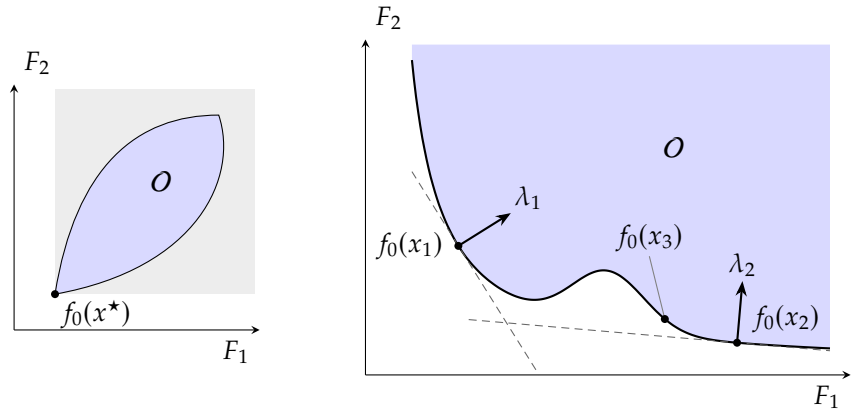


Figure 20: Left: $f_0(x^\star)$ is the minimum element of $\mathcal{O}$, so $x^\star$ is optimal. Right (schematic): $f_0(x_1)$ and $f_0(x_2)$ minimize $\lambda_1^T z$ and $\lambda_2^T z$ over $\mathcal{O}$, so x_1, x_2 are Pareto optimal. The point $f_0(x_3)$ is also minimal, but because $\mathcal{O}$ is not convex no weighted sum finds it.

For multicriterion problems, scalarization minimizes a positive weighted sum

$$\lambda^T f_0(x) = \lambda_1 F_1(x) + \cdots + \lambda_q F_q(x).$$

- Regularized least-squares: take $\lambda = (1, \gamma)$ with $\gamma > 0$ and minimize $\|Ax - b\|_2^2 + \gamma \|x\|_2^2$; for fixed γ this is a least-squares problem. Varying γ sweeps out the trade-off curve.
- Risk-return trade-off: minimize $-\bar{p}^T x + \gamma x^T \Sigma x$ subject to $\mathbf{1}^T x = 1$, $x \geq 0$; for fixed $\gamma > 0$ this is a quadratic program, with γ the risk aversion parameter.

7.3 The Lagrangian and the dual function

Consider the standard form problem, not necessarily convex,

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \le 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p, \end{array}$$

with variable $x \in \mathbb{R}^n$, domain $\mathcal{D}$, and optimal value $p^\star$.

Definition 7.3 [Lagrangian]. The Lagrangian is $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$ with $\operatorname{dom} L = \mathcal{D} \times \mathbb{R}^m \times \mathbb{R}^p$,

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x),$$

a weighted sum of the objective and constraint functions. λ_i is the Lagrange multiplier associated with $f_i(x) \le 0$ and ν_i the one associated with $h_i(x) = 0$.

Definition 7.4 [Lagrange dual function]. The Lagrange dual function $g : \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$ is

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = \inf_{x \in \mathcal{D}} \Big(f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x) \Big).$$

g is concave, as a pointwise infimum of affine functions of (λ, ν), and can be $-\infty$ for some λ, ν.

Proposition 7.5 [Lower bound property]. *If $\lambda \ge 0$, then $g(\lambda, \nu) \le p^\star$.*

Proof. If $\tilde{x}$ is feasible and $\lambda \ge 0$, then $\lambda_i f_i(\tilde{x}) \le 0$ and $h_i(\tilde{x}) = 0$, so

$$f_0(\tilde{x}) \ge L(\tilde{x}, \lambda, \nu) \ge \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = g(\lambda, \nu).$$

Minimizing over all feasible $\tilde{x}$ gives $p^\star \ge g(\lambda, \nu)$. □

7.4 Examples of dual functions

Example: *Least-norm solution of linear equations*: minimize $x^T x$ subject to $Ax = b$. The Lagrangian is $L(x, \nu) = x^T x + \nu^T (Ax - b)$. To minimize over x, set the gradient to zero: $\nabla_x L(x, \nu) = 2x + A^T \nu = 0$ gives $x = -(1/2) A^T \nu$. Substituting,

$$g(\nu) = L\big((-1/2) A^T \nu, \nu\big) = -\tfrac{1}{4} \nu^T A A^T \nu - b^T \nu,$$

a concave function of ν. Lower bound property: $p^\star \ge -(1/4) \nu^T A A^T \nu - b^T \nu$ for all ν.

Example: *Standard form LP*: minimize $c^T x$ subject to $Ax = b$, $x \succeq 0$. The Lagrangian

$$L(x, \lambda, \nu) = c^T x + \nu^T (Ax - b) - \lambda^T x = -b^T \nu + (c + A^T \nu - \lambda)^T x$$

is affine in x, hence

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) = \begin{cases} -b^T \nu & A^T \nu - \lambda + c = 0, \\ -\infty & \text{otherwise.} \end{cases}$$

g is linear on the affine domain $\{(\lambda, \nu) \mid A^T \nu - \lambda + c = 0\}$, hence concave. Lower bound property: $p^\star \geq -b^T \nu$ if $A^T \nu + c \succeq 0$.

Example: *Equality constrained norm minimization*: minimize $\|x\|$ subject to $Ax = b$. The dual function is

$$g(\nu) = \inf_x \left(\|x\| - \nu^T A x + b^T \nu \right) = \begin{cases} b^T \nu & \|A^T \nu\|_* \leq 1, \\ -\infty & \text{otherwise,} \end{cases}$$

where $\|v\|_* = \sup_{\|u\| \leq 1} u^T v$ is the DUAL NORM of $\|\cdot\|$. This follows from $\inf_x (\|x\| - y^T x) = 0$ if $\|y\|_* \leq 1$ and $-\infty$ otherwise: if $\|y\|_* \leq 1$, then $\|x\| - y^T x \geq 0$ for all x, with equality at $x = 0$; if $\|y\|_* > 1$, choose $x = tu$ with $\|u\| \leq 1$, $u^T y = \|y\|_* > 1$, so that $\|x\| - y^T x = t(\|u\| - \|y\|_*) \to -\infty$ as $t \to \infty$. Lower bound property: $p^\star \geq b^T \nu$ if $\|A^T \nu\|_* \leq 1$.

Example: *Two-way partitioning*:

$$\begin{array}{ll} \text{minimize} & x^T W x \\ \text{subject to} & x_i^2 = 1, \quad i = 1, \ldots, n. \end{array}$$

This is a nonconvex problem whose feasible set contains 2^n discrete points. Interpretation: partition $\{1, \ldots, n\}$ into two sets (by the sign of x_i), where W_{ij} is the cost of assigning i, j to the same set and $-W_{ij}$ the cost of assigning them to different sets. The dual function is

$$g(\nu) = \inf_x \left(x^T W x + \sum_i \nu_i (x_i^2 - 1) \right) = \inf_x x^T (W + \operatorname{diag}(\nu)) x - \mathbf{1}^T \nu = \begin{cases} -\mathbf{1}^T \nu & W + \operatorname{diag}(\nu) \succeq 0, \\ -\infty & \text{otherwise.} \end{cases}$$

Lower bound property: $p^\star \geq -\mathbf{1}^T \nu$ if $W + \operatorname{diag}(\nu) \succeq 0$. For example, $\nu = -\lambda_{\min}(W)\mathbf{1}$ gives the bound $p^\star \geq n \lambda_{\min}(W)$.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 7 (2023); original slides, *Convex optimization problems* and *Duality*.

Lecture 8: Duality (Continued)

8.1 Dual function and conjugate

For the problem of minimizing $f_0(x)$ subject to $Ax \preceq b$, $Cx = d$, the dual function is

$$g(\lambda, \nu) = \inf_{x \in \operatorname{dom} f_0} \left(f_0(x) + (A^T\lambda + C^T\nu)^T x - b^T\lambda - d^T\nu \right) = -f_0^*(-A^T\lambda - C^T\nu) - b^T\lambda - d^T\nu,$$

by the definition of the conjugate, $f^*(y) = \sup_{x \in \operatorname{dom} f}(y^T x - f(x))$. This simplifies the derivation of the dual whenever the conjugate of f_0 is known. For example, for entropy maximization,

$$f_0(x) = \sum_{i=1}^n x_i \log x_i, \qquad f_0^*(y) = \sum_{i=1}^n e^{y_i - 1}.$$

8.2 The dual problem

Definition 8.1 [Lagrange dual problem]. The Lagrange dual problem is

$$\begin{array}{ll} \text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \succeq 0. \end{array}$$

It finds the best lower bound on $p^\star$ obtainable from the dual function. It is a convex optimization problem (concave objective, convex constraint) whatever the primal; its optimal value is denoted $d^\star$. λ, ν are dual feasible if $\lambda \succeq 0$ and $(\lambda, \nu) \in \operatorname{dom} g$. The dual is often simplified by making the implicit constraint $(\lambda, \nu) \in \operatorname{dom} g$ explicit. For example, the standard form LP and its dual are

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b, \quad x \succeq 0, \end{array} \qquad\qquad \begin{array}{ll} \text{maximize} & -b^T\nu \\ \text{subject to} & A^T\nu + c \succeq 0. \end{array}$$

Definition 8.2 [Weak and strong duality]. Weak duality is the inequality $d^\star \leq p^\star$. Strong duality is the equality $d^\star = p^\star$.

Weak duality always holds, for convex and nonconvex problems, and can be used to find nontrivial lower bounds for difficult problems. For example, solving the SDP

$$\begin{array}{ll} \text{maximize} & -\mathbf{1}^T\nu \\ \text{subject to} & W + \operatorname{diag}(\nu) \succeq 0 \end{array}$$

gives a lower bound for the two-way partitioning problem (Lecture 7). Strong duality does not hold

in general, but (usually) holds for convex problems. Conditions that guarantee strong duality in convex problems are called CONSTRAINT QUALIFICATIONS.

Theorem 8.3 [SLATER'S CONSTRAINT QUALIFICATION]. *Strong duality holds for a convex problem*

$$\begin{array}{ll} \textit{minimize} & f_0(x) \\ \textit{subject to} & f_i(x) \le 0, \quad i = 1, \dots, m \\ & Ax = b \end{array}$$

if it is STRICTLY FEASIBLE, *that is, there is an* $x \in \operatorname{int} \mathcal{D}$ *with* $f_i(x) < 0$, $i = 1, \dots, m$, *and* $Ax = b$. *It also guarantees that the dual optimum is attained (if* $p^\star > -\infty$*).*

Slater's condition can be sharpened: for example, $\operatorname{int} \mathcal{D}$ can be replaced by $\operatorname{relint} \mathcal{D}$ (the interior relative to the affine hull), and linear inequalities do not need to hold with strict inequality. There are many other types of constraint qualifications.

8.3 Examples

Example: *Inequality form LP*: minimize $c^T x$ subject to $Ax \preceq b$. The dual function is

$$g(\lambda) = \inf_x \big((c + A^T\lambda)^T x - b^T\lambda\big) = \begin{cases} -b^T\lambda & A^T\lambda + c = 0, \\ -\infty & \text{otherwise}, \end{cases}$$

so the dual problem is to maximize $-b^T\lambda$ subject to $A^T\lambda + c = 0$, $\lambda \succeq 0$. By Slater's condition $p^\star = d^\star$ if $A\tilde{x} \prec b$ for some $\tilde{x}$; in fact $p^\star = d^\star$ except when the primal and dual are both infeasible.

Example: *Quadratic program*: minimize $x^T P x$ subject to $Ax \preceq b$, with $P \in \mathbf{S}^n_{++}$. Minimizing the Lagrangian at $x = -(1/2)P^{-1}A^T\lambda$ gives

$$g(\lambda) = \inf_x \big(x^T P x + \lambda^T(Ax - b)\big) = -\tfrac{1}{4}\lambda^T A P^{-1} A^T \lambda - b^T\lambda,$$

and the dual problem is to maximize $-(1/4)\lambda^T A P^{-1} A^T \lambda - b^T\lambda$ subject to $\lambda \succeq 0$. Slater's condition gives $p^\star = d^\star$ if $A\tilde{x} \prec b$ for some $\tilde{x}$; in fact $p^\star = d^\star$ always.

Example: *A nonconvex problem with strong duality*: minimize $x^T A x + 2b^T x$ subject to $x^T x \le 1$, where $A \not\succeq 0$, so the problem is nonconvex. The dual function is $g(\lambda) = \inf_x \big(x^T(A + \lambda I)x + 2b^T x - \lambda\big)$. It is unbounded below if $A + \lambda I \not\succeq 0$, or if $A + \lambda I \succeq 0$ and $b \notin \mathcal{R}(A + \lambda I)$; otherwise it is minimized by $x = -(A + \lambda I)^\dagger b$, giving $g(\lambda) = -b^T(A + \lambda I)^\dagger b - \lambda$. The dual problem and an equivalent SDP are

$$\begin{array}{ll} \text{maximize} & -b^T(A + \lambda I)^\dagger b - \lambda \\ \text{subject to} & A + \lambda I \succeq 0 \\ & b \in \mathcal{R}(A + \lambda I), \end{array} \qquad \begin{array}{ll} \text{maximize} & -t - \lambda \\ \text{subject to} & \begin{bmatrix} A + \lambda I & b \\ b^T & t \end{bmatrix} \succeq 0. \end{array}$$

Strong duality holds although the primal problem is not convex (not easy to show).

8.4 Geometric interpretation

For simplicity, consider a problem with one constraint $f_1(x) \le 0$, and let $\mathcal{G} = \{(f_1(x), f_0(x)) \mid x \in \mathcal{D}\}$. Then

$$g(\lambda) = \inf_{(u,t)\in\mathcal{G}} (t + \lambda u),$$

so $\lambda u + t = g(\lambda)$ is a (non-vertical) supporting hyperplane to $\mathcal{G}$, and it intersects the t-axis at $t = g(\lambda)$. Since $p^\star = \inf\{t \mid (u,t) \in \mathcal{G},\ u \le 0\}$, every such intercept lies below $p^\star$ when $\lambda \ge 0$; the best intercept is $d^\star$, and when $\mathcal{G}$ is not convex there can be a gap (Figure 21).

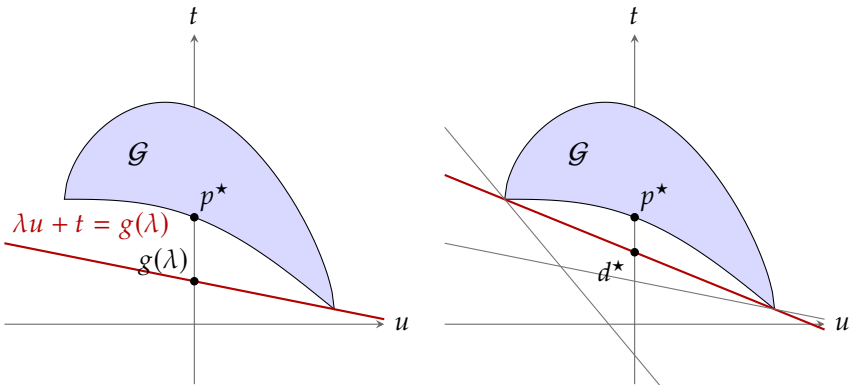


Figure 21: Left: the supporting line $\lambda u + t = g(\lambda)$ meets the t-axis at $g(\lambda) \le p^\star$. Right: supporting lines for three values of λ; the highest intercept is $d^\star$, which here is strictly below $p^\star$ because $\mathcal{G}$ is not convex (schematic).

Epigraph variation. The same interpretation holds if $\mathcal{G}$ is replaced with

$$\mathcal{A} = \{(u,t) \mid f_1(x) \le u,\ f_0(x) \le t \text{ for some } x \in \mathcal{D}\}.$$

Strong duality holds if there is a non-vertical supporting hyperplane to $\mathcal{A}$ at $(0, p^\star)$. For a convex problem, $\mathcal{A}$ is convex and hence has a supporting hyperplane at $(0, p^\star)$. Slater's condition ensures it is non-vertical: if there is $(\tilde{u}, \tilde{t}) \in \mathcal{A}$ with $\tilde{u} < 0$, supporting hyperplanes at $(0, p^\star)$ must be non-vertical (Figure 23).

8.5 Optimality conditions

Complementary slackness. Assume strong duality holds, $x^\star$ is primal optimal, and $(\lambda^\star, \nu^\star)$ is dual optimal. Then

$$\begin{aligned} f_0(x^\star) = g(\lambda^\star, \nu^\star) &= \inf_x \Big(f_0(x) + \sum_{i=1}^m \lambda_i^\star f_i(x) + \sum_{i=1}^p \nu_i^\star h_i(x) \Big) \\ &\le f_0(x^\star) + \sum_{i=1}^m \lambda_i^\star f_i(x^\star) + \sum_{i=1}^p \nu_i^\star h_i(x^\star) \le f_0(x^\star), \end{aligned}$$

so the two inequalities hold with equality. Hence $x^\star$ minimizes $L(x, \lambda^\star, \nu^\star)$, and $\lambda_i^\star f_i(x^\star) = 0$ for $i = 1, \dots, m$ (COMPLEMENTARY SLACKNESS):

$$\lambda_i^\star > 0 \implies f_i(x^\star) = 0, \qquad f_i(x^\star) < 0 \implies \lambda_i^\star = 0.$$

Definition 8.4 [KKT CONDITIONS]. For a problem with differentiable f_i, h_i, the KARUSH–KUHN–TUCKER (KKT) CONDITIONS are:

1. primal constraints: $f_i(x) \le 0$, $i = 1, \dots, m$, and $h_i(x) = 0$, $i = 1, \dots, p$;
2. dual constraints: $\lambda \succeq 0$;
3. complementary slackness: $\lambda_i f_i(x) = 0$, $i = 1, \dots, m$;
4. the gradient of the Lagrangian with respect to x vanishes:

$$\nabla f_0(x) + \sum_{i=1}^m \lambda_i \nabla f_i(x) + \sum_{i=1}^p \nu_i \nabla h_i(x) = 0.$$

By the argument above, if strong duality holds and x, λ, ν are optimal, they satisfy the KKT conditions (the fourth because x minimizes $L(\cdot, \lambda, \nu)$).

Theorem 8.5 [KKT CONDITIONS FOR CONVEX PROBLEMS]. *If $\tilde{x}, \tilde{\lambda}, \tilde{\nu}$ satisfy the KKT conditions for a convex problem, then they are optimal. If Slater's condition is satisfied, x is optimal if and only if there exist λ, ν that satisfy the KKT conditions.*

Proof. Complementary slackness gives $f_0(\tilde{x}) = L(\tilde{x}, \tilde{\lambda}, \tilde{\nu})$. The fourth condition and convexity of $L(\cdot, \tilde{\lambda}, \tilde{\nu})$ (since $\tilde{\lambda} \succeq 0$) give $g(\tilde{\lambda}, \tilde{\nu}) = L(\tilde{x}, \tilde{\lambda}, \tilde{\nu})$. Hence $f_0(\tilde{x}) = g(\tilde{\lambda}, \tilde{\nu})$, and weak duality makes both optimal. For the second statement, Slater's condition implies strong duality and that the dual optimum is attained, so an optimal x has multipliers satisfying KKT. □

This generalizes the optimality condition $\nabla f_0(x) = 0$ for unconstrained problems.

Example: *Water-filling* (assume $\alpha_i > 0$):

$$\begin{array}{ll} \text{minimize} & -\sum_{i=1}^n \log(x_i + \alpha_i) \\ \text{subject to} & x \succeq 0, \quad \mathbf{1}^T x = 1. \end{array}$$

x is optimal if and only if $x \succeq 0$, $\mathbf{1}^T x = 1$, and there exist $\lambda \in \mathbb{R}^n$, $\nu \in \mathbb{R}$ such that

$$\lambda \succeq 0, \qquad \lambda_i x_i = 0, \qquad \frac{1}{x_i + \alpha_i} + \lambda_i = \nu.$$

If $\nu < 1/\alpha_i$, then $\lambda_i = 0$ and $x_i = 1/\nu - \alpha_i$; if $\nu \geq 1/\alpha_i$, then $\lambda_i = \nu - 1/\alpha_i$ and $x_i = 0$. Determine ν from $\mathbf{1}^T x = \sum_{i=1}^n \max\{0, 1/\nu - \alpha_i\} = 1$. Interpretation: there are n patches, the ground level of patch i is at height α_i, and we flood the area with a unit amount of water; the resulting water level is $1/\nu^\star$ (Figure 22).

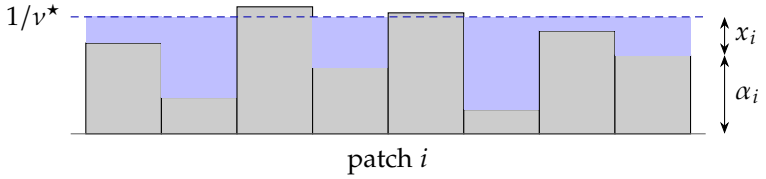


Figure 22: Water-filling with $n = 8$ patches of heights $\alpha = (0.30, 0.12, 0.42, 0.22, 0.40, 0.08, 0.34, 0.26)$. A unit of water settles at level $1/\nu^\star = 0.387$; patches 3 and 5 stay dry ($x_i = 0$).

8.6 Perturbation and sensitivity analysis

The unperturbed problem and its dual are

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p, \end{array} \qquad \begin{array}{ll} \text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \succeq 0, \end{array}$$

and the perturbed problem and its dual are

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq u_i, \quad i = 1, \ldots, m \\ & h_i(x) = v_i, \quad i = 1, \ldots, p, \end{array} \qquad \begin{array}{ll} \text{maximize} & g(\lambda, \nu) - u^T\lambda - v^T\nu \\ \text{subject to} & \lambda \succeq 0. \end{array}$$

Here x is the primal variable and u, v are parameters; $p^\star(u, v)$ is the optimal value as a function of u, v. We want information about $p^\star(u, v)$ from the solution of the unperturbed problem and its dual.

Proposition 8.6 [Global sensitivity]. *Assume strong duality holds for the unperturbed problem and $\lambda^\star, \nu^\star$ are dual optimal for it. Then*

$$p^\star(u, v) \geq p^\star(0, 0) - u^T\lambda^\star - v^T\nu^\star.$$

Proof. Weak duality for the perturbed problem gives $p^\star(u,v) \geq g(\lambda^\star,\nu^\star) - u^T\lambda^\star - v^T\nu^\star$, and $g(\lambda^\star,\nu^\star) = p^\star(0,0)$ by strong duality. □

Sensitivity interpretation:

- if $\lambda_i^\star$ is large, $p^\star$ increases greatly if we tighten constraint i ($u_i < 0$);
- if $\lambda_i^\star$ is small, $p^\star$ does not decrease much if we loosen constraint i ($u_i > 0$);
- if $\nu_i^\star$ is large and positive, $p^\star$ increases greatly if we take $v_i < 0$; if $\nu_i^\star$ is large and negative, $p^\star$ increases greatly if we take $v_i > 0$;
- if $\nu_i^\star$ is small and positive, $p^\star$ does not decrease much if we take $v_i > 0$; if $\nu_i^\star$ is small and negative, $p^\star$ does not decrease much if we take $v_i < 0$.

Proposition 8.7 [Local sensitivity]. *If, in addition, $p^\star(u,v)$ is differentiable at $(0,0)$, then*

$$\lambda_i^\star = -\frac{\partial p^\star(0,0)}{\partial u_i}, \qquad \nu_i^\star = -\frac{\partial p^\star(0,0)}{\partial v_i}.$$

Proof. For $\lambda_i^\star$, the global sensitivity result with $u = te_i$, $v = 0$ gives

$$\frac{\partial p^\star(0,0)}{\partial u_i} = \lim_{t\searrow 0} \frac{p^\star(te_i,0) - p^\star(0,0)}{t} \geq -\lambda_i^\star, \qquad \frac{\partial p^\star(0,0)}{\partial u_i} = \lim_{t\nearrow 0} \frac{p^\star(te_i,0) - p^\star(0,0)}{t} \leq -\lambda_i^\star,$$

where the inequality flips for $t < 0$; hence equality. □

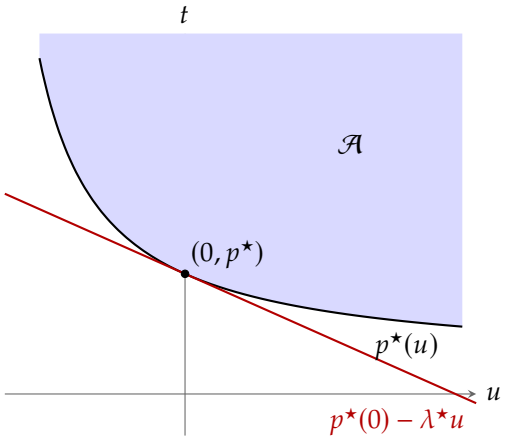


Figure 23: For one inequality constraint, the boundary of $\mathcal{A}$ is the graph of $p^\star(u)$. For a convex problem $\mathcal{A}$ is convex, and the line $p^\star(0) - \lambda^\star u$ supports it at $(0, p^\star)$, lying below $p^\star(u)$ everywhere (schematic).

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 8 (2023); original slides, *Duality*.

Lecture 9: Duality (Continued); Approximation and Fitting

9.1 Duality and problem reformulations

Equivalent formulations of a problem can lead to very different duals, so reformulating the primal problem can be useful when the dual is difficult to derive or uninteresting. Common reformulations: introduce new variables and equality constraints; make explicit constraints implicit or vice versa; and transform objective or constraint functions, for example replace $f_0(x)$ by $\phi(f_0(x))$ with ϕ convex and increasing.

Introducing new variables and equality constraints. For the unconstrained problem of minimizing $f_0(Ax+b)$, the dual function is constant, $g = \inf_x L(x) = \inf_x f_0(Ax+b) = p^\star$: strong duality holds, but the dual is useless. The reformulated problem and its dual are

$$\begin{array}{ll} \text{minimize} & f_0(y) \\ \text{subject to} & Ax+b-y=0, \end{array} \qquad \begin{array}{ll} \text{maximize} & b^T\nu - f_0^*(\nu) \\ \text{subject to} & A^T\nu = 0, \end{array}$$

since

$$g(\nu) = \inf_{x,y}\big(f_0(y) - \nu^T y + \nu^T A x + b^T\nu\big) = \begin{cases} -f_0^*(\nu) + b^T\nu & A^T\nu = 0, \\ -\infty & \text{otherwise.} \end{cases}$$

Example: *Norm approximation*: minimize $\|Ax-b\|$, reformulated as minimize $\|y\|$ subject to $y = Ax-b$. We can look up the conjugate of $\|\cdot\|$, or derive the dual directly:

$$\begin{aligned} g(\nu) = \inf_{x,y}\big(\|y\| + \nu^T y - \nu^T A x + b^T\nu\big) &= \begin{cases} b^T\nu + \inf_y(\|y\| + \nu^T y) & A^T\nu = 0, \\ -\infty & \text{otherwise} \end{cases} \\ &= \begin{cases} b^T\nu & A^T\nu = 0,\ \|\nu\|_* \le 1, \\ -\infty & \text{otherwise,} \end{cases} \end{aligned}$$

using $\inf_y(\|y\| - z^T y) = 0$ if $\|z\|_* \le 1$ and $-\infty$ otherwise (Lecture 7). The dual of the norm approximation problem is to maximize $b^T\nu$ subject to $A^T\nu = 0$, $\|\nu\|_* \le 1$.

Implicit constraints. The LP with box constraints and its dual are

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & -\mathbf{1} \le x \le \mathbf{1}, \end{array} \qquad \begin{array}{ll} \text{maximize} & -b^T\nu - \mathbf{1}^T\lambda_1 - \mathbf{1}^T\lambda_2 \\ \text{subject to} & c + A^T\nu + \lambda_1 - \lambda_2 = 0 \\ & \lambda_1 \ge 0, \quad \lambda_2 \ge 0. \end{array}$$

Making the box constraints implicit, minimize $f_0(x)$ subject to $Ax = b$, where $f_0(x) = c^T x$ if

$-\mathbf{1} \preceq x \preceq \mathbf{1}$ and $f_0(x) = \infty$ otherwise. The dual function is

$$g(\nu) = \inf_{-\mathbf{1} \preceq x \preceq \mathbf{1}} \big(c^T x + \nu^T (Ax - b)\big) = -b^T \nu - \|A^T \nu + c\|_1,$$

and the dual problem is the unconstrained problem of maximizing $-b^T \nu - \|A^T \nu + c\|_1$.

9.2 Problems with generalized inequalities

Consider

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \preceq_{K_i} 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p, \end{array}$$

where $\preceq_{K_i}$ is a generalized inequality on $\mathbb{R}^{k_i}$. The definitions parallel the scalar case: the Lagrange multiplier for $f_i(x) \preceq_{K_i} 0$ is a vector $\lambda_i \in \mathbb{R}^{k_i}$; the Lagrangian $L : \mathbb{R}^n \times \mathbb{R}^{k_1} \times \cdots \times \mathbb{R}^{k_m} \times \mathbb{R}^p \to \mathbb{R}$ is

$$L(x, \lambda_1, \ldots, \lambda_m, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i^T f_i(x) + \sum_{i=1}^p \nu_i h_i(x);$$

and the dual function is $g(\lambda_1, \ldots, \lambda_m, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda_1, \ldots, \lambda_m, \nu)$.

Proposition 9.1 [Lower bound property]. *If $\lambda_i \succeq_{K_i^*} 0$, then $g(\lambda_1, \ldots, \lambda_m, \nu) \le p^\star$.*

Proof. If $\tilde{x}$ is feasible and $\lambda_i \succeq_{K_i^*} 0$, then $\lambda_i^T f_i(\tilde{x}) \le 0$ by the definition of the dual cone, so

$$f_0(\tilde{x}) \ge f_0(\tilde{x}) + \sum_{i=1}^m \lambda_i^T f_i(\tilde{x}) + \sum_{i=1}^p \nu_i h_i(\tilde{x}) \ge \inf_{x \in \mathcal{D}} L(x, \lambda_1, \ldots, \lambda_m, \nu) = g(\lambda_1, \ldots, \lambda_m, \nu).$$

Minimizing over all feasible $\tilde{x}$ gives $p^\star \ge g(\lambda_1, \ldots, \lambda_m, \nu)$. □

The dual problem is to maximize $g(\lambda_1, \ldots, \lambda_m, \nu)$ subject to $\lambda_i \succeq_{K_i^*} 0$, $i = 1, \ldots, m$. Weak duality $p^\star \ge d^\star$ always holds, and strong duality $p^\star = d^\star$ holds for convex problems with a constraint qualification (for example Slater's: the primal problem is strictly feasible).

Example: *Semidefinite program.* The primal SDP (with $F_i, G \in \mathbf{S}^k$) is to minimize $c^T x$ subject to $x_1 F_1 + \cdots + x_n F_n \preceq G$. The Lagrange multiplier is a matrix $Z \in \mathbf{S}^k$, with Lagrangian $L(x, Z) = c^T x + \operatorname{tr}\big(Z(x_1 F_1 + \cdots + x_n F_n - G)\big)$ and dual function

$$g(Z) = \inf_x L(x, Z) = \begin{cases} -\operatorname{tr}(GZ) & \operatorname{tr}(F_i Z) + c_i = 0, \ i = 1, \ldots, n, \\ -\infty & \text{otherwise.} \end{cases}$$

The dual SDP is

$$\begin{array}{ll} \text{maximize} & -\operatorname{tr}(GZ) \\ \text{subject to} & Z \succeq 0, \quad \operatorname{tr}(F_i Z) + c_i = 0, \quad i = 1, \ldots, n, \end{array}$$

and $p^\star = d^\star$ if the primal SDP is strictly feasible (there is an x with $x_1 F_1 + \cdots + x_n F_n \prec G$).

9.3 Norm approximation

Definition 9.2 [Norm approximation]. The norm approximation problem is to minimize $\|Ax - b\|$, with $A \in \mathbb{R}^{m \times n}$, $m \geq n$, and $\|\cdot\|$ a norm on $\mathbb{R}^m$.

Interpretations of the solution $x^\star = \arg\min_x \|Ax - b\|$:

- *geometric*: $Ax^\star$ is the point in $\mathcal{R}(A)$ closest to b;
- *estimation*: in the linear measurement model $y = Ax + v$, y are measurements, x is unknown, and v is measurement error; given $y = b$, the best guess of x is $x^\star$;
- *optimal design*: x are design variables (input) and Ax is the result (output); $x^\star$ is the design that best approximates the desired result b.

Examples:

- least-squares approximation ($\|\cdot\|_2$): the solution satisfies the normal equations $A^T A x = A^T b$ ($x^\star = (A^T A)^{-1} A^T b$ if $\operatorname{rank} A = n$);
- Chebyshev approximation ($\|\cdot\|_\infty$): solved as the LP minimize t subject to $-t\mathbf{1} \preceq Ax - b \preceq t\mathbf{1}$;
- sum of absolute residuals approximation ($\|\cdot\|_1$): solved as the LP minimize $\mathbf{1}^T y$ subject to $-y \preceq Ax - b \preceq y$.

9.4 Penalty function approximation

Definition 9.3 [Penalty function approximation]. Penalty function approximation is the problem

$$\begin{array}{ll} \text{minimize} & \phi(r_1) + \cdots + \phi(r_m) \\ \text{subject to} & r = Ax - b, \end{array}$$

with $A \in \mathbb{R}^{m \times n}$ and $\phi : \mathbb{R} \to \mathbb{R}$ a convex penalty function.

Examples (Figure 24, left):

- quadratic: $\phi(u) = u^2$;
- deadzone-linear with width a: $\phi(u) = \max\{0, |u| - a\}$;

- log-barrier with limit a: $\phi(u) = -a^2 \log(1 - (u/a)^2)$ for $|u| < a$, and ∞ otherwise.

The shape of the penalty function has a large effect on the distribution of the residuals. In an example with $m = 100$, $n = 30$, the ℓ_1 penalty $\phi(u) = |u|$ puts many residuals exactly at zero; the quadratic penalty spreads them out with few large ones; the deadzone penalty pushes many residuals to the edges $\pm a$ of the dead zone; and the log barrier with $a = 1$ keeps all residuals inside $(-1, 1)$.

Definition 9.4 [Huber penalty]. The Huber penalty with parameter M is

$$\phi_{\text{hub}}(u) = \begin{cases} u^2 & |u| \le M, \\ M(2|u| - M) & |u| > M. \end{cases}$$

Its linear growth for large u makes the approximation less sensitive to outliers (Figure 24, right). For example, fitting an affine function $f(t) = \alpha + \beta t$ to points containing a couple of outliers, the quadratic penalty lets the outliers tilt the fitted line, while the Huber penalty essentially ignores them.

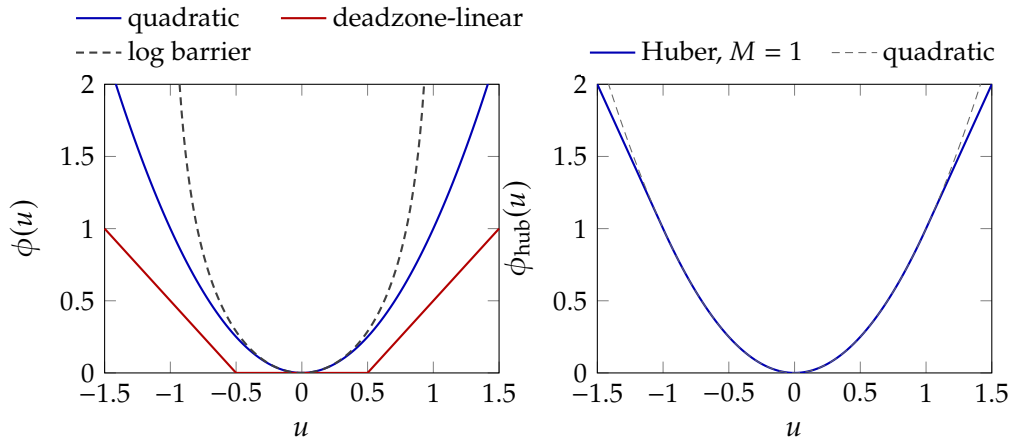


Figure 24: Left: the quadratic, deadzone-linear ($a = 1/2$), and log-barrier ($a = 1$) penalties. Right: the Huber penalty with $M = 1$ agrees with u^2 on $|u| \le 1$ and grows linearly beyond.

9.5 Least-norm problems

Definition 9.5 [Least-norm problem]. The least-norm problem is to minimize $\|x\|$ subject to $Ax = b$, with $A \in \mathbb{R}^{m \times n}$, $m \le n$, and $\|\cdot\|$ a norm on $\mathbb{R}^n$.

Interpretations of the solution $x^\star = \arg\min_{Ax=b} \|x\|$:

- *geometric*: $x^\star$ is the point in the affine set $\{x \mid Ax = b\}$ with minimum distance to 0;
- *estimation*: $b = Ax$ are (perfect) measurements of x, and $x^\star$ is the smallest ("most plausible")

estimate consistent with the measurements;

- *design*: x are design variables (inputs) and b are required results (outputs); $x^\star$ is the smallest ("most efficient") design that satisfies the requirements.

Examples:

- least-squares solution of linear equations ($\|\cdot\|_2$): solved via the optimality conditions $2x + A^T\nu = 0$, $Ax = b$;
- minimum sum of absolute values ($\|\cdot\|_1$): solved as the LP minimize $\mathbf{1}^T y$ subject to $-y \preceq x \preceq y$, $Ax = b$; it tends to produce a sparse solution $x^\star$.

The extension to a LEAST-PENALTY PROBLEM minimizes $\phi(x_1) + \cdots + \phi(x_n)$ subject to $Ax = b$, with $\phi : \mathbb{R} \to \mathbb{R}$ a convex penalty function.

9.6 Regularized approximation

Definition 9.6 [REGULARIZED APPROXIMATION]. REGULARIZED APPROXIMATION is the bi-criterion problem

$$\text{minimize (w.r.t. } \mathbb{R}^2_+) \quad \big(\|Ax - b\|,\ \|x\|\big),$$

with $A \in \mathbb{R}^{m\times n}$; the norms on $\mathbb{R}^m$ and $\mathbb{R}^n$ can be different.

The aim is a good approximation $Ax \approx b$ with small x. Interpretations:

- *estimation*: linear measurement model $y = Ax + v$, with prior knowledge that $\|x\|$ is small;
- *optimal design*: small x is cheaper or more efficient, or the linear model $y = Ax$ is only valid for small x;
- *robust approximation*: a good approximation $Ax \approx b$ with small x is less sensitive to errors in A than one with large x.

The scalarized problem minimizes $\|Ax - b\| + \gamma\|x\|$; its solutions for $\gamma > 0$ trace out the optimal trade-off curve. Another common method minimizes $\|Ax - b\|^2 + \delta\|x\|^2$ with $\delta > 0$.

Example: *Tikhonov regularization*, minimize $\|Ax - b\|_2^2 + \delta\|x\|_2^2$, is the least-squares problem

$$\text{minimize} \quad \left\| \begin{bmatrix} A \\ \sqrt{\delta} I \end{bmatrix} x - \begin{bmatrix} b \\ 0 \end{bmatrix} \right\|_2^2,$$

with solution $x^\star = (A^TA + \delta I)^{-1}A^Tb$.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 9 (2023); original slides, *Duality* and *Approximation and fitting*.

Lecture 10: Approximation and Fitting (Continued); Statistical Estimation

10.1 Optimal input design

A linear dynamical system with impulse response h maps an input u to the output

$$y(t) = \sum_{\tau=0}^{t} h(\tau)u(t-\tau), \qquad t = 0, 1, \ldots, N.$$

The input design problem is a multicriterion problem with three objectives:

1. tracking error with desired output y_{des}: $J_{\text{track}} = \sum_{t=0}^{N}(y(t) - y_{\text{des}}(t))^2$;
2. input magnitude: $J_{\text{mag}} = \sum_{t=0}^{N} u(t)^2$;
3. input variation: $J_{\text{der}} = \sum_{t=0}^{N-1}(u(t+1) - u(t))^2$.

The goal is to track the desired output using a small and slowly varying input signal. The regularized least-squares formulation

$$\text{minimize} \quad J_{\text{track}} + \delta J_{\text{der}} + \eta J_{\text{mag}}$$

is, for fixed δ, η, a least-squares problem in $u(0), \ldots, u(N)$. Three solutions on the optimal trade-off surface, tracking a desired output made of steps, illustrate the trade-off: with $\delta = 0$ and small η the output tracks almost perfectly but the input is large, with sharp oscillating spikes at the transitions; with $\delta = 0$ and larger η the input is smaller but still oscillatory; with large δ the input is smooth, at the price of rounded transitions in the output.

10.2 Signal reconstruction

Definition 10.1 [Signal reconstruction]. Signal reconstruction is the bi-criterion problem

$$\text{minimize (w.r.t. } \mathbb{R}^2_+) \quad \left(\|\hat{x} - x_{\text{cor}}\|_2,\ \phi(\hat{x})\right),$$

where $x \in \mathbb{R}^n$ is an unknown signal, $x_{\text{cor}} = x + v$ is a known version corrupted by additive noise v, the variable $\hat{x}$ (the reconstructed signal) is an estimate of x, and $\phi : \mathbb{R}^n \to \mathbb{R}$ is a regularization function or smoothing objective.

Examples are quadratic smoothing and total variation smoothing:

$$\phi_{\text{quad}}(\hat{x}) = \sum_{i=1}^{n-1}(\hat{x}_{i+1} - \hat{x}_i)^2, \qquad \phi_{\text{tv}}(\hat{x}) = \sum_{i=1}^{n-1}|\hat{x}_{i+1} - \hat{x}_i|.$$

On a smoothly varying signal, quadratic smoothing works well: along the trade-off curve, too little smoothing leaves noise and too much flattens the signal, with a good reconstruction in between. On a signal that is smooth except for a few jumps, quadratic smoothing smooths out the noise *and* the sharp transitions: along its trade-off curve one can remove the noise only by rounding off the jumps. Total variation smoothing preserves the sharp transitions in the signal while removing the noise.

10.3 Robust approximation

To minimize $\|Ax - b\|$ with uncertain A, there are two approaches:

- *stochastic*: assume A is random and minimize $\mathbb{E}\|Ax - b\|$;
- *worst-case*: given a set $\mathcal{A}$ of possible values of A, minimize $\sup_{A\in\mathcal{A}}\|Ax - b\|$.

Both are tractable only in special cases (certain norms $\|\cdot\|$, distributions, and sets $\mathcal{A}$).

Example: Let $A(u) = A_0 + uA_1$. Let x_{nom} minimize $\|A_0x - b\|_2^2$, x_{stoch} minimize $\mathbb{E}\|A(u)x - b\|_2^2$ with u uniform on $[-1, 1]$, and x_{wc} minimize $\sup_{-1\le u\le 1}\|A(u)x - b\|_2^2$. Plotting $r(u) = \|A(u)x - b\|_2$ for each: x_{nom} has the smallest residual near $u = 0$ but its residual grows fastest as $|u|$ grows; x_{wc} has the flattest residual over $[-1, 1]$; and x_{stoch} lies in between.

Stochastic robust least-squares. Let $A = \bar{A} + U$ with U random, $\mathbb{E}U = 0$, and $\mathbb{E}U^TU = P$, and minimize $\mathbb{E}\|(\bar{A} + U)x - b\|_2^2$. The objective has the explicit expression

$$\mathbb{E}\|Ax - b\|_2^2 = \mathbb{E}\|\bar{A}x - b + Ux\|_2^2 = \|\bar{A}x - b\|_2^2 + \mathbb{E}x^TU^TUx = \|\bar{A}x - b\|_2^2 + x^TPx,$$

so the robust LS problem is equivalent to the LS problem of minimizing $\|\bar{A}x - b\|_2^2 + \|P^{1/2}x\|_2^2$. For $P = \delta I$ this is the Tikhonov regularized problem, minimize $\|\bar{A}x - b\|_2^2 + \delta\|x\|_2^2$.

Worst-case robust least-squares. Let $\mathcal{A} = \{\bar{A} + u_1A_1 + \cdots + u_pA_p \mid \|u\|_2 \le 1\}$ and minimize

$$\sup_{A\in\mathcal{A}}\|Ax - b\|_2^2 = \sup_{\|u\|_2\le 1}\|P(x)u + q(x)\|_2^2, \qquad P(x) = \begin{bmatrix} A_1x & A_2x & \cdots & A_px \end{bmatrix}, \quad q(x) = \bar{A}x - b.$$

By the nonconvex problem with strong duality (Lecture 8), strong duality holds between

$$\begin{array}{ll} \text{maximize} & \|Pu + q\|_2^2 \\ \text{subject to} & \|u\|_2^2 \le 1 \end{array} \qquad \text{and} \qquad \begin{array}{ll} \text{minimize} & t + \lambda \\ \text{subject to} & \begin{bmatrix} I & P & q \\ P^T & \lambda I & 0 \\ q^T & 0 & t \end{bmatrix} \succeq 0. \end{array}$$

Hence the robust LS problem is equivalent to the SDP in x, t, λ

$$\begin{array}{ll} \text{minimize} & t + \lambda \\ \text{subject to} & \begin{bmatrix} I & P(x) & q(x) \\ P(x)^T & \lambda I & 0 \\ q(x)^T & 0 & t \end{bmatrix} \geq 0, \end{array}$$

which is an LMI because $P(x)$ and $q(x)$ are affine in x.

Example: Compare the residuals $r(u) = \|(A_0 + u_1 A_1 + u_2 A_2)x - b\|_2$, with u uniformly distributed on the unit disk, for three choices of x: x_{ls} minimizes $\|A_0 x - b\|_2$; x_{tik} minimizes $\|A_0 x - b\|_2^2 + \delta\|x\|_2^2$ (the Tikhonov solution); and x_{rls} minimizes $\sup_{A \in \mathcal{A}} \|Ax - b\|_2^2 + \|x\|_2^2$. The histogram of $r(u)$ is widest for x_{ls}, narrower for x_{tik}, and sharply concentrated for x_{rls}, whose worst residual is far smaller than that of x_{ls}.

10.4 Maximum likelihood estimation

The DISTRIBUTION ESTIMATION PROBLEM is to estimate the probability density $p(y)$ of a random variable from observed values. In PARAMETRIC DISTRIBUTION ESTIMATION we choose from a family of densities $p_x(y)$ indexed by a parameter x.

Definition 10.2 [MAXIMUM LIKELIHOOD ESTIMATION**].** MAXIMUM LIKELIHOOD ESTIMATION solves

$$\text{maximize (over } x) \quad \log p_x(y),$$

where y is the observed value and $l(x) = \log p_x(y)$ is the LOG-LIKELIHOOD FUNCTION.

Constraints $x \in C$ can be added explicitly, or by defining $p_x(y) = 0$ for $x \notin C$. The problem is a convex optimization problem if $\log p_x(y)$ is concave in x for fixed y.

Linear measurements with IID noise. In the linear measurement model

$$y_i = a_i^T x + v_i, \qquad i = 1, \ldots, m,$$

$x \in \mathbb{R}^n$ is a vector of unknown parameters, v_i is IID measurement noise with density $p(z)$, and the measurement $y \in \mathbb{R}^m$ has density $p_x(y) = \prod_{i=1}^m p(y_i - a_i^T x)$. A maximum likelihood estimate is any solution x of

$$\text{maximize} \quad l(x) = \sum_{i=1}^m \log p(y_i - a_i^T x)$$

(y is the observed value). Examples:

- Gaussian noise $\mathcal{N}(0, \sigma^2)$, $p(z) = (2\pi\sigma^2)^{-1/2} e^{-z^2/(2\sigma^2)}$:

$$l(x) = -\frac{m}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{m}(a_i^T x - y_i)^2,$$

  so the ML estimate is the least-squares solution;

- Laplacian noise, $p(z) = (1/(2a))e^{-|z|/a}$:

$$l(x) = -m\log(2a) - \frac{1}{a}\sum_{i=1}^{m}|a_i^T x - y_i|,$$

  so the ML estimate is the ℓ_1-norm solution;

- uniform noise on $[-a, a]$: $l(x) = -m\log(2a)$ if $|a_i^T x - y_i| \leq a$ for $i = 1, \ldots, m$, and $-\infty$ otherwise, so the ML estimate is any x with $|a_i^T x - y_i| \leq a$.

10.5 Logistic regression

A random variable $y \in \{0, 1\}$ has the distribution

$$p = \text{prob}(y = 1) = \frac{\exp(a^T u + b)}{1 + \exp(a^T u + b)},$$

where a, b are parameters and $u \in \mathbb{R}^n$ are (observable) explanatory variables. The estimation problem is to estimate a, b from m observations (u_i, y_i). Ordering the data so that $y_1 = \cdots = y_k = 1$ and $y_{k+1} = \cdots = y_m = 0$, the log-likelihood is

$$\begin{aligned} l(a, b) &= \log\left(\prod_{i=1}^{k}\frac{\exp(a^T u_i + b)}{1 + \exp(a^T u_i + b)}\prod_{i=k+1}^{m}\frac{1}{1 + \exp(a^T u_i + b)}\right) \\ &= \sum_{i=1}^{k}(a^T u_i + b) - \sum_{i=1}^{m}\log\bigl(1 + \exp(a^T u_i + b)\bigr), \end{aligned}$$

which is concave in a, b (a linear function minus a sum of log-sum-exp terms of affine functions). For $n = 1$ and $m = 50$ measurements, the ML estimate of $p = \exp(au + b)/(1 + \exp(au + b))$ is a sigmoid curve in u that rises from near 0, where most observations are $y_i = 0$, to near 1, where most are $y_i = 1$.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 10 (2023); original slides, *Approximation and fitting* and *Statistical estimation*.

Lecture 11: Statistical Estimation (Continued); Geometric Problems

11.1 Binary hypothesis testing

In the detection (hypothesis testing) problem we observe a random variable $X \in \{1, \ldots, n\}$ and choose between

- hypothesis 1: X was generated by the distribution $p = (p_1, \ldots, p_n)$;
- hypothesis 2: X was generated by the distribution $q = (q_1, \ldots, q_n)$.

Definition 11.1 [Randomized detector]. A randomized detector is a nonnegative matrix $T \in \mathbb{R}^{2\times n}$ with $\mathbf{1}^T T = \mathbf{1}^T$: if we observe $X = k$, we choose hypothesis 1 with probability t_{1k} and hypothesis 2 with probability t_{2k}. If all elements of T are 0 or 1, it is a deterministic detector.

The detection probability matrix is

$$D = \begin{bmatrix} Tp & Tq \end{bmatrix} = \begin{bmatrix} 1 - P_{\text{fp}} & P_{\text{fn}} \\ P_{\text{fp}} & 1 - P_{\text{fn}} \end{bmatrix},$$

where P_{fp} is the probability of selecting hypothesis 2 when X is generated by distribution 1 (false positive) and P_{fn} the probability of selecting hypothesis 1 when X is generated by distribution 2 (false negative). Detector design is the multicriterion LP in the variable $T \in \mathbb{R}^{2\times n}$

$$\begin{array}{ll} \text{minimize (w.r.t. } \mathbb{R}^2_+) & (P_{\text{fp}}, P_{\text{fn}}) = ((Tp)_2, (Tq)_1) \\ \text{subject to} & t_{1k} + t_{2k} = 1, \quad k = 1, \ldots, n \\ & t_{ik} \geq 0, \quad i = 1, 2, \quad k = 1, \ldots, n. \end{array}$$

Scalarization. With weight $\lambda > 0$, minimize $(Tp)_2 + \lambda(Tq)_1$ subject to the same constraints. This LP has a simple analytical solution: since the objective is $\sum_k (p_k t_{2k} + \lambda q_k t_{1k})$, each column is chosen separately,

$$(t_{1k}, t_{2k}) = \begin{cases} (1, 0) & p_k \geq \lambda q_k, \\ (0, 1) & p_k < \lambda q_k. \end{cases}$$

This is a deterministic detector given by a likelihood ratio test. If $p_k = \lambda q_k$ for some k, any value $0 \leq t_{1k} \leq 1$, $t_{1k} = 1 - t_{2k}$ is optimal, so the Pareto optimal detectors include non-deterministic ones.

Minimax detector. Minimizing $\max\{P_{\text{fp}}, P_{\text{fn}}\} = \max\{(Tp)_2, (Tq)_1\}$ subject to the same constraints is an LP whose solution is usually not deterministic.

Example: For

$$P = \begin{bmatrix} p & q \end{bmatrix} = \begin{bmatrix} 0.70 & 0.10 \\ 0.20 & 0.10 \\ 0.05 & 0.70 \\ 0.05 & 0.10 \end{bmatrix},$$

the optimal trade-off curve is shown in Figure 25. Solutions 1, 2, 3 (and the endpoints) are deterministic; solution 4 is the minimax detector.

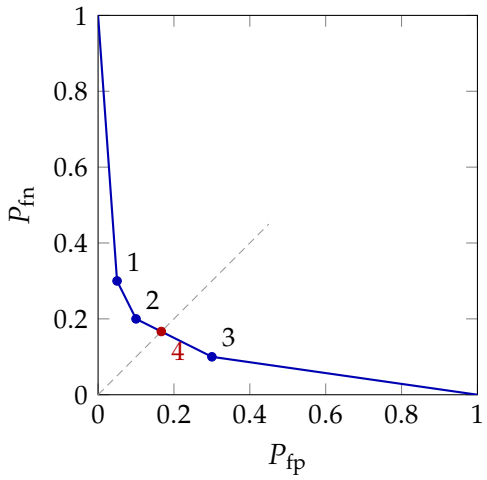


Figure 25: Optimal trade-off between P_{fp} and P_{fn} for the example. The vertices $1, 2, 3$ are likelihood ratio tests; the minimax detector 4, at $P_{\text{fp}} = P_{\text{fn}} = 1/6$, randomizes between detectors 2 and 3.

11.2 Experiment design

Consider m linear measurements $y_i = a_i^T x + w_i$, $i = 1, \ldots, m$, of an unknown $x \in \mathbb{R}^n$, with IID measurement errors $w_i \sim \mathcal{N}(0, 1)$. The ML (least-squares) estimate is

$$\hat{x} = \left(\sum_{i=1}^{m} a_i a_i^T \right)^{-1} \sum_{i=1}^{m} y_i a_i,$$

and the error $e = \hat{x} - x$ has zero mean and covariance

$$E = \mathbb{E} e e^T = \left(\sum_{i=1}^{m} a_i a_i^T \right)^{-1};$$

confidence ellipsoids are $\{x \mid (x - \hat{x})^T E^{-1} (x - \hat{x}) \leq \beta\}$. Experiment design chooses $a_i \in \{v_1, \ldots, v_p\}$ (a set of possible test vectors) to make E "small".

Vector optimization formulation.

$$\begin{array}{ll} \text{minimize (w.r.t. } \mathbf{S}_+^n) & E = \left(\sum_{k=1}^p m_k v_k v_k^T\right)^{-1} \\ \text{subject to} & m_k \geq 0, \quad m_1 + \cdots + m_p = m, \quad m_k \in \mathbb{Z}, \end{array}$$

with variables m_k, the number of vectors a_i equal to v_k. This is difficult in general, due to the integer constraint.

Relaxed experiment design. Assume $m \gg p$ and use $\lambda_k = m_k/m$ as a continuous variable:

$$\begin{array}{ll} \text{minimize (w.r.t. } \mathbf{S}_+^n) & E = (1/m)\left(\sum_{k=1}^p \lambda_k v_k v_k^T\right)^{-1} \\ \text{subject to} & \lambda \geq 0, \quad \mathbf{1}^T\lambda = 1. \end{array}$$

Common scalarizations minimize $\log\det E$, $\operatorname{tr} E$, $\lambda_{\max}(E)$, and so on; other convex constraints can be added, for example a bound $c^T\lambda \leq B$ on the experiment cost.

Definition 11.2 [D-optimal design]. D-optimal design is

$$\begin{array}{ll} \text{minimize} & \log\det\left(\sum_{k=1}^p \lambda_k v_k v_k^T\right)^{-1} \\ \text{subject to} & \lambda \geq 0, \quad \mathbf{1}^T\lambda = 1. \end{array}$$

It minimizes the volume of the confidence ellipsoids.

Its dual problem is

$$\begin{array}{ll} \text{maximize} & \log\det W + n\log n \\ \text{subject to} & v_k^T W v_k \leq 1, \quad k = 1, \ldots, p. \end{array}$$

Interpretation: $\{x \mid x^T W x \leq 1\}$ is the minimum volume ellipsoid centered at the origin that includes all test vectors v_k. Complementary slackness for primal and dual optimal λ, W gives

$$\lambda_k(1 - v_k^T W v_k) = 0, \qquad k = 1, \ldots, p,$$

so the optimal experiment uses only vectors v_k on the boundary of the ellipsoid defined by W. For example, with $p = 20$ candidate vectors in $\mathbb{R}^2$, the optimal design uses just two of them, each with $\lambda_k = 0.5$, both on the boundary of the ellipse defined by the optimal W.

Derivation of the dual. Reformulate the primal problem with a new variable X: minimize $\log\det X^{-1}$ subject to $X = \sum_{k=1}^p \lambda_k v_k v_k^T$, $\lambda \geq 0$, $\mathbf{1}^T\lambda = 1$. The Lagrangian is

$$L(X, \lambda, Z, z, \nu) = \log\det X^{-1} + \operatorname{tr}\left(Z\left(X - \sum_{k=1}^p \lambda_k v_k v_k^T\right)\right) - z^T\lambda + \nu(\mathbf{1}^T\lambda - 1).$$

Minimizing over X by setting the gradient to zero gives $-X^{-1} + Z = 0$, and the minimum over

λ_k is $-\infty$ unless $-v_k^T Z v_k - z_k + \nu = 0$. The dual problem is therefore

$$\begin{array}{ll} \text{maximize} & n + \log\det Z - \nu \\ \text{subject to} & v_k^T Z v_k \le \nu, \quad k = 1, \ldots, p. \end{array}$$

Changing variables to $W = Z/\nu$ and optimizing over ν (at $\nu = n$) gives the dual above. □

11.3 Extremal volume ellipsoids

Definition 11.3 [Löwner–John ellipsoid]. The Löwner–John ellipsoid of a set C is the minimum volume ellipsoid $\mathcal{E}$ with $C \subseteq \mathcal{E}$.

Parametrize $\mathcal{E} = \{v \mid \|Av + b\|_2 \le 1\}$, where without loss of generality $A \in \mathbf{S}^n_{++}$. Since $\operatorname{vol}\mathcal{E}$ is proportional to $\det A^{-1}$, the minimum volume ellipsoid solves

$$\begin{array}{ll} \text{minimize (over } A, b) & \log\det A^{-1} \\ \text{subject to} & \sup_{v \in C} \|Av + b\|_2 \le 1, \end{array}$$

which is convex, but evaluating the constraint can be hard for general C. For a finite set $C = \{x_1, \ldots, x_m\}$ it becomes

$$\begin{array}{ll} \text{minimize (over } A, b) & \log\det A^{-1} \\ \text{subject to} & \|Ax_i + b\|_2 \le 1, \quad i = 1, \ldots, m, \end{array}$$

which also gives the Löwner–John ellipsoid of the polyhedron $\operatorname{conv}\{x_1, \ldots, x_m\}$.

Maximum volume inscribed ellipsoid. To find the maximum volume ellipsoid $\mathcal{E}$ inside a convex set $C \subseteq \mathbb{R}^n$, parametrize $\mathcal{E} = \{Bu + d \mid \|u\|_2 \le 1\}$ with, without loss of generality, $B \in \mathbf{S}^n_{++}$. Since $\operatorname{vol}\mathcal{E}$ is proportional to $\det B$, solve

$$\begin{array}{ll} \text{maximize} & \log\det B \\ \text{subject to} & \sup_{\|u\|_2 \le 1} I_C(Bu + d) \le 0, \end{array}$$

where $I_C(x) = 0$ for $x \in C$ and $I_C(x) = \infty$ for $x \notin C$. Again this is convex, but the constraint can be hard to evaluate. For a polyhedron $\{x \mid a_i^T x \le b_i,\ i = 1, \ldots, m\}$ it becomes

$$\begin{array}{ll} \text{maximize} & \log\det B \\ \text{subject to} & \|Ba_i\|_2 + a_i^T d \le b_i, \quad i = 1, \ldots, m, \end{array}$$

since $\sup_{\|u\|_2 \le 1} a_i^T(Bu + d) = \|Ba_i\|_2 + a_i^T d$.

Efficiency of ellipsoidal approximations. Let $C \subseteq \mathbb{R}^n$ be convex, bounded, and with nonempty interior. The Löwner–John ellipsoid, shrunk by a factor n (about its center), lies inside C, and

the maximum volume inscribed ellipsoid, expanded by a factor n, covers C. The factor n can be improved to $\sqrt{n}$ if C is symmetric.

11.4 Centering

Some possible definitions of the "center" of a convex set C:

- the center of the largest inscribed ball, the Chebyshev center; for a polyhedron it can be computed via linear programming (Lecture 6);
- the center of the maximum volume inscribed ellipsoid, the MVE center.

The MVE center is invariant under affine coordinate transformations, while the Chebyshev center is not.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 11 (2023); original slides, *Statistical estimation* and *Geometric problems*.

Lecture 12: Geometric Problems (Continued); Numerical Linear Algebra

12.1 Analytic center

Definition 12.1 [Analytic center]**.** The analytic center of a set of convex inequalities and linear equations $f_i(x) \leq 0$, $i = 1, \ldots, m$, $Fx = g$, is the optimal point of

$$\begin{array}{ll} \text{minimize} & -\sum_{i=1}^m \log(-f_i(x)) \\ \text{subject to} & Fx = g. \end{array}$$

It is more easily computed than the MVE or Chebyshev center (see the algorithms in later lectures). It is not just a property of the feasible set: two sets of inequalities can describe the same set but have different analytic centers.

For linear inequalities $a_i^T x \leq b_i$, $i = 1, \ldots, m$, the analytic center x_{ac} minimizes $\phi(x) = -\sum_{i=1}^m \log(b_i - a_i^T x)$, and the Hessian at the center gives inner and outer ellipsoids (Figure 26):

$$\mathcal{E}_{\text{inner}} \subseteq \{x \mid a_i^T x \leq b_i,\ i = 1, \ldots, m\} \subseteq \mathcal{E}_{\text{outer}},$$

where

$$\begin{aligned} \mathcal{E}_{\text{inner}} &= \{x \mid (x - x_{\text{ac}})^T \nabla^2 \phi(x_{\text{ac}})(x - x_{\text{ac}}) \leq 1\}, \\ \mathcal{E}_{\text{outer}} &= \{x \mid (x - x_{\text{ac}})^T \nabla^2 \phi(x_{\text{ac}})(x - x_{\text{ac}}) \leq m(m-1)\}. \end{aligned}$$

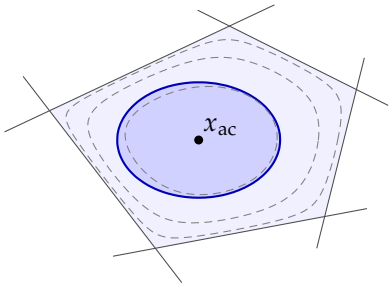


Figure 26: The analytic center x_{ac} of five linear inequalities, level curves of ϕ (dashed), and the inner ellipsoid $\mathcal{E}_{\text{inner}}$ (shaded). The outer ellipsoid, scaled by $\sqrt{m(m-1)} = \sqrt{20}$, is too large to show.

12.2 Linear discrimination

To separate two sets of points $\{x_1, \ldots, x_N\}$ and $\{y_1, \ldots, y_M\}$ by a hyperplane, we need

$$a^T x_i + b > 0, \quad i = 1, \ldots, N, \qquad a^T y_i + b < 0, \quad i = 1, \ldots, M.$$

These are homogeneous in a, b, hence equivalent to the set of linear inequalities in a, b

$$a^T x_i + b \geq 1, \quad i = 1, \ldots, N, \qquad a^T y_i + b \leq -1, \quad i = 1, \ldots, M.$$

Robust linear discrimination. The Euclidean distance between the hyperplanes $\mathcal{H}_1 = \{z \mid a^T z + b = 1\}$ and $\mathcal{H}_2 = \{z \mid a^T z + b = -1\}$ is $\mathrm{dist}(\mathcal{H}_1, \mathcal{H}_2) = 2/\|a\|_2$. To separate two sets of points by maximum margin (Figure 27), solve

$$\begin{array}{ll} \text{minimize} & (1/2)\|a\|_2 \\ \text{subject to} & a^T x_i + b \geq 1, \quad i = 1, \ldots, N \\ & a^T y_i + b \leq -1, \quad i = 1, \ldots, M, \end{array} \tag{12.1}$$

which (after squaring the objective) is a QP in a, b.

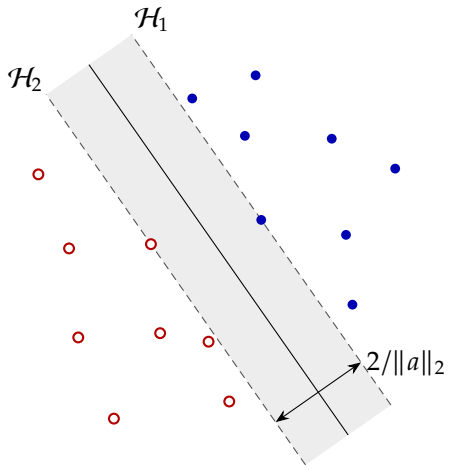


Figure 27: Maximum margin separation of the points x_i (filled) and y_i (open), schematic. The slab between $\mathcal{H}_1$ and $\mathcal{H}_2$ contains no points, and its width $2/\|a\|_2$ is as large as possible.

The Lagrange dual of the maximum margin separation problem (12.1) is

$$\begin{array}{ll} \text{maximize} & \mathbf{1}^T\lambda + \mathbf{1}^T\mu \\ \text{subject to} & 2\left\|\sum_{i=1}^N \lambda_i x_i - \sum_{i=1}^M \mu_i y_i\right\|_2 \leq 1 \\ & \mathbf{1}^T\lambda = \mathbf{1}^T\mu, \quad \lambda \geq 0, \quad \mu \geq 0. \end{array}$$

By duality, its optimal value is the inverse of the maximum margin of separation. To interpret it, change variables to $\theta_i = \lambda_i/\mathbf{1}^T\lambda$, $\gamma_i = \mu_i/\mathbf{1}^T\mu$, and $t = 1/(\mathbf{1}^T\lambda + \mathbf{1}^T\mu)$, and invert the objective to

minimize $1/(\mathbf{1}^T\lambda + \mathbf{1}^T\mu) = t$:

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & \left\| \sum_{i=1}^N \theta_i x_i - \sum_{i=1}^M \gamma_i y_i \right\|_2 \leq t \\ & \theta \succeq 0, \quad \mathbf{1}^T\theta = 1, \quad \gamma \succeq 0, \quad \mathbf{1}^T\gamma = 1. \end{array}$$

The optimal value is the distance between the convex hulls of the two sets.

Approximate linear separation of non-separable sets.

$$\begin{array}{ll} \text{minimize} & \mathbf{1}^T u + \mathbf{1}^T v \\ \text{subject to} & a^T x_i + b \geq 1 - u_i, \quad i = 1, \ldots, N \\ & a^T y_i + b \leq -1 + v_i, \quad i = 1, \ldots, M \\ & u \succeq 0, \quad v \succeq 0 \end{array}$$

is an LP in a, b, u, v. At the optimum, $u_i = \max\{0, 1 - a^T x_i - b\}$ and $v_i = \max\{0, 1 + a^T y_i + b\}$, and the problem can be interpreted as a heuristic for minimizing the number of misclassified points.

Definition 12.2 [Support vector classifier]. The support vector classifier solves

$$\begin{array}{ll} \text{minimize} & \|a\|_2 + \gamma(\mathbf{1}^T u + \mathbf{1}^T v) \\ \text{subject to} & a^T x_i + b \geq 1 - u_i, \quad i = 1, \ldots, N \\ & a^T y_i + b \leq -1 + v_i, \quad i = 1, \ldots, M \\ & u \succeq 0, \quad v \succeq 0. \end{array}$$

It produces a point on the trade-off curve between the inverse of the margin $2/\|a\|_2$ and the classification error, measured by the total slack $\mathbf{1}^T u + \mathbf{1}^T v$.

Nonlinear discrimination. To separate two sets of points by a nonlinear function, $f(x_i) > 0$, $i = 1, \ldots, N$, and $f(y_i) < 0$, $i = 1, \ldots, M$, choose a linearly parametrized family of functions $f(z) = \theta^T F(z)$, where $F = (F_1, \ldots, F_k) : \mathbb{R}^n \to \mathbb{R}^k$ are basis functions, and solve the linear inequalities in θ

$$\theta^T F(x_i) \geq 1, \quad i = 1, \ldots, N, \qquad \theta^T F(y_i) \leq -1, \quad i = 1, \ldots, M.$$

For quadratic discrimination, $f(z) = z^T P z + q^T z + r$, the conditions $x_i^T P x_i + q^T x_i + r \geq 1$ and $y_i^T P y_i + q^T y_i + r \leq -1$ are linear in (P, q, r), and additional constraints can be added (for example $P \preceq -I$ to separate by an ellipsoid). For polynomial discrimination, $F(z)$ consists of all monomials up to a given degree.

12.3 Placement and facility location

There are N points with coordinates $x_i \in \mathbb{R}^2$ (or $\mathbb{R}^3$); some positions x_i are given and the others are variables. For each pair of points there is a cost function $f_{ij}(x_i, x_j)$. The PLACEMENT PROBLEM is

$$\text{minimize} \quad \sum_{i \neq j} f_{ij}(x_i, x_j)$$

over the positions of the free points. Interpretations: the points represent plants or warehouses, and f_{ij} is the transportation cost between facilities i and j; or the points represent cells on an IC, and f_{ij} represents wirelength.

Example: Minimize $\sum_{(i,j)\in\mathcal{A}} h(\|x_i - x_j\|_2)$ with 6 free points and 27 links, for $h(z) = z$, $h(z) = z^2$, and $h(z) = z^4$. With $h(z) = z$ many links become very short while a few remain long (up to about 2); as the power grows, long links are penalized more heavily, and with $h(z) = z^4$ the connection lengths are more uniform and the longest is shorter (about 1.3).

12.4 Matrix structure and algorithm complexity

The cost (execution time) of solving $Ax = b$ with $A \in \mathbb{R}^{n\times n}$ grows as n^3 for general methods, and is less if A is structured (banded, sparse, Toeplitz, ...).

Definition 12.3 [FLOP COUNT]. A FLOP (floating-point operation) is one addition, subtraction, multiplication, or division of two floating-point numbers. To estimate the complexity of an algorithm, express its number of flops as a (polynomial) function of the problem dimensions and simplify by keeping only the leading terms.

Flop counts are not an accurate predictor of computation time on modern computers, but they are useful as a rough estimate of complexity.

- Vector-vector operations ($x, y \in \mathbb{R}^n$): inner product $x^T y$ costs $2n - 1$ flops (or $2n$ if n is large); the sum $x + y$ and scalar multiplication αx cost n flops.
- Matrix-vector product $y = Ax$ with $A \in \mathbb{R}^{m\times n}$: $m(2n - 1)$ flops (or $2mn$ if n is large); $2N$ if A is sparse with N nonzero elements; $2p(n + m)$ if A is given as $A = UV^T$ with $U \in \mathbb{R}^{m\times p}$, $V \in \mathbb{R}^{n\times p}$.
- Matrix-matrix product $C = AB$ with $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$: $mp(2n - 1)$ flops (or $2mnp$ if n is large); less if A and/or B are sparse; $(1/2)m(m + 1)(2n - 1) \approx m^2 n$ if $m = p$ and C is symmetric.

12.5 Linear equations that are easy to solve

- *Diagonal* matrices ($a_{ij} = 0$ if $i \neq j$): n flops, $x = A^{-1}b = (b_1/a_{11}, \ldots, b_n/a_{nn})$.

- *Lower triangular* ($a_{ij} = 0$ if $j > i$): n^2 flops by FORWARD SUBSTITUTION,

$$x_1 := b_1/a_{11}, \quad x_2 := (b_2 - a_{21}x_1)/a_{22}, \quad \ldots, \quad x_n := (b_n - a_{n1}x_1 - \cdots - a_{n,n-1}x_{n-1})/a_{nn}.$$

- *Upper triangular* ($a_{ij} = 0$ if $j < i$): n^2 flops via BACKWARD SUBSTITUTION.
- *Orthogonal* matrices, $A^{-1} = A^T$: $2n^2$ flops to compute $x = A^T b$ for general A; less with structure, for example if $A = I - 2uu^T$ with $\|u\|_2 = 1$, then $x = A^T b = b - 2(u^T b)u$ costs $4n$ flops.
- *Permutation* matrices, $a_{ij} = 1$ if $j = \pi_i$ and 0 otherwise, where $\pi = (\pi_1, \ldots, \pi_n)$ is a permutation of $(1, 2, \ldots, n)$. Then $Ax = (x_{\pi_1}, \ldots, x_{\pi_n})$ and $A^{-1} = A^T$, so solving $Ax = b$ costs 0 flops. For example,

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \qquad A^{-1} = A^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

12.6 The factor-solve method

To solve $Ax = b$, factor A as a product of simple matrices (usually 2 or 3), $A = A_1 A_2 \cdots A_k$ with A_i diagonal, upper or lower triangular, and so on. Then compute $x = A^{-1}b = A_k^{-1} \cdots A_2^{-1} A_1^{-1} b$ by solving k "easy" equations

$$A_1 x_1 = b, \qquad A_2 x_2 = x_1, \qquad \ldots, \qquad A_k x = x_{k-1}.$$

The cost of the factorization step usually dominates the cost of the solve step. For equations with multiple righthand sides, $Ax_1 = b_1$, $Ax_2 = b_2$, $\ldots$, $Ax_m = b_m$, the cost is one factorization plus m solves.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 12 (2023); original slides, *Geometric problems* and *Numerical linear algebra background*.

Lecture 13: Numerical Linear Algebra (Continued); Unconstrained Minimization

13.1 LU, Cholesky, and LDL^T factorization

Every nonsingular matrix A can be factored as $A = PLU$ with P a permutation matrix, L lower triangular, and U upper triangular, at a cost of $(2/3)n^3$ flops.

Algorithm 13.1: Solving linear equations by LU factorization

Input: A set of linear equations $Ax = b$, with A nonsingular.

1. **LU factorization.** Factor A as $A = PLU$ // $(2/3)n^3$ *flops*
2. **Permutation.** Solve $Pz_1 = b$ // 0 *flops*
3. **Forward substitution.** Solve $Lz_2 = z_1$ // n^2 *flops*
4. **Backward substitution.** Solve $Ux = z_2$ // n^2 *flops*

The cost is $(2/3)n^3 + 2n^2 \approx (2/3)n^3$ for large n. In a sparse LU factorization $A = P_1LUP_2$, the extra permutation P_2 offers the possibility of sparser L, U, hence cheaper factor and solve steps. P_1 and P_2 are chosen (heuristically) to yield sparse L, U, and the choice depends on the sparsity pattern and the values of A. The cost is usually much less than $(2/3)n^3$; its exact value depends in a complicated way on n, the number of zeros in A, and the sparsity pattern.

Every positive definite A can be factored as $A = LL^T$ with L lower triangular (the Cholesky factorization), at a cost of $(1/3)n^3$ flops.

Algorithm 13.2: Solving linear equations by Cholesky factorization

Input: A set of linear equations $Ax = b$, with $A \in \mathbf{S}^n_{++}$.

1. **Cholesky factorization.** Factor A as $A = LL^T$ // $(1/3)n^3$ *flops*
2. **Forward substitution.** Solve $Lz_1 = b$ // n^2 *flops*
3. **Backward substitution.** Solve $L^Tx = z_1$ // n^2 *flops*

The cost is $(1/3)n^3 + 2n^2 \approx (1/3)n^3$ for large n. In a sparse Cholesky factorization $A = PLL^TP^T$, the permutation P is chosen (heuristically) to yield a sparse L; unlike sparse LU, the choice of P depends only on the sparsity pattern of A. The cost is usually much less than $(1/3)n^3$.

Every nonsingular symmetric matrix A can be factored as $A = PLDL^TP^T$ with P a permutation matrix, L lower triangular, and D block diagonal with 1×1 or 2×2 diagonal blocks (the LDL^T factorization), at a cost of $(1/3)n^3$. Solving symmetric sets of linear equations this way costs $(1/3)n^3 + 2n^2 \approx (1/3)n^3$ for large n; for sparse A, P can be chosen to yield a sparse L, with cost $\ll (1/3)n^3$.

13.2 Block elimination

Consider equations with structured sub-blocks,

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, \tag{13.1}$$

with variables $x_1 \in \mathbb{R}^{n_1}$, $x_2 \in \mathbb{R}^{n_2}$ and blocks $A_{ij} \in \mathbb{R}^{n_i \times n_j}$. If A_{11} is nonsingular we can eliminate $x_1 = A_{11}^{-1}(b_1 - A_{12}x_2)$; to compute x_2, solve

$$(A_{22} - A_{21}A_{11}^{-1}A_{12})x_2 = b_2 - A_{21}A_{11}^{-1}b_1.$$

Algorithm 13.3: Solving linear equations by block elimination

Input: A nonsingular set of linear equations (13.1), with A_{11} nonsingular.

1. Form $A_{11}^{-1}A_{12}$ and $A_{11}^{-1}b_1$
2. Form $S = A_{22} - A_{21}A_{11}^{-1}A_{12}$ and $\tilde{b} = b_2 - A_{21}A_{11}^{-1}b_1$
3. Determine x_2 by solving $Sx_2 = \tilde{b}$
4. Determine x_1 by solving $A_{11}x_1 = b_1 - A_{12}x_2$

The dominant terms in the flop count are $f + n_2 s$ for step 1 (f is the cost of factoring A_{11} and s the cost of a solve step), $2n_2^2 n_1$ for step 2 (dominated by the product of A_{21} and $A_{11}^{-1}A_{12}$), and $(2/3)n_2^3$ for step 3, for a total of $f + n_2 s + 2n_2^2 n_1 + (2/3)n_2^3$. For general A_{11} ($f = (2/3)n_1^3$, $s = 2n_1^2$) there is no gain over the standard method:

$$\#\text{flops} = (2/3)n_1^3 + 2n_1^2 n_2 + 2n_2^2 n_1 + (2/3)n_2^3 = (2/3)(n_1 + n_2)^3.$$

Block elimination is useful for structured A_{11} ($f \ll n_1^3$); for example, for diagonal A_{11} ($f = 0$, $s = n_1$) the cost is $\approx 2n_2^2 n_1 + (2/3)n_2^3$.

Structured matrix plus low rank term. To solve $(A + BC)x = b$ with $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times p}$, $C \in \mathbb{R}^{p \times n}$, where A has structure ($Ax = b$ is easy to solve), first write

$$\begin{bmatrix} A & B \\ C & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix}.$$

Block elimination of x gives $(I + CA^{-1}B)y = CA^{-1}b$, and then $Ax = b - By$. This proves the MATRIX INVERSION LEMMA: if A and $A + BC$ are nonsingular,

$$(A + BC)^{-1} = A^{-1} - A^{-1}B(I + CA^{-1}B)^{-1}CA^{-1}.$$

Example: For A diagonal and B, C dense: forming $D = A + BC$ and solving $Dx = b$ costs $(2/3)n^3 + 2pn^2$. Via the matrix inversion lemma, solve $(I + CA^{-1}B)y = CA^{-1}b$, then compute $x = A^{-1}b - A^{-1}By$; the total cost is dominated by the first step, $2p^2 n + (2/3)p^3$, which is linear in n.

Underdetermined linear equations. If $A \in \mathbb{R}^{p\times n}$ with $p < n$ and $\operatorname{rank} A = p$, then

$$\{x \mid Ax = b\} = \{Fz + \hat{x} \mid z \in \mathbb{R}^{n-p}\},$$

where $\hat{x}$ is any particular solution and the columns of $F \in \mathbb{R}^{n\times(n-p)}$ span the nullspace of A. There are several numerical methods for computing F (QR factorization, rectangular LU factorization, ...).

13.3 Unconstrained minimization

We consider

$$\text{minimize} \quad f(x)$$

with f convex and twice continuously differentiable (hence $\operatorname{dom} f$ open), and assume the optimal value $p^\star = \inf_x f(x)$ is attained and finite. Unconstrained minimization methods produce a sequence of points $x^{(k)} \in \operatorname{dom} f$, $k = 0, 1, \ldots$, with $f(x^{(k)}) \to p^\star$; they can be interpreted as iterative methods for solving the optimality condition $\nabla f(x^\star) = 0$.

Initial point and sublevel set. The algorithms require a starting point $x^{(0)} \in \operatorname{dom} f$ whose sublevel set $S = \{x \mid f(x) \le f(x^{(0)})\}$ is closed. The second condition is hard to verify, except when all sublevel sets are closed: this is equivalent to $\operatorname{epi} f$ being closed, and holds if $\operatorname{dom} f = \mathbb{R}^n$ or if $f(x) \to \infty$ as $x \to \operatorname{bd}\operatorname{dom} f$. Examples of differentiable functions with closed sublevel sets are $f(x) = \log(\sum_{i=1}^m \exp(a_i^T x + b_i))$ and $f(x) = -\sum_{i=1}^m \log(b_i - a_i^T x)$.

Definition 13.1 [Strong convexity]. f is strongly convex on S if there exists an $m > 0$ such that $\nabla^2 f(x) \ge mI$ for all $x \in S$.

Proposition 13.2 [Implications of strong convexity]. *If f is strongly convex on S, then for $x, y \in S$*

$$f(y) \ge f(x) + \nabla f(x)^T (y - x) + \frac{m}{2}\|x - y\|_2^2,$$

hence S is bounded; moreover $p^\star > -\infty$ and, for $x \in S$,

$$f(x) - p^\star \le \frac{1}{2m}\|\nabla f(x)\|_2^2.$$

Proof. The first inequality follows from Taylor's theorem with the bound $\nabla^2 f \ge mI$ on the segment between x and y. Minimizing its right-hand side over y (at $y = x - (1/m)\nabla f(x)$) gives $f(y) \ge f(x) - \|\nabla f(x)\|_2^2/(2m)$ for all y, hence the second inequality. □

The second inequality is useful as a stopping criterion (if m is known).

13.4 Descent methods

A DESCENT METHOD generates

$$x^{(k+1)} = x^{(k)} + t^{(k)}\Delta x^{(k)} \qquad \text{with } f(x^{(k+1)}) < f(x^{(k)}),$$

also written $x^+ = x + t\Delta x$ or $x := x + t\Delta x$. Here Δx is the STEP or SEARCH DIRECTION and t the STEP SIZE or step length. By convexity, $f(x^+) < f(x)$ implies $\nabla f(x)^T \Delta x < 0$: Δx is a DESCENT DIRECTION.

Algorithm 13.4: General descent method

Input: A starting point $x \in \operatorname{dom} f$.

repeat

1. Determine a descent direction Δx
2. **Line search.** Choose a step size $t > 0$
3. **Update.** $x := x + t\Delta x$

until stopping criterion is satisfied

Line search types. EXACT LINE SEARCH takes $t = \arg\min_{t>0} f(x + t\Delta x)$. BACKTRACKING LINE SEARCH, with parameters $\alpha \in (0, 1/2)$ and $\beta \in (0, 1)$, starts at $t = 1$ and repeats $t := \beta t$ until

$$f(x + t\Delta x) < f(x) + \alpha t \nabla f(x)^T \Delta x.$$

Graphically, it backtracks until $t \le t_0$, where t_0 is the point at which the graph of $f(x + t\Delta x)$ crosses the line $f(x) + \alpha t \nabla f(x)^T \Delta x$ (Figure 28).

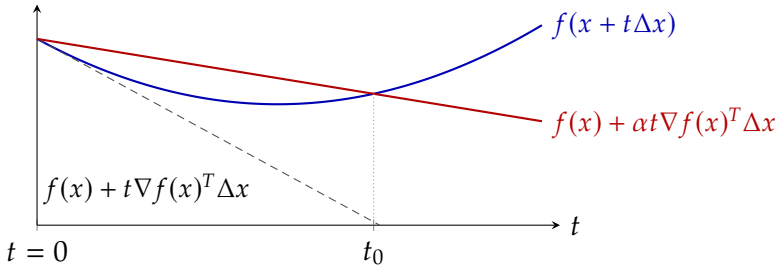


Figure 28: Backtracking line search. The condition $f(x + t\Delta x) < f(x) + \alpha t \nabla f(x)^T \Delta x$ holds for $0 < t < t_0$; the dashed line is the tangent $f(x) + t \nabla f(x)^T \Delta x$.

13.5 Gradient descent

Algorithm 13.5: Gradient descent method

Input: A starting point $x \in \operatorname{dom} f$.

repeat

- $\Delta x := -\nabla f(x)$
- **Line search.** Choose step size t via exact or backtracking line search
- **Update.** $x := x + t\Delta x$

until stopping criterion is satisfied

Theorem 13.3 [Convergence of gradient descent]. *For strongly convex f,*

$$f(x^{(k)}) - p^\star \le c^k \big(f(x^{(0)}) - p^\star\big),$$

where $c \in (0,1)$ depends on m, $x^{(0)}$, and the line search type.

The stopping criterion is usually of the form $\|\nabla f(x)\|_2 \le \epsilon$. Gradient descent is very simple, but often very slow, and rarely used in practice.

Example: *Quadratic problem in $\mathbb{R}^2$:* $f(x) = (1/2)(x_1^2 + \gamma x_2^2)$ with $\gamma > 0$. With exact line search, starting at $x^{(0)} = (\gamma, 1)$,

$$x_1^{(k)} = \gamma \left(\frac{\gamma - 1}{\gamma + 1}\right)^k, \qquad x_2^{(k)} = \left(-\frac{\gamma - 1}{\gamma + 1}\right)^k,$$

which is very slow if $\gamma \gg 1$ or $\gamma \ll 1$ (Figure 29).

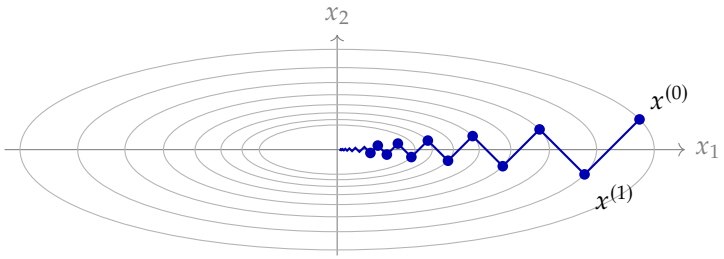


Figure 29: Gradient descent with exact line search for $\gamma = 10$, from $x^{(0)} = (10, 1)$, with level sets of f: the iterates zigzag across the narrow valley, shrinking by the factor $9/11$ per step.

On the nonquadratic example $f(x_1, x_2) = e^{x_1 + 3x_2 - 0.1} + e^{x_1 - 3x_2 - 0.1} + e^{-x_1 - 0.1}$, backtracking line search zigzags toward the minimizer in many short steps, while exact line search gets close in a few. On a problem in $\mathbb{R}^{100}$, $f(x) = c^T x - \sum_{i=1}^{500} \log(b_i - a_i^T x)$, the error $f(x^{(k)}) - p^\star$ decreases geometrically with both line searches ("linear" convergence, a straight line on a semilog plot), taking well over a hundred iterations to go from about 10^2 to 10^{-4}.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 13 (2023); original slides, *Numerical linear algebra background* and *Unconstrained minimization*.

Lecture 14: Unconstrained Minimization (Continued)

14.1 Steepest descent

Definition 14.1 [Steepest descent direction]. The normalized steepest descent direction at x for a norm $\|\cdot\|$ is

$$\Delta x_{\text{nsd}} = \arg\min\{\nabla f(x)^T v \mid \|v\| = 1\}.$$

The (unnormalized) steepest descent direction

$$\Delta x_{\text{sd}} = \|\nabla f(x)\|_* \Delta x_{\text{nsd}}$$

satisfies $\nabla f(x)^T \Delta x_{\text{sd}} = -\|\nabla f(x)\|_*^2$.

Interpretation: for small v, $f(x+v) \approx f(x) + \nabla f(x)^T v$, so Δx_{nsd} is the unit-norm step with the most negative directional derivative. The steepest descent method is the general descent method with $\Delta x = \Delta x_{\text{sd}}$; its convergence properties are similar to those of gradient descent.

Examples (Figure 30):

- Euclidean norm: $\Delta x_{\text{sd}} = -\nabla f(x)$;
- quadratic norm $\|x\|_P = (x^T P x)^{1/2}$ ($P \in \mathbf{S}^n_{++}$): $\Delta x_{\text{sd}} = -P^{-1}\nabla f(x)$;
- ℓ_1-norm: $\Delta x_{\text{sd}} = -(\partial f(x)/\partial x_i) e_i$, where $|\partial f(x)/\partial x_i| = \|\nabla f(x)\|_\infty$.

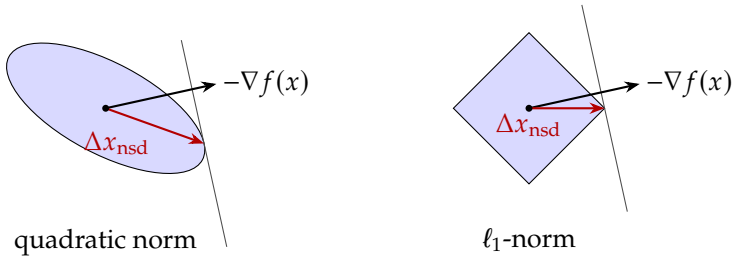


Figure 30: Unit balls and normalized steepest descent directions for a quadratic norm and the ℓ_1-norm. Δx_{nsd} is the point of the unit ball extreme in the direction $-\nabla f(x)$, where a hyperplane normal to $\nabla f(x)$ supports the ball.

Choice of norm. Steepest descent with the quadratic norm $\|\cdot\|_P$ is gradient descent after the change of variables $\bar{x} = P^{1/2} x$. The choice of P has a strong effect on the speed of convergence: with a P whose unit ellipsoid $\{x \mid \|x - x^{(k)}\|_P = 1\}$ is aligned with the sublevel sets of f, the iterates head almost directly to the minimizer, while a badly chosen P makes them zigzag slowly.

14.2 Newton step

Definition 14.2 [Newton step]. The Newton step is $\Delta x_{\text{nt}} = -\nabla^2 f(x)^{-1} \nabla f(x)$.

Interpretations:

- $x + \Delta x_{\text{nt}}$ minimizes the second-order approximation

$$\widehat{f}(x+v) = f(x) + \nabla f(x)^T v + \frac{1}{2} v^T \nabla^2 f(x) v;$$

- $x + \Delta x_{\text{nt}}$ solves the linearized optimality condition

$$\nabla f(x+v) \approx \nabla \widehat{f}(x+v) = \nabla f(x) + \nabla^2 f(x) v = 0;$$

- Δx_{nt} is the steepest descent direction at x in the local Hessian norm $\|u\|_{\nabla^2 f(x)} = (u^T \nabla^2 f(x) u)^{1/2}$.

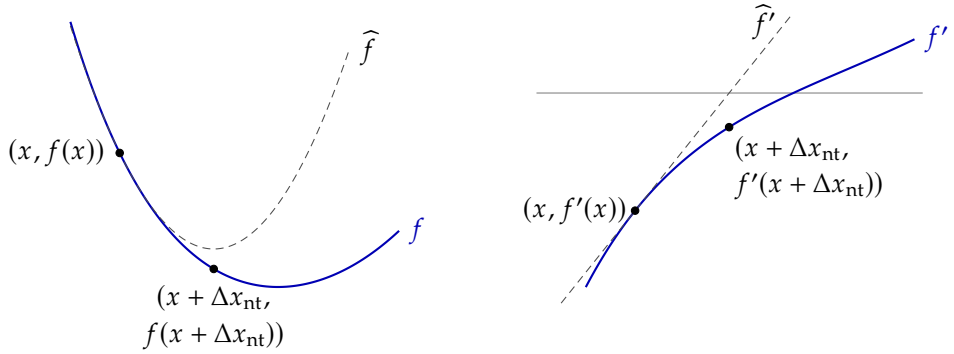


Figure 31: The Newton step for a function of one variable. Left: $x + \Delta x_{\text{nt}}$ minimizes the quadratic approximation $\widehat{f}$ (dashed). Right: it is the zero of the linearization $\widehat{f}'$ (dashed) of f'.

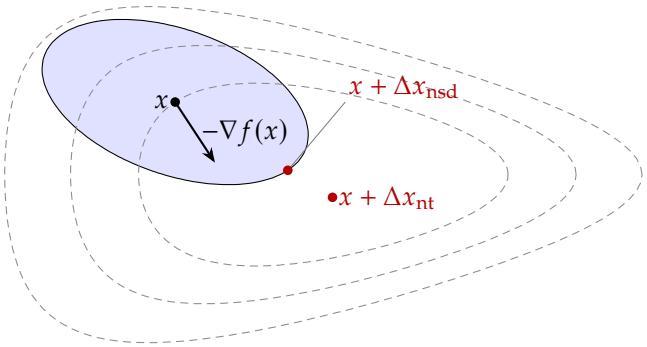


Figure 32: The Newton step as steepest descent in the Hessian norm, for the function $f(x) = e^{x_1+3x_2-0.1} + e^{x_1-3x_2-0.1} + e^{-x_1-0.1}$. Dashed curves are level sets of f; the ellipse is $\{x + v \mid v^T \nabla^2 f(x) v = 1\}$, and Δx_{nsd} (normalized in the Hessian norm) points in the direction of Δx_{nt}.

14.3 Newton decrement

Definition 14.3 [Newton decrement]. The Newton decrement is

$$\lambda(x) = \left(\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x)\right)^{1/2},$$

a measure of the proximity of x to $x^\star$.

Properties:

- it gives an estimate of $f(x) - p^\star$ using the quadratic approximation $\widehat{f}$: $f(x) - \inf_y \widehat{f}(y) = \frac{1}{2}\lambda(x)^2$;
- it is the norm of the Newton step in the quadratic Hessian norm:

$$\lambda(x) = \left(\Delta x_{\text{nt}}^T \nabla^2 f(x) \Delta x_{\text{nt}}\right)^{1/2};$$

- it gives the directional derivative in the Newton direction, $\nabla f(x)^T \Delta x_{\text{nt}} = -\lambda(x)^2$;
- it is affine invariant (unlike $\|\nabla f(x)\|_2$).

14.4 Newton's method

Algorithm 14.1: Newton's method

Input: A starting point $x \in \text{dom}\, f$, tolerance $\epsilon > 0$.

Compute the Newton step and decrement. $\Delta x_{\text{nt}} := -\nabla^2 f(x)^{-1} \nabla f(x)$; $\lambda^2 := \nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x)$

while $\lambda^2/2 > \epsilon$ **do** *// stopping criterion*

  Line search. Choose step size t by backtracking line search

  Update. $x := x + t\Delta x_{\text{nt}}$

  Compute the Newton step Δx_{nt} and decrement λ at the new x

Newton's method is affine invariant, that is, independent of linear changes of coordinates: the Newton iterates for $\tilde{f}(y) = f(Ty)$ with starting point $y^{(0)} = T^{-1}x^{(0)}$ are $y^{(k)} = T^{-1}x^{(k)}$.

14.5 Classical convergence analysis

Assume f is strongly convex on S with constant m, and $\nabla^2 f$ is Lipschitz continuous on S with constant $L > 0$,

$$\|\nabla^2 f(x) - \nabla^2 f(y)\|_2 \leq L\|x - y\|_2$$

(L measures how well f can be approximated by a quadratic function).

Theorem 14.4 [Two phases of Newton's method]. *There exist constants $\eta \in (0, m^2/L)$ and $\gamma > 0$ such that*

- *if* $\|\nabla f(x^{(k)})\|_2 \geq \eta$, *then* $f(x^{(k+1)}) - f(x^{(k)}) \leq -\gamma$;
- *if* $\|\nabla f(x^{(k)})\|_2 < \eta$, *then*

$$\frac{L}{2m^2}\|\nabla f(x^{(k+1)})\|_2 \leq \left(\frac{L}{2m^2}\|\nabla f(x^{(k)})\|_2\right)^2.$$

In the DAMPED NEWTON PHASE ($\|\nabla f(x)\|_2 \geq \eta$), most iterations require backtracking steps and the function value decreases by at least γ; if $p^\star > -\infty$, this phase ends after at most $(f(x^{(0)}) - p^\star)/\gamma$ iterations. In the QUADRATICALLY CONVERGENT PHASE ($\|\nabla f(x)\|_2 < \eta$), all iterations use step size $t = 1$, and $\|\nabla f(x)\|_2$ converges to zero quadratically: if $\|\nabla f(x^{(k)})\|_2 < \eta$, then for $l \geq k$

$$\frac{L}{2m^2}\|\nabla f(x^{(l)})\|_2 \leq \left(\frac{L}{2m^2}\|\nabla f(x^{(k)})\|_2\right)^{2^{l-k}} \leq \left(\frac{1}{2}\right)^{2^{l-k}}.$$

Combining this with $f(x) - p^\star \leq \|\nabla f(x)\|_2^2/(2m)$ gives the conclusion.

Corollary 14.5 [ITERATION BOUND]. *The number of iterations until* $f(x) - p^\star \leq \epsilon$ *is bounded above by*

$$\frac{f(x^{(0)}) - p^\star}{\gamma} + \log_2 \log_2(\epsilon_0/\epsilon),$$

where γ *and* ϵ_0 *are constants that depend on* m, L, *and* $x^{(0)}$.

The second term is small (of the order of 6) and almost constant for practical purposes. In practice the constants m, L (hence γ, ϵ_0) are usually unknown, so the bound mainly provides qualitative insight into the convergence properties: it explains the two phases of the algorithm.

Examples. On the nonquadratic example in $\mathbb{R}^2$, Newton's method with backtracking parameters $\alpha = 0.1$, $\beta = 0.7$ converges in only 5 steps, with quadratic local convergence. On the problem in $\mathbb{R}^{100}$ (with $\alpha = 0.01$, $\beta = 0.5$) it reaches very high accuracy in fewer than ten iterations; backtracking line search is almost as fast as exact line search (and much simpler), and the two phases are clearly visible, with the step size settling at $t = 1$ once the quadratic phase starts. On a problem in $\mathbb{R}^{10000}$ with sparse a_i, $f(x) = -\sum_{i=1}^{10000} \log(1 - x_i^2) - \sum_{i=1}^{100000} \log(b_i - a_i^T x)$, the performance is similar to the small examples: about 20 iterations.

14.6 Self-concordance

The classical analysis has shortcomings: it depends on unknown constants (m, L, ...), and the bound is not affinely invariant, although Newton's method is. The convergence analysis via SELF-CONCORDANCE (Nesterov and Nemirovski) does not depend on any unknown constants and gives an affine-invariant bound. It applies to a special class of convex functions, the self-concordant functions, and was developed to analyze polynomial-time interior-point methods for convex optimization.

Definition 14.6 [SELF-CONCORDANT FUNCTION]. A convex $f : \mathbb{R} \to \mathbb{R}$ is SELF-CONCORDANT if $|f'''(x)| \leq$

$2f''(x)^{3/2}$ for all $x \in \operatorname{dom} f$. A function $f : \mathbb{R}^n \to \mathbb{R}$ is self-concordant if $g(t) = f(x + tv)$ is self-concordant for all $x \in \operatorname{dom} f$, $v \in \mathbb{R}^n$.

Examples on $\mathbb{R}$: linear and quadratic functions; the negative logarithm $f(x) = -\log x$; and the negative entropy plus negative logarithm $f(x) = x \log x - \log x$. *Affine invariance*: if $f : \mathbb{R} \to \mathbb{R}$ is self-concordant, then so is $\tilde{f}(y) = f(ay + b)$, since $\tilde{f}'''(y) = a^3 f'''(ay + b)$ and $\tilde{f}''(y) = a^2 f''(ay + b)$.

Self-concordant calculus. Self-concordance is preserved under positive scaling by $\alpha \geq 1$ and under sums, and under composition with an affine function. If g is convex with $\operatorname{dom} g = \mathbb{R}_{++}$ and $|g'''(x)| \leq 3g''(x)/x$, then $f(x) = \log(-g(x)) - \log x$ is self-concordant. These properties show that the following are self-concordant:

- $f(x) = -\sum_{i=1}^m \log(b_i - a_i^T x)$ on $\{x \mid a_i^T x < b_i,\ i = 1, \ldots, m\}$;
- $f(X) = -\log \det X$ on $\mathbf{S}_{++}^n$;
- $f(x) = -\log(y^2 - x^T x)$ on $\{(x, y) \mid \|x\|_2 < y\}$.

Theorem 14.7 [Convergence for self-concordant functions]. *There exist constants $\eta \in (0, 1/4]$ and $\gamma > 0$, depending only on the backtracking parameters α, β, such that*

- *if $\lambda(x^{(k)}) > \eta$, then $f(x^{(k+1)}) - f(x^{(k)}) \leq -\gamma$;*
- *if $\lambda(x^{(k)}) \leq \eta$, then $2\lambda(x^{(k+1)}) \leq \left(2\lambda(x^{(k)})\right)^2$.*

Hence the number of Newton iterations is bounded by

$$\frac{f(x^{(0)}) - p^\star}{\gamma} + \log_2 \log_2(1/\epsilon).$$

For $\alpha = 0.1$, $\beta = 0.8$, $\epsilon = 10^{-10}$, the bound evaluates to $375(f(x^{(0)}) - p^\star) + 6$. In a numerical example with 150 randomly generated instances of minimizing $f(x) = -\sum_{i=1}^m \log(b_i - a_i^T x)$ (sizes $m = 100$, $n = 50$; $m = 1000$, $n = 500$; $m = 1000$, $n = 50$), the number of iterations is much smaller than $375(f(x^{(0)}) - p^\star) + 6$: never more than about 25, for $f(x^{(0)}) - p^\star$ up to about 35. Empirically, a bound of the form $c(f(x^{(0)}) - p^\star) + 6$ with a much smaller c is valid.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 14 (2023); original slides, *Unconstrained minimization*.

Lecture 15: Unconstrained Minimization (Continued); Equality Constrained Minimization

15.1 Implementation of Newton's method

The main effort in each iteration is evaluating derivatives and solving the Newton system

$$H\Delta x = -g, \qquad H = \nabla^2 f(x), \quad g = \nabla f(x).$$

Via the Cholesky factorization $H = LL^T$,

$$\Delta x_{\text{nt}} = -L^{-T}L^{-1}g, \qquad \lambda(x) = \|L^{-1}g\|_2,$$

at a cost of $(1/3)n^3$ flops for an unstructured system, and $\ll (1/3)n^3$ if H is sparse or banded.

Example: *Dense Newton system with structure.* Let

$$f(x) = \sum_{i=1}^{n} \psi_i(x_i) + \psi_0(Ax + b), \qquad H = D + A^T H_0 A,$$

with $A \in \mathbb{R}^{p\times n}$ dense and $p \ll n$, D diagonal with diagonal elements $\psi_i''(x_i)$, and $H_0 = \nabla^2\psi_0(Ax+b)$.

- Method 1: form H and solve via a dense Cholesky factorization, at cost $(1/3)n^3$.
- Method 2 (block elimination, as for a structured matrix plus low rank term): factor $H_0 = L_0 L_0^T$ and write the Newton system as

  $$D\Delta x + A^T L_0 w = -g, \qquad L_0^T A\Delta x - w = 0.$$

  Eliminating Δx from the first equation, compute w and Δx from

  $$(I + L_0^T AD^{-1}A^T L_0)w = -L_0^T AD^{-1}g, \qquad D\Delta x = -g - A^T L_0 w,$$

  at cost $2p^2 n$ (dominated by the computation of $L_0^T AD^{-1}A^T L_0$).

15.2 Equality constrained minimization

Consider

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & Ax = b, \end{array}$$

with f convex and twice continuously differentiable, $A \in \mathbb{R}^{p\times n}$ with $\operatorname{rank} A = p$, and $p^\star$ finite and attained. The optimality conditions are: $x^\star$ is optimal if and only if there exists a $\nu^\star$ such that

$$\nabla f(x^\star) + A^T\nu^\star = 0, \qquad Ax^\star = b.$$

Equality constrained quadratic minimization. For $f(x) = (1/2)x^TPx + q^Tx + r$ with $P \in \mathbf{S}^n_+$, the optimality condition is the linear system

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix}\begin{bmatrix} x^\star \\ \nu^\star \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}.$$

The coefficient matrix is called the KKT matrix. It is nonsingular if and only if

$$Ax = 0,\ x \neq 0 \implies x^TPx > 0;$$

an equivalent condition for nonsingularity is $P + A^TA > 0$.

Eliminating equality constraints. Represent the solution set as $\{x \mid Ax = b\} = \{Fz + \hat{x} \mid z \in \mathbb{R}^{n-p}\}$, where $\hat{x}$ is any particular solution and the range of $F \in \mathbb{R}^{n\times(n-p)}$ is the nullspace of A ($\operatorname{rank} F = n - p$ and $AF = 0$). The reduced or eliminated problem, minimize $f(Fz + \hat{x})$, is an unconstrained problem with variable $z \in \mathbb{R}^{n-p}$. From its solution $z^\star$ we obtain

$$x^\star = Fz^\star + \hat{x}, \qquad \nu^\star = -(AA^T)^{-1}A\nabla f(x^\star).$$

Example: *Optimal allocation with a resource constraint*: minimize $f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n)$ subject to $x_1 + x_2 + \cdots + x_n = b$. Eliminate $x_n = b - x_1 - \cdots - x_{n-1}$, that is, choose

$$\hat{x} = be_n, \qquad F = \begin{bmatrix} I \\ -\mathbf{1}^T \end{bmatrix} \in \mathbb{R}^{n\times(n-1)}.$$

The reduced problem is to minimize $f_1(x_1) + \cdots + f_{n-1}(x_{n-1}) + f_n(b - x_1 - \cdots - x_{n-1})$ over $x_1, \ldots, x_{n-1}$.

15.3 Newton's method with equality constraints

Definition 15.1 [Newton step with equality constraints]. The Newton step Δx_{nt} of f at a feasible x is given by the solution v of

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}.$$

Interpretations:

- Δx_{nt} solves the second-order approximation (with variable v): minimize $\widehat{f}(x+v) = f(x) + \nabla f(x)^T v + (1/2) v^T \nabla^2 f(x) v$ subject to $A(x+v) = b$;
- the equations follow from linearizing the optimality conditions:

$$\nabla f(x+v) + A^T w \approx \nabla f(x) + \nabla^2 f(x) v + A^T w = 0, \qquad A(x+v) = b.$$

The Newton decrement is

$$\lambda(x) = \left(\Delta x_{\mathrm{nt}}^T \nabla^2 f(x) \Delta x_{\mathrm{nt}}\right)^{1/2} = \left(-\nabla f(x)^T \Delta x_{\mathrm{nt}}\right)^{1/2}.$$

It gives an estimate of $f(x) - p^\star$ using the quadratic approximation, $f(x) - \inf_{Ay=b} \widehat{f}(y) = \frac{1}{2}\lambda(x)^2$, and the directional derivative in the Newton direction, $\frac{d}{dt} f(x + t\Delta x_{\mathrm{nt}})\big|_{t=0} = -\lambda(x)^2$. In general, $\lambda(x) \neq (\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x))^{1/2}$.

Algorithm 15.1: Newton's method with equality constraints

Input: A starting point $x \in \operatorname{dom} f$ with $Ax = b$, tolerance $\epsilon > 0$.

Compute the Newton step and decrement Δx_{nt}, $\lambda(x)$

while $\lambda^2/2 > \epsilon$ **do** *// stopping criterion*

  Line search. Choose step size t by backtracking line search

  Update. $x := x + t\Delta x_{\mathrm{nt}}$

  Compute the Newton step and decrement Δx_{nt}, $\lambda(x)$

This is a feasible descent method ($x^{(k)}$ is feasible and $f(x^{(k+1)}) < f(x^{(k)})$), and it is affine invariant.

Newton's method and elimination. Newton's method for the reduced problem, minimize $\tilde{f}(z) = f(Fz + \hat{x})$ with $z \in \mathbb{R}^{n-p}$ (where $A\hat{x} = b$, $\operatorname{rank} F = n - p$, $AF = 0$), started at $z^{(0)}$, generates iterates $z^{(k)}$. Newton's method with equality constraints, started at $x^{(0)} = Fz^{(0)} + \hat{x}$, generates exactly the iterates $x^{(k)} = Fz^{(k)} + \hat{x}$. Hence no separate convergence analysis is needed.

15.4 Infeasible start Newton method

The second interpretation of the Newton step extends to infeasible x ($Ax \neq b$, but $x \in \operatorname{dom} f$): linearizing the optimality conditions gives

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\mathrm{nt}} \\ w \end{bmatrix} = - \begin{bmatrix} \nabla f(x) \\ Ax - b \end{bmatrix}. \tag{15.1}$$

Primal-dual interpretation. Write the optimality condition as $r(y) = 0$, where

$$y = (x, \nu), \qquad r(y) = (\nabla f(x) + A^T \nu,\ Ax - b).$$

Linearizing $r(y) = 0$ gives $r(y + \Delta y) \approx r(y) + Dr(y)\Delta y = 0$:

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\text{nt}} \\ \Delta \nu_{\text{nt}} \end{bmatrix} = - \begin{bmatrix} \nabla f(x) + A^T \nu \\ Ax - b \end{bmatrix},$$

which is the same as (15.1) with $w = \nu + \Delta \nu_{\text{nt}}$.

Algorithm 15.2: Infeasible start Newton method

Input: A starting point $x \in \operatorname{dom} f$, ν, tolerance $\epsilon > 0$, $\alpha \in (0, 1/2)$, $\beta \in (0, 1)$.

repeat

- Compute primal and dual Newton steps Δx_{nt}, $\Delta \nu_{\text{nt}}$
- **Backtracking line search on** $\|r\|_2$. $t := 1$
- **while** $\|r(x + t\Delta x_{\text{nt}}, \nu + t\Delta \nu_{\text{nt}})\|_2 > (1 - \alpha t)\|r(x, \nu)\|_2$ **do** $t := \beta t$
- **Update.** $x := x + t\Delta x_{\text{nt}}$, $\nu := \nu + t\Delta \nu_{\text{nt}}$

until $Ax = b$ **and** $\|r(x, \nu)\|_2 \leq \epsilon$

This is not a descent method: $f(x^{(k+1)}) > f(x^{(k)})$ is possible. The directional derivative of $\|r(y)\|_2$ in the direction $\Delta y = (\Delta x_{\text{nt}}, \Delta \nu_{\text{nt}})$ is

$$\frac{d}{dt}\|r(y + t\Delta y)\|_2\Big|_{t=0} = -\|r(y)\|_2,$$

which is why the line search on $\|r\|_2$ terminates.

15.5 Solving KKT systems

To solve

$$\begin{bmatrix} H & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = - \begin{bmatrix} g \\ h \end{bmatrix},$$

use

- an LDL^T factorization;
- elimination (if H is nonsingular): $AH^{-1}A^T w = h - AH^{-1}g$, then $Hv = -(g + A^T w)$;
- elimination with singular H: write the system as

$$\begin{bmatrix} H + A^T Q A & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = - \begin{bmatrix} g + A^T Q h \\ h \end{bmatrix}$$

  with $Q \succeq 0$ for which $H + A^T Q A > 0$, and apply elimination.

Example: *Equality constrained analytic centering.* The primal problem is to minimize $-\sum_{i=1}^n \log x_i$ subject to $Ax = b$, and the dual problem is to maximize $-b^T \nu + \sum_{i=1}^n \log(A^T \nu)_i + n$. Three methods were compared on an example with $A \in \mathbb{R}^{100 \times 500}$ and several starting points:

1. Newton's method with equality constraints (requires $x^{(0)} \succ 0$, $Ax^{(0)} = b$): $f(x^{(k)}) - p^\star$ reaches about 10^{-10} in 15 to 20 iterations;
2. Newton's method applied to the dual problem (requires $A^T\nu^{(0)} \succ 0$): $p^\star - g(\nu^{(k)})$ falls below 10^{-6} in 9 iterations;
3. infeasible start Newton method (requires $x^{(0)} \succ 0$): $\|r(x^{(k)}, \nu^{(k)})\|_2$ reaches about 10^{-10} in 10 to 21 iterations, depending on the starting point.

The complexity per iteration of the three methods is identical:

1. block elimination for the KKT system

$$\begin{bmatrix} \operatorname{diag}(x)^{-2} & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ w \end{bmatrix} = \begin{bmatrix} \operatorname{diag}(x)^{-1}\mathbf{1} \\ 0 \end{bmatrix}$$

   reduces to solving $A \operatorname{diag}(x)^2 A^T w = b$;
2. the dual Newton system is $A \operatorname{diag}(A^T\nu)^{-2} A^T \Delta\nu = -b + A \operatorname{diag}(A^T\nu)^{-1}\mathbf{1}$;
3. block elimination for the KKT system

$$\begin{bmatrix} \operatorname{diag}(x)^{-2} & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta\nu \end{bmatrix} = \begin{bmatrix} \operatorname{diag}(x)^{-1}\mathbf{1} - A^T\nu \\ b - Ax \end{bmatrix}$$

   reduces to solving $A \operatorname{diag}(x)^2 A^T w = 2Ax - b$.

In each case we solve $ADA^T w = h$ with D positive diagonal.

15.6 Network flow optimization

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^n \phi_i(x_i) \\ \text{subject to} & Ax = b \end{array}$$

models a directed graph with n arcs and $p+1$ nodes: x_i is the flow through arc i, and ϕ_i is the cost flow function for arc i (with $\phi_i''(x) > 0$). The NODE-INCIDENCE MATRIX $\tilde{A} \in \mathbb{R}^{(p+1)\times n}$ has $\tilde{A}_{ij} = 1$ if arc j leaves node i, -1 if arc j enters node i, and 0 otherwise. The reduced node-incidence matrix $A \in \mathbb{R}^{p\times n}$ is $\tilde{A}$ with the last row removed, and $b \in \mathbb{R}^p$ is the (reduced) source vector; rank $A = p$ if the graph is connected.

In the KKT system, $H = \operatorname{diag}(\phi_1''(x_1), \ldots, \phi_n''(x_n))$ is positive diagonal, so we solve via elimination, $AH^{-1}A^T w = h - AH^{-1}g$ and $Hv = -(g + A^T w)$. The sparsity pattern of the coefficient matrix is given by the graph connectivity:

$$(AH^{-1}A^T)_{ij} \neq 0 \iff (AA^T)_{ij} \neq 0 \iff \text{nodes } i \text{ and } j \text{ are connected by an arc.}$$

15.7 Analytic center of a linear matrix inequality

$$\begin{array}{ll} \text{minimize} & -\log\det X \\ \text{subject to} & \operatorname{tr}(A_i X) = b_i, \quad i = 1, \ldots, p, \end{array}$$

with variable $X \in \mathbf{S}^n$. The optimality conditions are

$$X^\star \succ 0, \qquad -(X^\star)^{-1} + \sum_{j=1}^{p} \nu_j^\star A_j = 0, \qquad \operatorname{tr}(A_i X^\star) = b_i, \quad i = 1, \ldots, p.$$

The Newton equation at a feasible X,

$$X^{-1}\Delta X X^{-1} + \sum_{j=1}^{p} w_j A_j = X^{-1}, \qquad \operatorname{tr}(A_i \Delta X) = 0, \quad i = 1, \ldots, p,$$

follows from the linear approximation $(X + \Delta X)^{-1} \approx X^{-1} - X^{-1}\Delta X X^{-1}$; it has $n(n+1)/2 + p$ variables $\Delta X, w$.

Solution by block elimination. Eliminate ΔX from the first equation, $\Delta X = X - \sum_{j=1}^{p} w_j X A_j X$, and substitute into the second:

$$\sum_{j=1}^{p} \operatorname{tr}(A_i X A_j X) w_j = b_i, \qquad i = 1, \ldots, p, \tag{15.2}$$

a dense positive definite set of linear equations in $w \in \mathbb{R}^p$. With the Cholesky factorization $X = LL^T$, the dominant terms of the flop count are: forming the p products $L^T A_j L$, $(3/2)pn^3$; forming the $p(p+1)/2$ inner products $\operatorname{tr}((L^T A_i L)(L^T A_j L))$, $(1/2)p^2 n^2$; and solving (15.2) via Cholesky factorization, $(1/3)p^3$.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 15 (2023); original slides, *Unconstrained minimization* and *Equality constrained minimization*.

Lecture 16: Interior-Point Methods

16.1 Inequality constrained minimization

Consider

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \le 0, \quad i = 1, \dots, m \\ & Ax = b, \end{array} \tag{16.1}$$

with f_i convex and twice continuously differentiable, $A \in \mathbb{R}^{p \times n}$ with $\operatorname{rank} A = p$, and $p^\star$ finite and attained. We assume the problem is strictly feasible: there exists $\tilde{x}$ with

$$\tilde{x} \in \operatorname{dom} f_0, \qquad f_i(\tilde{x}) < 0, \quad i = 1, \dots, m, \qquad A\tilde{x} = b,$$

hence strong duality holds and the dual optimum is attained.

Examples are LP, QP, QCQP, and GP, and entropy maximization with linear inequality constraints, minimize $\sum_{i=1}^n x_i \log x_i$ subject to $Fx \preceq g$, $Ax = b$, with $\operatorname{dom} f_0 = \mathbb{R}^n_{++}$. Differentiability may require reformulating the problem, for example piecewise-linear minimization or ℓ_∞-norm approximation via an LP. SDPs and SOCPs are better handled as problems with generalized inequalities (Lecture 17).

16.2 Logarithmic barrier and central path

Using the indicator function $I_-(u) = 0$ if $u \le 0$ and $I_-(u) = \infty$ otherwise (the indicator function of $\mathbb{R}_-$), problem (16.1) is

$$\begin{array}{ll} \text{minimize} & f_0(x) + \sum_{i=1}^m I_-(f_i(x)) \\ \text{subject to} & Ax = b. \end{array}$$

Approximating via the logarithmic barrier gives

$$\begin{array}{ll} \text{minimize} & f_0(x) - (1/t) \sum_{i=1}^m \log(-f_i(x)) \\ \text{subject to} & Ax = b, \end{array}$$

an equality constrained problem. For $t > 0$, $-(1/t)\log(-u)$ is a smooth approximation of I_-, and the approximation improves as $t \to \infty$ (Figure 33).

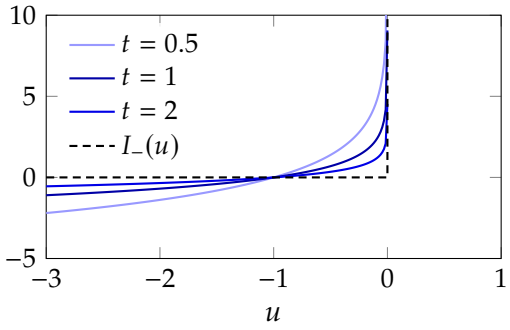


Figure 33: The indicator I_- and its approximations $-(1/t)\log(-u)$ for $t = 0.5, 1, 2$.

Definition 16.1 [Logarithmic barrier]. The logarithmic barrier function is

$$\phi(x) = -\sum_{i=1}^{m} \log(-f_i(x)), \qquad \operatorname{dom}\phi = \{x \mid f_1(x) < 0, \ldots, f_m(x) < 0\}.$$

ϕ is convex (by the composition rules) and twice continuously differentiable, with

$$\nabla\phi(x) = \sum_{i=1}^{m} \frac{1}{-f_i(x)}\nabla f_i(x), \qquad \nabla^2\phi(x) = \sum_{i=1}^{m} \frac{1}{f_i(x)^2}\nabla f_i(x)\nabla f_i(x)^T + \sum_{i=1}^{m} \frac{1}{-f_i(x)}\nabla^2 f_i(x).$$

Definition 16.2 [Central path]. For $t > 0$, let $x^\star(t)$ be the solution of

$$\begin{array}{ll} \text{minimize} & t f_0(x) + \phi(x) \\ \text{subject to} & Ax = b \end{array}$$

(assumed for now to exist and be unique for each $t > 0$). The central path is $\{x^\star(t) \mid t > 0\}$.

For an LP, minimize $c^T x$ subject to $a_i^T x \le b_i$, $i = 1, \ldots, 6$, the hyperplane $c^T x = c^T x^\star(t)$ is tangent to the level curve of ϕ through $x^\star(t)$ (Figure 34), since the centering condition $tc + \nabla\phi(x^\star(t)) = 0$ makes c normal to that level curve.

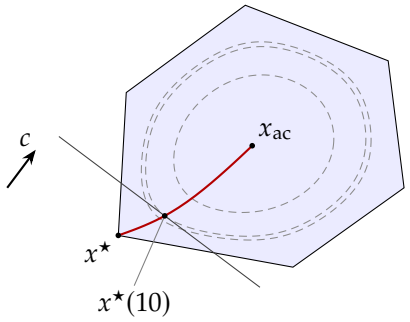


Figure 34: The central path of an LP with six inequalities, from the analytic center x_{ac} (as $t \to 0$) to the solution $x^\star$ (as $t \to \infty$). Dashed curves are level sets of ϕ; the line $c^T x = c^T x^\star(10)$ is tangent to the one through $x^\star(10)$.

Dual points on the central path. $x = x^\star(t)$ if and only if there exists a w such that

$$t\nabla f_0(x) + \sum_{i=1}^m \frac{1}{-f_i(x)}\nabla f_i(x) + A^T w = 0, \qquad Ax = b.$$

Therefore $x^\star(t)$ minimizes the Lagrangian

$$L(x, \lambda^\star(t), \nu^\star(t)) = f_0(x) + \sum_{i=1}^m \lambda_i^\star(t) f_i(x) + \nu^\star(t)^T(Ax - b),$$

where we define $\lambda_i^\star(t) = 1/(-t f_i(x^\star(t)))$ and $\nu^\star(t) = w/t$. This confirms the intuitive idea that $f_0(x^\star(t)) \to p^\star$ as $t \to \infty$:

$$p^\star \geq g(\lambda^\star(t), \nu^\star(t)) = L(x^\star(t), \lambda^\star(t), \nu^\star(t)) = f_0(x^\star(t)) - m/t.$$

Interpretation via KKT conditions. $x = x^\star(t)$, $\lambda = \lambda^\star(t)$, $\nu = \nu^\star(t)$ satisfy

1. primal constraints: $f_i(x) \leq 0$, $i = 1, \ldots, m$, $Ax = b$;
2. dual constraints: $\lambda \geq 0$;
3. APPROXIMATE COMPLEMENTARY SLACKNESS: $-\lambda_i f_i(x) = 1/t$, $i = 1, \ldots, m$;
4. the gradient of the Lagrangian with respect to x vanishes: $\nabla f_0(x) + \sum_{i=1}^m \lambda_i \nabla f_i(x) + A^T\nu = 0$.

The difference from the KKT conditions is that condition 3 replaces $\lambda_i f_i(x) = 0$.

Force field interpretation. For a problem with no equality constraints, the centering problem is to minimize $t f_0(x) - \sum_{i=1}^m \log(-f_i(x))$. Here $t f_0(x)$ is the potential of the force field $F_0(x) = -t\nabla f_0(x)$, and $-\log(-f_i(x))$ is the potential of the force field $F_i(x) = (1/f_i(x))\nabla f_i(x)$. The forces balance at $x^\star(t)$:

$$F_0(x^\star(t)) + \sum_{i=1}^m F_i(x^\star(t)) = 0.$$

For the LP minimize $c^T x$ subject to $a_i^T x \leq b_i$, the objective force field is constant, $F_0(x) = -tc$, and the constraint force field decays as the inverse distance to the constraint hyperplane:

$$F_i(x) = \frac{-a_i}{b_i - a_i^T x}, \qquad \|F_i(x)\|_2 = \frac{1}{\mathrm{dist}(x, \mathcal{H}_i)}, \qquad \mathcal{H}_i = \{x \mid a_i^T x = b_i\}$$

(Figure 35).

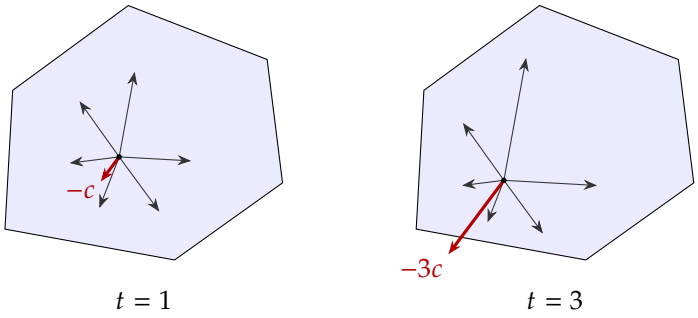


Figure 35: Force balance at $x^\star(t)$ for the LP of Figure 34, for $t = 1$ and $t = 3$. A larger objective force $-tc$ pushes $x^\star(t)$ toward the constraints, where their repelling forces grow.

16.3 Barrier method

Algorithm 16.1: Barrier method

Input: A strictly feasible x, $t := t^{(0)} > 0$, $\mu > 1$, tolerance $\epsilon > 0$.

Centering step. Compute $x^\star(t)$ by minimizing $tf_0 + \phi$, subject to $Ax = b$

Update. $x := x^\star(t)$

while $m/t \geq \epsilon$ **do** *// stopping criterion*

  Increase t**.** $t := \mu t$

  Centering step. Compute $x^\star(t)$, starting from the current x

  Update. $x := x^\star(t)$

- The method terminates with $f_0(x) - p^\star \leq \epsilon$; the stopping criterion follows from $f_0(x^\star(t)) - p^\star \leq m/t$.
- Centering is usually done using Newton's method, starting at the current x.
- The choice of μ involves a trade-off: large μ means fewer outer iterations but more inner (Newton) iterations; typical values are $\mu = 10$–20.
- There are several heuristics for the choice of $t^{(0)}$.

Theorem 16.3 [Outer iterations]. *The number of outer (centering) iterations is exactly*

$$\left\lceil \frac{\log(m/(\epsilon t^{(0)}))}{\log \mu} \right\rceil$$

plus the initial centering step (to compute $x^\star(t^{(0)})$).

Since $t = \mu^k t^{(0)}$ after k updates, the criterion $m/t < \epsilon$ first holds at that k. For the centering problem, minimize $tf_0(x) + \phi(x)$, we use the convergence analysis of Newton's method: $tf_0 + \phi$ must have closed sublevel sets for $t \geq t^{(0)}$; the classical analysis requires strong convexity and a Lipschitz condition, and the analysis via self-concordance requires self-concordance of $tf_0 + \phi$.

Examples. For an inequality form LP with $m = 100$ inequalities and $n = 50$ variables, starting with x on the central path ($t^{(0)} = 1$, duality gap 100) and terminating when $t = 10^8$ (gap 10^{-6}), with centering by Newton's method with backtracking, the duality gap decreases in a staircase pattern, one step per centering. With $\mu = 50$ the gap reaches 10^{-6} after about 35 Newton iterations in total and with $\mu = 150$ after about 40, while $\mu = 2$ needs about 80. The total number of Newton iterations is not very sensitive to μ for $\mu \geq 10$ (roughly 25 to 40), and grows rapidly as μ approaches 1. A geometric program with $m = 100$ inequalities and $n = 50$ variables behaves similarly. For a family of standard form LPs (minimize $c^T x$ subject to $Ax = b$, $x \geq 0$, with $A \in \mathbb{R}^{m \times 2m}$), solving 100 randomly generated instances for each $m = 10, \ldots, 1000$, the number of Newton iterations grows very slowly, from about 20 to about 27, as m ranges over a 100 : 1 ratio.

16.4 Feasibility and phase I methods

The FEASIBILITY PROBLEM is to find x such that

$$f_i(x) \leq 0, \quad i = 1, \ldots, m, \qquad Ax = b. \tag{16.2}$$

PHASE I computes a strictly feasible starting point for the barrier method. The BASIC PHASE I METHOD solves

$$\begin{array}{lll} \text{minimize (over } x, s) & s & \\ \text{subject to} & f_i(x) \leq s, \quad i = 1, \ldots, m & \\ & Ax = b. & \end{array} \tag{16.3}$$

- If x, s are feasible with $s < 0$, then x is strictly feasible for (16.2).
- If the optimal value $\bar{p}^\star$ of (16.3) is positive, then (16.2) is infeasible.
- If $\bar{p}^\star = 0$ and is attained, then (16.2) is feasible (but not strictly); if $\bar{p}^\star = 0$ and is not attained, then (16.2) is infeasible.

The SUM OF INFEASIBILITIES phase I method solves

$$\begin{array}{lll} \text{minimize} & \mathbf{1}^T s & \\ \text{subject to} & s \geq 0, \quad f_i(x) \leq s_i, & i = 1, \ldots, m \\ & Ax = b. & \end{array}$$

For infeasible problems it produces a solution that satisfies many more inequalities than the basic phase I method. For an infeasible set of 100 linear inequalities in 50 variables, the basic phase I solution satisfies 39 inequalities, while the sum of infeasibilities solution satisfies 79.

Example: Consider the family of linear inequalities $Ax \preceq b + \gamma \Delta b$, with data chosen to be strictly feasible for $\gamma > 0$ and infeasible for $\gamma \leq 0$. Using basic phase I, terminated when $s < 0$ or the dual objective is positive, the number of Newton iterations stays moderate (about 30 to 50) away from $\gamma = 0$ and grows as $\gamma \to 0$: it is roughly proportional to $\log(1/|\gamma|)$, on both the feasible and the

infeasible side.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 16 (2023); original slides, *Interior-point methods*.

Lecture 17: Interior-Point Methods (Continued); Convex-Cardinality Problems

17.1 Complexity analysis via self-concordance

In addition to the assumptions of Lecture 16, assume that the sublevel sets (of f_0, on the feasible set) are bounded, and that $tf_0 + \phi$ is self-concordant with closed sublevel sets. The second condition holds for LP, QP, and QCQP, and may require reformulating the problem, for example

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^n x_i \log x_i \\ \text{subject to} & Fx \preceq g \end{array} \quad \longrightarrow \quad \begin{array}{ll} \text{minimize} & \sum_{i=1}^n x_i \log x_i \\ \text{subject to} & Fx \preceq g, \quad x \succeq 0. \end{array}$$

The condition is needed for the complexity analysis; the barrier method works even when the self-concordance assumption does not apply.

Newton iterations per centering step. Self-concordance theory bounds the effort of computing $x^+ = x^\star(\mu t)$ starting at $x = x^\star(t)$:

$$\#\text{Newton iterations} \leq \frac{\mu t f_0(x) + \phi(x) - \mu t f_0(x^+) - \phi(x^+)}{\gamma} + c,$$

where γ, c are constants that depend only on the Newton algorithm parameters. By duality, with $\lambda = \lambda^\star(t)$, $\nu = \nu^\star(t)$ (so that $-f_i(x) = 1/(t\lambda_i)$),

$$\begin{aligned} \mu t f_0(x) + \phi(x) - \mu t f_0(x^+) - \phi(x^+) &= \mu t f_0(x) - \mu t f_0(x^+) + \sum_{i=1}^m \log(-\mu t \lambda_i f_i(x^+)) - m \log \mu \\ &\leq \mu t f_0(x) - \mu t f_0(x^+) - \mu t \sum_{i=1}^m \lambda_i f_i(x^+) - m - m \log \mu \\ &\leq \mu t f_0(x) - \mu t g(\lambda, \nu) - m - m \log \mu \\ &= m(\mu - 1 - \log \mu), \end{aligned}$$

using $\log y \leq y - 1$ in the first inequality, $f_0(x^+) + \sum_i \lambda_i f_i(x^+) \geq g(\lambda, \nu)$ (since $Ax^+ = b$) in the second, and $f_0(x) - g(\lambda, \nu) = m/t$ in the last step.

Theorem 17.1 [Total Newton iterations]. *The total number of Newton iterations, excluding the first centering step, is at most*

$$N = \left\lceil \frac{\log(m/(t^{(0)}\epsilon))}{\log \mu} \right\rceil \left(\frac{m(\mu - 1 - \log \mu)}{\gamma} + c \right).$$

For typical values of γ and c, with $m = 100$ and $m/(t^{(0)}\epsilon) = 10^5$, N is very large as $\mu \to 1$, is smallest for μ slightly above 1, and then grows roughly linearly in μ: this confirms the trade-off in the choice of μ. In practice, the number of iterations is in the tens, and not very sensitive to μ for $\mu \geq 10$.

Corollary 17.2 [Polynomial-time complexity]. *For $\mu = 1 + 1/\sqrt{m}$,*

$$N = O\Big(\sqrt{m} \log \frac{m/t^{(0)}}{\epsilon}\Big).$$

The number of Newton iterations for a fixed gap reduction is $O(\sqrt{m})$; multiplying by the cost of one Newton iteration (a polynomial function of the problem dimensions) gives a bound on the number of flops. This choice of μ optimizes worst-case complexity; in practice we choose μ fixed ($\mu = 10, \ldots, 20$).

17.2 Generalized inequalities

Consider

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \preceq_{K_i} 0, \quad i = 1, \ldots, m \\ & Ax = b, \end{array}$$

with f_0 convex; $f_i : \mathbb{R}^n \to \mathbb{R}^{k_i}$ convex with respect to proper cones $K_i \subseteq \mathbb{R}^{k_i}$ and twice continuously differentiable; $A \in \mathbb{R}^{p \times n}$ with $\operatorname{rank} A = p$; $p^\star$ finite and attained; and the problem strictly feasible, so that strong duality holds and the dual optimum is attained. The examples of greatest interest are SOCP and SDP.

Definition 17.3 [Generalized logarithm]. $\psi : \mathbb{R}^q \to \mathbb{R}$ is a generalized logarithm for a proper cone $K \subseteq \mathbb{R}^q$ if $\operatorname{dom} \psi = \operatorname{int} K$, $\nabla^2 \psi(y) \prec 0$ for $y \succ_K 0$, and

$$\psi(sy) = \psi(y) + \theta \log s \qquad \text{for } y \succ_K 0,\ s > 0,$$

where θ is the degree of ψ.

Examples:

- nonnegative orthant $K = \mathbb{R}^n_+$: $\psi(y) = \sum_{i=1}^n \log y_i$, with degree $\theta = n$;
- positive semidefinite cone $K = \mathbf{S}^n_+$: $\psi(Y) = \log\det Y$ ($\theta = n$);
- second-order cone $K = \{y \in \mathbb{R}^{n+1} \mid (y_1^2 + \cdots + y_n^2)^{1/2} \leq y_{n+1}\}$: $\psi(y) = \log(y_{n+1}^2 - y_1^2 - \cdots - y_n^2)$ ($\theta = 2$).

Properties (without proof): for $y \succ_K 0$,

$$\nabla\psi(y) \succeq_{K^*} 0, \qquad y^T \nabla\psi(y) = \theta.$$

For the nonnegative orthant, $\nabla\psi(y) = (1/y_1, \ldots, 1/y_n)$ and $y^T\nabla\psi(y) = n$; for the positive semidefinite cone, $\nabla\psi(Y) = Y^{-1}$ and $\mathrm{tr}(Y\nabla\psi(Y)) = n$; for the second-order cone,

$$\nabla\psi(y) = \frac{2}{y_{n+1}^2 - y_1^2 - \cdots - y_n^2}\begin{bmatrix} -y_1 \\ \vdots \\ -y_n \\ y_{n+1} \end{bmatrix}, \qquad y^T\nabla\psi(y) = 2.$$

Logarithmic barrier and central path. The logarithmic barrier for $f_1(x) \preceq_{K_1} 0, \ldots, f_m(x) \preceq_{K_m} 0$ is

$$\phi(x) = -\sum_{i=1}^m \psi_i(-f_i(x)), \qquad \mathrm{dom}\,\phi = \{x \mid f_i(x) \prec_{K_i} 0,\ i = 1, \ldots, m\},$$

where ψ_i is a generalized logarithm for K_i with degree θ_i; ϕ is convex and twice continuously differentiable. The central path is $\{x^\star(t) \mid t > 0\}$, where $x^\star(t)$ solves

$$\begin{array}{ll} \text{minimize} & tf_0(x) + \phi(x) \\ \text{subject to} & Ax = b. \end{array}$$

Dual points on the central path. $x = x^\star(t)$ if there exists $w \in \mathbb{R}^p$ with

$$t\nabla f_0(x) + \sum_{i=1}^m Df_i(x)^T\nabla\psi_i(-f_i(x)) + A^Tw = 0,$$

where $Df_i(x) \in \mathbb{R}^{k_i \times n}$ is the derivative matrix of f_i. Therefore $x^\star(t)$ minimizes the Lagrangian $L(x, \lambda^\star(t), \nu^\star(t))$, where

$$\lambda_i^\star(t) = \frac{1}{t}\nabla\psi_i(-f_i(x^\star(t))), \qquad \nu^\star(t) = \frac{w}{t}.$$

By the properties of ψ_i, $\lambda_i^\star(t) \succ_{K_i^*} 0$, and the duality gap is

$$f_0(x^\star(t)) - g(\lambda^\star(t), \nu^\star(t)) = (1/t)\sum_{i=1}^m \theta_i.$$

Example: *Semidefinite programming* (with $F_i \in \mathbf{S}^p$): minimize c^Tx subject to $F(x) = \sum_{i=1}^n x_iF_i + G \preceq 0$. The logarithmic barrier is $\phi(x) = \log\det(-F(x)^{-1})$. On the central path, $x^\star(t)$ minimizes $tc^Tx - \log\det(-F(x))$, hence

$$tc_i - \mathrm{tr}(F_iF(x^\star(t))^{-1}) = 0, \qquad i = 1, \ldots, n.$$

The dual point $Z^\star(t) = -(1/t)F(x^\star(t))^{-1}$ is feasible for the dual

$$\begin{array}{ll} \text{maximize} & \operatorname{tr}(GZ) \\ \text{subject to} & \operatorname{tr}(F_i Z) + c_i = 0, \quad i = 1, \ldots, n \\ & Z \succeq 0, \end{array}$$

and the duality gap on the central path is $c^T x^\star(t) - \operatorname{tr}(GZ^\star(t)) = p/t$.

Barrier method. The barrier method is the same as before; the only difference is that the duality gap m/t on the central path is replaced by $\sum_i \theta_i/t$, so the stopping criterion becomes $(\sum_i \theta_i)/t < \epsilon$. The number of outer iterations is

$$\left\lceil \frac{\log((\sum_i \theta_i)/(\epsilon t^{(0)}))}{\log \mu} \right\rceil,$$

and the complexity analysis via self-concordance applies to SDP and SOCP.

Examples. For an SOCP with 50 variables and 50 SOC constraints in $\mathbb{R}^6$, and an SDP with 100 variables and an LMI constraint in $\mathbf{S}^{100}$, the duality gap again decreases in a staircase pattern: with $\mu = 50$ or $\mu = 150$ to 200 the gap reaches 10^{-6} in about 30 to 40 Newton iterations, while $\mu = 2$ needs about 80; the total is roughly 20 to 40 Newton iterations for a wide range of μ. For a family of SDPs, minimize $\mathbf{1}^T x$ subject to $A + \operatorname{diag}(x) \succeq 0$ with $A \in \mathbf{S}^n$, solving 100 randomly generated instances for each $n = 10, \ldots, 1000$, the number of Newton iterations grows only from about 20 to about 26.

17.3 Primal-dual interior-point methods

Primal-dual interior-point methods are more efficient than the barrier method when high accuracy is needed. They update the primal and dual variables at each iteration, with no distinction between inner and outer iterations; they often exhibit superlinear asymptotic convergence; their search directions can be interpreted as Newton directions for modified KKT conditions; they can start at infeasible points; and their cost per iteration is the same as for the barrier method.

17.4 Cardinality

Cardinality problems arise often but are hard to solve exactly. A simple heuristic that relies on the ℓ_1-norm seems to work well, and has been used for many years in many fields: sparse design; LASSO and robust estimation in statistics; the support vector machine in machine learning; total variation reconstruction in signal processing and geophysics; and compressed sensing. New theoretical results guarantee that the method works, at least for a few problems.

Definition 17.4 [Cardinality]. The cardinality of $x \in \mathbb{R}^n$, denoted $\operatorname{card}(x)$, is the number of nonzero components of x.

card is separable: for scalar x, $\operatorname{card}(x) = 0$ if $x = 0$ and 1 otherwise. It is quasiconcave on $\mathbb{R}^n_+$ (but

not on $\mathbb{R}^n$), since $\text{card}(x+y) \geq \min\{\text{card}(x), \text{card}(y)\}$ holds for $x, y \geq 0$, but otherwise it has no convexity properties.

Definition 17.5 [Convex-cardinality problem]. A convex-cardinality problem is one that would be convex, except for the appearance of card in the objective or constraints.

Examples, with C and f convex, are the convex minimum cardinality problem, minimize $\text{card}(x)$ subject to $x \in C$, and the convex problem with a cardinality constraint, minimize $f(x)$ subject to $x \in C$, $\text{card}(x) \leq k$.

Solving convex-cardinality problems. If we fix the sparsity pattern of $x \in \mathbb{R}^n$ (which entries are zero or nonzero), we get a convex problem. By solving the 2^n convex problems associated with all possible sparsity patterns we can solve the convex-cardinality problem; this is possibly practical for $n \leq 10$, but not for $n > 15$ or so. The general convex-cardinality problem is (NP-)hard. It can be solved globally by branch-and-bound, which can work for particular problem instances (with some luck) but in the worst case reduces to checking all (or many) of the 2^n sparsity patterns.

Example: *Boolean LP as a convex-cardinality problem.* The Boolean LP, minimize $c^T x$ subject to $Ax \preceq b$, $x_i \in \{0,1\}$, includes many famous hard problems (3-SAT, traveling salesman). It can be expressed as

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax \preceq b, \quad \text{card}(x) + \text{card}(\mathbf{1} - x) \leq n, \end{array}$$

since $\text{card}(x) + \text{card}(\mathbf{1} - x) \leq n \iff x_i \in \{0,1\}$. Hence the general convex-cardinality problem is hard.

17.5 Applications of convex-cardinality problems

- *Sparse design*: minimize $\text{card}(x)$ subject to $x \in C$, that is, find the sparsest design vector that satisfies a set of specifications. Zero values of x simplify the design or correspond to components that aren't even needed: zero coefficients in FIR filter design reduce the required hardware; in antenna array beamforming they correspond to unneeded antenna elements; in truss design, to bars that are not needed; in wire sizing, to wires that are not needed.
- *Sparse modeling or regressor selection*: fit $b \in \mathbb{R}^m$ as a linear combination of k regressors chosen from n possible regressors, minimize $\|Ax - b\|_2$ subject to $\text{card}(x) \leq k$. This gives a k-term model, choosing the subset of k regressors that together best fit or explain b; it can be solved (in principle) by trying all $\binom{n}{k}$ choices. Variations: minimize $\text{card}(x)$ subject to $\|Ax - b\|_2 \leq \epsilon$, or minimize $\|Ax - b\|_2 + \lambda \,\text{card}(x)$.
- *Sparse signal reconstruction*: estimate a signal x given a noisy measurement $y = Ax + v$, $v \sim \mathcal{N}(0, \sigma^2 I)$ (A known, v not), and the prior information $\text{card}(x) \leq k$. The maximum likelihood estimate $\hat{x}_{\text{ml}}$ solves minimize $\|Ax - y\|_2$ subject to $\text{card}(x) \leq k$.
- *Estimation with outliers*: measurements $y_i = a_i^T x + v_i + w_i$, $i = 1, \ldots, m$, with independent

noises $v_i \sim \mathcal{N}(0, \sigma^2)$; the only assumption on w is sparsity, $\mathrm{card}(w) \leq k$, and $\mathcal{B} = \{i \mid w_i \neq 0\}$ is the set of bad measurements or outliers. The maximum likelihood estimate of x minimizes $\sum_{i \notin \mathcal{B}} (y_i - a_i^T x)^2$ subject to $|\mathcal{B}| \leq k$, over x and $\mathcal{B} \subseteq \{1, \ldots, m\}$; equivalently, minimize $\|y - Ax - w\|_2^2$ subject to $\mathrm{card}(w) \leq k$.

- *Minimum number of violations*: given convex inequalities $f_1(x) \leq 0, \ldots, f_m(x) \leq 0$, $x \in C$, choose x to minimize the number of violated inequalities: minimize $\mathrm{card}(t)$ subject to $f_i(x) \leq t_i$, $x \in C$, $t \geq 0$. Determining whether zero inequalities can be violated is an (easy) convex feasibility problem.

- *Linear classifier with fewest errors*: given data $(x_1, y_1), \ldots, (x_m, y_m) \in \mathbb{R}^n \times \{-1, 1\}$, we seek an affine classifier $y \approx \mathrm{sign}(w^T x + v)$; a classification error corresponds to $y_i(w^T x_i + v) \leq 0$. To find w, v giving the fewest errors, minimize $\mathrm{card}(t)$ subject to $y_i(w^T x_i + v) + t_i \geq 1$, $i = 1, \ldots, m$, with variables w, v, t (using homogeneity in w, v).

- *Smallest set of mutually infeasible inequalities*: given mutually infeasible convex inequalities $f_1(x) \leq 0, \ldots, f_m(x) \leq 0$, find the smallest subset of them that is infeasible. A certificate of infeasibility is $g(\lambda) = \inf_x \left(\sum_{i=1}^m \lambda_i f_i(x) \right) \geq 1, \lambda \geq 0$, so (assuming some constraint qualifications) we solve: minimize $\mathrm{card}(\lambda)$ subject to $g(\lambda) \geq 1, \lambda \geq 0$.

- *Portfolio investment with linear and fixed costs*: we use a budget B to purchase (dollar) amounts $x_i \geq 0$ of stock i. The trading fee is a fixed cost plus a linear cost, $\beta\, \mathrm{card}(x) + \alpha^T x$, so the budget constraint is $\mathbf{1}^T x + \beta\, \mathrm{card}(x) + \alpha^T x \leq B$. The mean return on the investment is $\mu^T x$ and its variance $x^T \Sigma x$. To minimize the investment variance (risk) with mean return at least $R_{\min}$,

$$\begin{array}{ll} \text{minimize} & x^T \Sigma x \\ \text{subject to} & \mu^T x \geq R_{\min}, \quad x \geq 0 \\ & \mathbf{1}^T x + \beta\, \mathrm{card}(x) + \alpha^T x \leq B. \end{array}$$

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 17 (2023); original slides, *Interior-point methods*; EE364b slides, *ℓ_1-norm methods for convex-cardinality problems*.

Lecture 18: Convex-Cardinality Problems (Continued); Conclusions

18.1 Piecewise constant and piecewise linear fitting

To fit a corrupted signal x_{cor} by a piecewise constant signal $\hat{x}$ with k or fewer jumps: the problem is convex once the locations (indices) of the jumps are fixed, and $\hat{x}$ is piecewise constant with at most k jumps if and only if $\text{card}(D\hat{x}) \leq k$, where

$$D = \begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & \ddots & \ddots & \\ & & & 1 & -1 \end{bmatrix} \in \mathbb{R}^{(n-1)\times n}.$$

As a convex-cardinality problem: minimize $\|\hat{x} - x_{\text{cor}}\|_2$ subject to $\text{card}(D\hat{x}) \leq k$. Similarly, to fit x_{cor} by a piecewise linear signal with k or fewer kinks, minimize $\|\hat{x} - x_{\text{cor}}\|_2$ subject to $\text{card}(\nabla^2\hat{x}) \leq k$, where ∇^2 is the second difference matrix with rows $(\ldots, -1, 2, -1, \ldots)$.

18.2 The ℓ_1-norm heuristic

The ℓ_1-norm heuristic replaces $\text{card}(z)$ with $\gamma\|z\|_1$, or adds a regularization term $\gamma\|z\|_1$ to the objective; $\gamma > 0$ is a parameter used to achieve the desired sparsity (when card appears in a constraint, or as a term in the objective). More sophisticated versions use $\sum_i w_i|z_i|$ or $\sum_i w_i(z_i)_+ + \sum_i v_i(z_i)_-$, where w, v are positive weights.

- *Minimum cardinality problem.* The (hard) problem minimize $\text{card}(x)$ subject to $x \in C$ (C convex) becomes the (easy) ℓ_1-norm minimization problem, minimize $\|x\|_1$ subject to $x \in C$.
- *Cardinality constrained problem.* The (hard) problem minimize $f(x)$ subject to $x \in C$, $\text{card}(x) \leq k$ (f, C convex) becomes the (easy) ℓ_1-constrained problem, minimize $f(x)$ subject to $x \in C$, $\|x\|_1 \leq \beta$, or the ℓ_1-regularized problem, minimize $f(x) + \gamma\|x\|_1$ subject to $x \in C$, with β, γ adjusted so that $\text{card}(x) \leq k$.

Polishing. Use the ℓ_1 heuristic to find $\hat{x}$ with the required sparsity, fix the sparsity pattern of $\hat{x}$, and re-solve the (convex) optimization problem with this sparsity pattern to obtain the final (heuristic) solution.

Interpretation as convex relaxation. The problem

$$\begin{array}{ll} \text{minimize} & \text{card}(x) \\ \text{subject to} & x \in C, \quad \|x\|_\infty \leq R \end{array}$$

is equivalent to the mixed Boolean convex problem in the variables x, z

$$\begin{array}{ll} \text{minimize} & \mathbf{1}^T z \\ \text{subject to} & |x_i| \leq R z_i, \quad i = 1, \ldots, n \\ & x \in C, \quad z_i \in \{0, 1\}, \quad i = 1, \ldots, n. \end{array}$$

Relaxing $z_i \in \{0, 1\}$ to $z_i \in [0, 1]$ gives a problem equivalent to

$$\begin{array}{ll} \text{minimize} & (1/R)\|x\|_1 \\ \text{subject to} & x \in C, \quad \|x\|_\infty \leq R, \end{array}$$

the ℓ_1 heuristic (at the optimum $z_i = |x_i|/R$). The optimal value of this problem is a lower bound on the original problem.

Definition 18.1 [Convex envelope]. The convex envelope f^{env} of a function f on a set C is the largest convex function that is an underestimator of f on C.

Equivalently, $\text{epi}(f^{\text{env}}) = \text{conv}(\text{epi}\, f)$, and $f^{\text{env}} = (f^*)^*$ (under some technical conditions). For scalar x, $|x|$ is the convex envelope of $\text{card}(x)$ on $[-1, 1]$ (Figure 36); for $x \in \mathbb{R}^n$, $(1/R)\|x\|_1$ is the convex envelope of $\text{card}(x)$ on $\{z \mid \|z\|_\infty \leq R\}$.

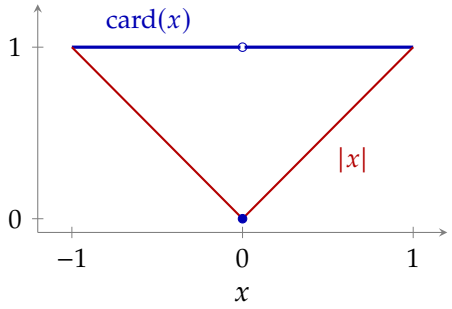


Figure 36: $\text{card}(x)$ on $[-1, 1]$ and its convex envelope $|x|$.

Weighted and asymmetric ℓ_1 heuristics. To minimize $\text{card}(x)$ over a convex set C, suppose we know lower and upper bounds on x_i over C, $x \in C \implies l_i \leq x_i \leq u_i$ (the best values can be found by solving $2n$ convex problems). If $u_i < 0$ or $l_i > 0$, then $\text{card}(x_i) = 1$ ($x_i \neq 0$) for all $x \in C$. Assuming $l_i < 0 < u_i$, the convex relaxation and convex envelope interpretations suggest using

$$\sum_{i=1}^{n} \left(\frac{(x_i)_+}{u_i} + \frac{(x_i)_-}{-l_i} \right)$$

as a surrogate (and also a lower bound) for $\text{card}(x)$.

18.3 Regressor selection and sparse signal reconstruction

Regressor selection. For minimize $\|Ax - b\|_2$ subject to $\text{card}(x) \leq k$, the heuristic is to minimize $\|Ax - b\|_2 + \gamma\|x\|_1$, find the smallest value of γ that gives $\text{card}(x) \leq k$, fix the associated sparsity pattern (the subset of selected regressors), and find the x that minimizes $\|Ax - b\|_2$. In an example with $A \in \mathbb{R}^{10\times 20}$, the trade-off curve of $\text{card}(x)$ versus $\|Ax - b\|_2$ obtained by the ℓ_1 heuristic with polishing lies close to the exact optimal trade-off curve computed by enumeration, and never below it.

Sparse signal reconstruction. The convex-cardinality problem minimize $\|Ax - y\|_2$ subject to $\text{card}(x) \leq k$ has the ℓ_1 heuristic minimize $\|Ax - y\|_2$ subject to $\|x\|_1 \leq \beta$, called the LASSO; another form, minimize $\|Ax - y\|_2 + \gamma\|x\|_1$, is called BASIS PURSUIT DENOISING.

Example: A signal $x \in \mathbb{R}^n$ with $n = 1000$ and $\text{card}(x) = 30$ is measured by $m = 200$ random noisy measurements $y = Ax + v$, $v \sim \mathcal{N}(0, \sigma^2 I)$, $A_{ij} \sim \mathcal{N}(0, 1)$. The ℓ_1 reconstruction with $\gamma = 10^{-3}$ recovers the 30 spikes almost exactly, while the ℓ_2 reconstruction, minimizing $\|Ax - y\|_2 + \gamma\|x\|_2$ with $\gamma = 10^{-3}$, is a dense, noisy signal that bears little resemblance to the original.

Some recent theoretical results. Suppose $y = Ax$ with $A \in \mathbb{R}^{m\times n}$ and $\text{card}(x) \leq k$. To reconstruct x we clearly need $m \geq k$, and if $m \geq n$ and A is full rank we can reconstruct x without the cardinality assumption. When does the ℓ_1 heuristic (minimizing $\|x\|_1$ subject to $Ax = y$) reconstruct x exactly? Results by Candès, Donoho, Romberg, Tao, and others show that (for some choices of A) if $m \geq (C \log n)k$, the ℓ_1 heuristic reconstructs x exactly, with overwhelming probability. Here C is an absolute constant, and valid A's include $A_{ij} \sim \mathcal{N}(0, \sigma^2)$, and A such that Ax gives the Fourier transform of x at m frequencies chosen from the uniform distribution.

18.4 Total variation reconstruction

To fit x_{cor} with a piecewise constant $\hat{x}$ with no more than k jumps, the convex-cardinality problem is to minimize $\|\hat{x} - x_{\text{cor}}\|_2$ subject to $\text{card}(D\hat{x}) \leq k$ (D the first order difference matrix). The heuristic is to minimize $\|\hat{x} - x_{\text{cor}}\|_2 + \gamma\|D\hat{x}\|_1$, varying γ to adjust the number of jumps. $\|D\hat{x}\|_1$ is the TOTAL VARIATION of the signal $\hat{x}$, and the method is called TOTAL VARIATION RECONSTRUCTION. Unlike ℓ_2-based reconstruction, it filters high-frequency noise out while preserving sharp jumps: on a signal in $\mathbb{R}^{2000}$ with a few jumps, total variation reconstruction for three values of γ keeps the jumps sharp, while ℓ_2 reconstruction either leaves noise or rounds off the jumps.

Example: *2D total variation reconstruction.* Let $x \in \mathbb{R}^n$ be the values of pixels on an $N \times N$ grid ($N = 31$, so $n = 961$), with the assumption that x has relatively few big changes in value (boundaries). We have $m = 120$ linear measurements $y = Fx$ ($F_{ij} \sim \mathcal{N}(0, 1)$). As a convex-cardinality problem,

$$\begin{array}{ll} \text{minimize} & \sum \text{card}(x_{i,j} - x_{i+1,j}) + \text{card}(x_{i,j} - x_{i,j+1}) \\ \text{subject to} & y = Fx, \end{array}$$

and the ℓ_1 heuristic, whose objective is a 2D version of total variation, is

$$\begin{array}{ll} \text{minimize} & \sum |x_{i,j} - x_{i+1,j}| + \sum |x_{i,j} - x_{i,j+1}| \\ \text{subject to} & y = Fx. \end{array}$$

For an image consisting of a raised disk, the TV reconstruction is nearly perfect, which is not bad for 8× more variables than measurements; the ℓ_2 reconstruction is poor, as one would expect with 8× more variables than measurements.

18.5 Iterated weighted ℓ_1 heuristic

Algorithm 18.1: Iterated weighted ℓ_1 heuristic

Input: A convex set C and $\epsilon > 0$; minimize $\text{card}(x)$ over $x \in C$.

$w := \mathbf{1}$

repeat

  Minimize $\|\text{diag}(w)x\|_1$ over $x \in C$

  $w_i := 1/(\epsilon + |x_i|)$, $i = 1, \ldots, n$

until convergence

The first iteration is the basic ℓ_1 heuristic. The reweighting increases the relative weight on small x_i; the method typically converges in 5 or fewer steps, and often gives a modest improvement (reduction in $\text{card}(x)$) over the basic ℓ_1 heuristic.

Interpretation. Without loss of generality take $x \geq 0$ (by writing $x = x^+ - x^-$ with $x^+, x^- \geq 0$ and replacing $\text{card}(x)$ with $\text{card}(x^+) + \text{card}(x^-)$). Use the approximation $\text{card}(z) \approx \log(1 + z/\epsilon)$ for $\epsilon > 0$, $z \in \mathbb{R}_+$, giving the (nonconvex) problem

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^n \log(1 + x_i/\epsilon) \\ \text{subject to} & x \in C, \quad x \geq 0. \end{array}$$

We find a local solution by linearizing the objective at the current point,

$$\sum_{i=1}^n \log(1 + x_i/\epsilon) \approx \sum_{i=1}^n \log(1 + x_i^{(k)}/\epsilon) + \sum_{i=1}^n \frac{x_i - x_i^{(k)}}{\epsilon + x_i^{(k)}},$$

and solving the resulting convex problem, minimize $\sum_{i=1}^n w_i x_i$ subject to $x \in C$, $x \geq 0$, with $w_i = 1/(\epsilon + x_i)$, to get the next iterate; repeating until convergence gives a local solution.

Example: *Sparse solution of linear inequalities.* To minimize $\text{card}(x)$ over the polyhedron $\{x \mid Ax \preceq b\}$ with $A \in \mathbb{R}^{100\times 50}$, the ℓ_1 heuristic finds $x \in \mathbb{R}^{50}$ with $\text{card}(x) = 44$, and the iterated weighted ℓ_1 heuristic finds x with $\text{card}(x) = 36$; the global solution, via branch and bound, has $\text{card}(x) = 32$.

Example: *Detecting changes in a time series model.* Consider the AR(2) scalar time-series model

$$y(t+2) = a(t)y(t+1) + b(t)y(t) + v(t), \qquad v(t) \text{ IID } \mathcal{N}(0, 0.5^2),$$

with the assumption that $a(t)$ and $b(t)$ are piecewise constant and change infrequently. Given $y(t)$, $t = 1, \ldots, T$, we estimate $a(t), b(t)$, $t = 1, \ldots, T-2$, by the heuristic

$$\text{minimize} \quad \sum_{t=1}^{T-2} \big(y(t+2) - a(t)y(t+1) - b(t)y(t)\big)^2 + \gamma \sum_{t=1}^{T-2} \big(|a(t+1) - a(t)| + |b(t+1) - b(t)|\big),$$

varying γ to trade off fit against the number of changes in a, b. On an example with $T = 300$ in which a and b each change once, total variation with $\gamma = 10$ locates the changes but leaves the estimates biased and slightly smeared, while the iterated (weighted) TV heuristic, with 5 iterations and $\epsilon = 0.005$, recovers the piecewise constant coefficients almost exactly.

18.6 Extension to matrices

Rank is the natural analog of cardinality for matrices. A CONVEX-RANK PROBLEM is convex, except for rank in the objective or constraints. The rank problem reduces to the cardinality problem when the matrices are diagonal, $\operatorname{rank}(\operatorname{diag}(x)) = \operatorname{card}(x)$. The analog of the ℓ_1 heuristic is to use the NUCLEAR NORM $\|X\|_* = \sum_i \sigma_i(X)$ (the sum of singular values, the dual of the spectral norm); for $X \succeq 0$ it reduces to $\operatorname{tr} X$ (just as, for $x \geq 0$, $\|x\|_1$ reduces to $\mathbf{1}^T x$).

Factor modeling. Given a matrix $\Sigma \in \mathbf{S}^n_+$, find an approximation of the form $\hat{\Sigma} = FF^T + D$, with $F \in \mathbb{R}^{n \times r}$ and D diagonal and nonnegative. This gives an underlying factor model with r factors, $x = Fz + v$ with $v \sim \mathcal{N}(0, D)$ and $z \sim \mathcal{N}(0, I)$. The model with the fewest factors solves

$$\begin{array}{ll} \text{minimize} & \operatorname{rank} X \\ \text{subject to} & X \succeq 0, \quad D \geq 0 \text{ diagonal}, \quad X + D \in C, \end{array}$$

with variables $D, X \in \mathbf{S}^n$, where C is a convex set of acceptable approximations to Σ, for example via the KL divergence,

$$C = \{\hat{\Sigma} \mid -\log\det(\Sigma^{-1/2}\hat{\Sigma}\Sigma^{-1/2}) + \operatorname{tr}(\Sigma^{-1/2}\hat{\Sigma}\Sigma^{-1/2}) - n \leq \epsilon\}.$$

The TRACE HEURISTIC minimizes $\operatorname{tr} X$ subject to $X \succeq 0$, $D \geq 0$ diagonal, $X + D \in C$, with variables $d \in \mathbb{R}^n$ (the diagonal of D) and $X \in \mathbf{S}^n$.

Example: Let $x = Fz + v$ with $z \sim \mathcal{N}(0, I)$, $v \sim \mathcal{N}(0, D)$, D diagonal, and $F \in \mathbb{R}^{20 \times 3}$, and let Σ be the empirical covariance matrix from $N = 3000$ samples. With the set of acceptable approximations $C = \{\hat{\Sigma} \mid \|\Sigma^{-1/2}(\hat{\Sigma} - \Sigma)\Sigma^{-1/2}\| \leq \beta\}$, the trace heuristic is

$$\begin{array}{ll} \text{minimize} & \operatorname{tr} X \\ \text{subject to} & X \succeq 0, \quad d \geq 0 \\ & \|\Sigma^{-1/2}(X + \operatorname{diag}(d) - \Sigma)\Sigma^{-1/2}\| \leq \beta. \end{array}$$

As β decreases, $\operatorname{rank} X$ increases. For $\beta = 0.1357$, the knee of the trade-off curve, the angle between $\mathcal{R}(X)$ and $\mathcal{R}(FF^T)$ is 6.8° and $\|d - \operatorname{diag}(D)\| / \|\operatorname{diag}(D)\| = 0.07$: the factor model has been recovered

from the empirical covariance.

18.7 Conclusions

Modeling. Problems in engineering design, data analysis and statistics, economics, management, and other fields can often be expressed as mathematical optimization problems, and techniques exist to take into account multiple objectives or uncertainty in the data. Roughly speaking, tractability in optimization requires convexity: algorithms for nonconvex optimization find local (suboptimal) solutions, or are very expensive. Surprisingly many applications can be formulated as convex problems.

Theoretical consequences of convexity. Local optima are global; there is an extensive duality theory, giving a systematic way of deriving lower bounds on the optimal value, necessary and sufficient optimality conditions, certificates of infeasibility, and sensitivity analysis; and there are solution methods with polynomial worst-case complexity theory (with self-concordance).

Practical consequences of convexity. (Most) convex problems can be solved globally and efficiently: interior-point methods require 20 to 80 steps in practice; the basic algorithms (Newton, the barrier method, . . .) are easy to implement and work well for small and medium size problems (larger problems if structure is exploited); more and more high-quality implementations of advanced algorithms and modeling tools are becoming available; and high-level modeling tools like CVX ease modeling and problem specification.

How to use convex optimization in an applied context:

- use rapid prototyping and approximate modeling: start with simple models, small problem instances, and inefficient solution methods; if you don't like the results, there is no need to expend further effort on more accurate models or efficient algorithms;
- work out, simplify, and interpret the optimality conditions and the dual;
- even if the problem is quite nonconvex, convex optimization can be used in subproblems (for example, to find a search direction), or by repeatedly forming and solving a convex approximation at the current point.

Further topics. Methods for very large scale problems; subgradient calculus and convex analysis; localization, subgradient, and related methods; distributed convex optimization; and applications that build on or use convex optimization.

Source. Stephen Boyd, *EE364A: Convex Optimization I*, Lecture 18 (2023); EE364b slides, *ℓ_1-norm methods for convex-cardinality problems*, part I and part II; original slides, *Conclusions*.